Manual

Advanced Configurations: MES Terminal AIP

EAT-AIP 8.2

Version 1.1.23049

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Contents

1 Extended Application Training: MES Terminal

Purpose

The configuration options described in this function package enable varied modifications to dialogs, dialog fields, formats, buttons of the HYDRA shop floor program AIP 8.2 (Acquisition and Information Panel).

Implementation notes

The function package is used if you would like to change

- the dialog structure
- dialog fields, labeling and units
- data types of dialogs including value ranges
- buttons and labeling
- the presentation of columns and if you would like to add further columns to lists

Integration

The AIP terminal provides various options to change dialogs. Changes are either carried out directly at the shop floor client or in the dynamic dialog configuration of the MOC if the presentation of input dialogs should be changed.

Features

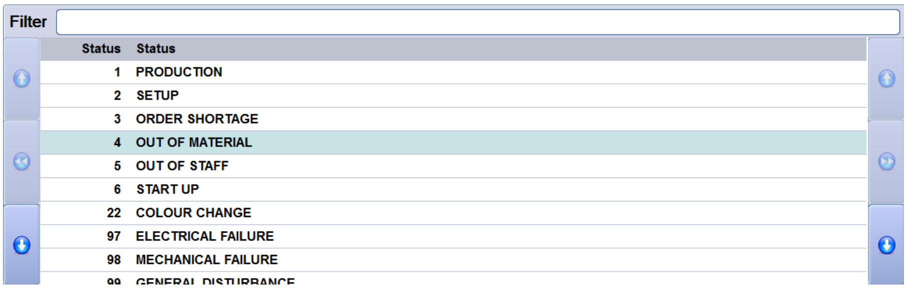
- General terminal configurations
- Configuration of grid layout
- Button configuration (basic screen)
- Configurations for the virtual keyboard
- Configurations for barcode input
- Dynamic dialog configuration for input fields and dialog buttons

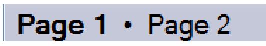
2 AIP2 Operation

2.1 Special Control and Display Elements on the AIP2

Tables

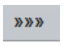
Uniform selection lists are used in AIP 8.2 posting dialogs:





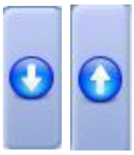
If information is available for more than one page, the page numbers are displayed below the table. The current page is highlighted in bold letters. If the user clicks/touches a page, the display directly changes to this page.

If more pages are available than the page numbers displayed, the following buttons can be displayed on the left or right hand side depending on the context (available as of SP10/2016):

-  : If you click this button, the system jumps to the first page of the next page navigation. This means: If *Page 1 ... Page 9* were displayed for the page navigation, the system jumps to *Page 10*.
-  : If you click this button, the system jumps to the first page of the next page navigation. This means: If *Page 10 ... Page* 18 were displayed for the page navigation, the system jumps to *Page* 9.
-  : If you click this button, the system directly jumps to *Page 1*.
-  : If you click this button, the system directly jumps to the last page.

You can select an operation using the mouse, touch screen, keyboard (arrow keys:'↑' or '↓'), scanner or by entering it manually.

The content of tables or lists depends on the respective context. Example: When you log on an operation, those operations are available that are included in the sequencing list or planned for the respective workplace or group. When you interrupt an operation, only running operations are available for selection.

 Scrolling page by page (up or down) in the table.

 Scrolling to the left or right. Only those buttons are activated that make sense for the current situation (context sensitive). This figure shows that scrolling to the left has been deactivated.

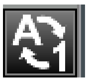
Optionally you can display a “table filter” (customization). This is an automatic filter that, once it has been entered, directly affects the table without having to update it. This process is realized through full-text search for (defined) columns. The search is *case-insensitive*.

Virtual keyboard

Using the virtual keyboard, you can enter data manually via touch screen or a connected mouse. The virtual keyboard is displayed automatically as soon as the focus is on an input field. The keyboard layout, which is installed and activated in the Windows language settings, specifies the layout of the virtual keyboard.

 Moving the virtual keyboard

 Hiding the keyboard for 10 seconds

 Switching between the alphanumeric and numeric keyboard

 Selecting the keyboard layout (language)

 Changing the scaling/size of the keyboard

To move the keyboard, you must configure the driver accordingly (configuration in the control panel of the terminal/PC)!

If you do not want to display the virtual keyboard in general, you must enter the parameter –t in the entry *parameters*= of the configuration file *ctaip.ini.*

Date display

AIP supports a country-specific date format in dynamic dialogs. The option "short date" has to be selected in the "regional settings" of the Windows "control panel" of the terminal/PC. Please note:

- Years are always four characters long.
- Months and days are always 2 characters long.
- Allowed separators are: '-' (minus), '/' (backslash) and '.' (dot).
- Blanks must not be included in the "short date" format, i.e. the <BLANK> separator is not allowed.
- The date separator "." (dot) is only allowed in connection with the DD.MM.YYYY format.
- The date format, which might possibly be configured in dynamic dialogs, is ignored.

Examples

- English(USA) MM/DD/YYYY
- Danish MM-DD-YYYY
- Customer-specific 1 YYYY-MM-DD
- Customer-specific 2 YYYY/MM/DD

- Customer-specific 3 MM/YYYY/DD

Note

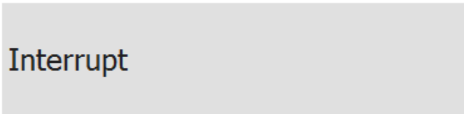
If the date format used is other than the permitted formats, a note appears when the program is started and the date format is set to MM/DD/YYYY.

In the status bar, the year format is shortened and displayed only with two characters.

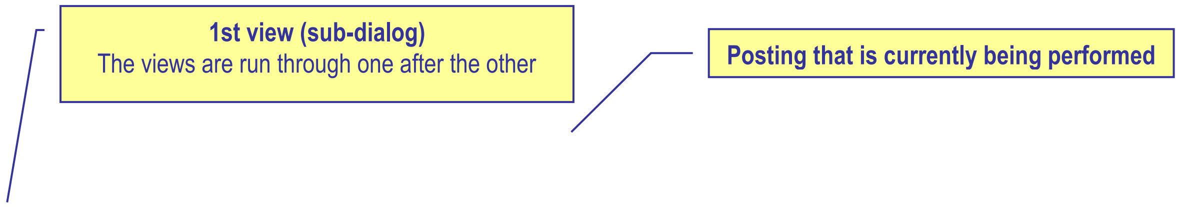
2.2 General description of the posting process on the AIP2

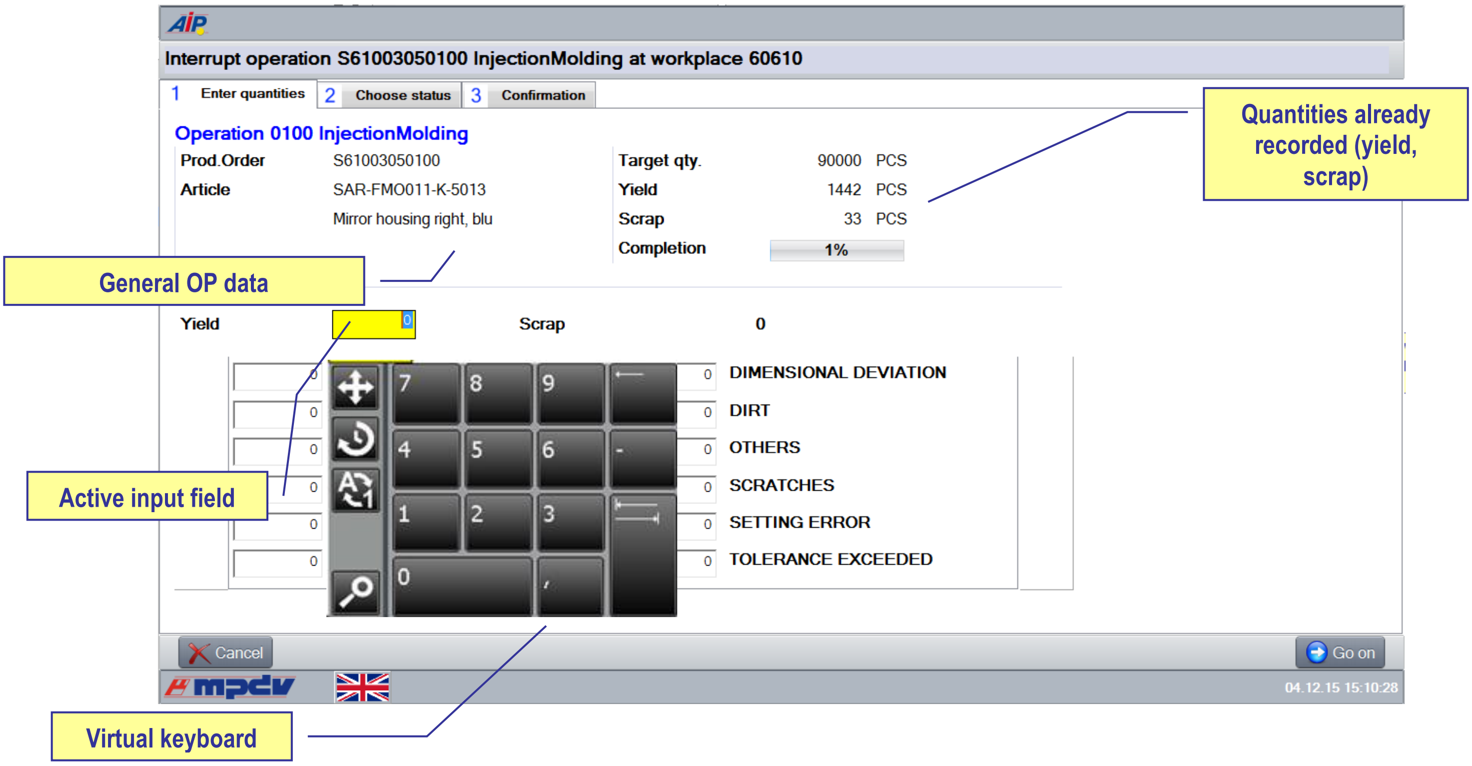
Many AIP posting dialogs are divided into several views (sub-dialogs). These views (sub-dialogs) cover the entire screen so that only one dialog is visible at a time. In a “workflow concept” the user is navigated through the posting dialog step by step. In the following, this process is described using the example *Interrupt operation*. Other posting dialogs are operated in the same way.

The action *Interrupt operation* is performed. To start this action, you click the button *Interrupt* when you have selected an operation:



The dialog *Interrupt operation* opens and the first view (sub-dialog) is displayed. The header displays the function that is currently being executed (here: *Interrupt operation).*





In the first dialog *Enter quantities,* the user can enter the produced yield or scrap quantities. Subject to the active input field, the virtual keyboard is shown or hidden automatically.

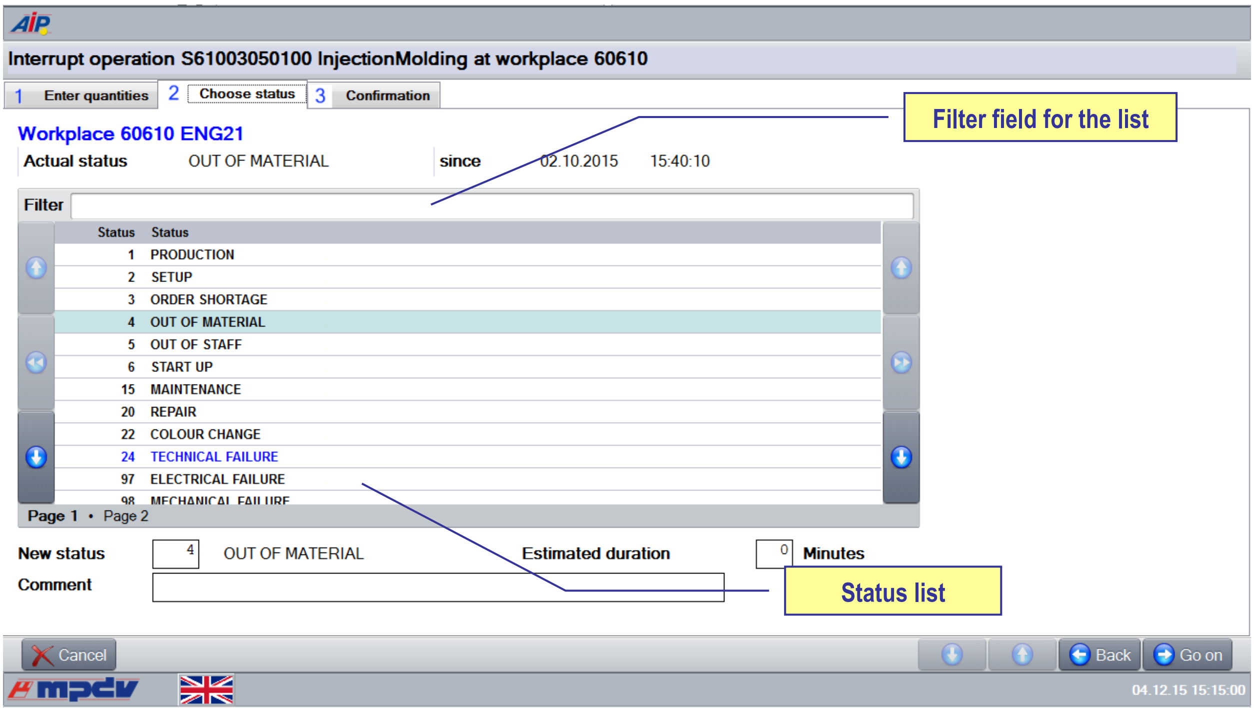
Quantities can be entered using the virtual or real keyboard. The user can go to the next field using the tabulator key (which can also be found on the virtual keyboard). When the user has entered all values in the first view, the next view (sub-dialog) can be opened by clicking ***Next***.

The ***Cancel*** button is displayed in all sub-dialogs. Click this button to cancel/close the entire process at any time.

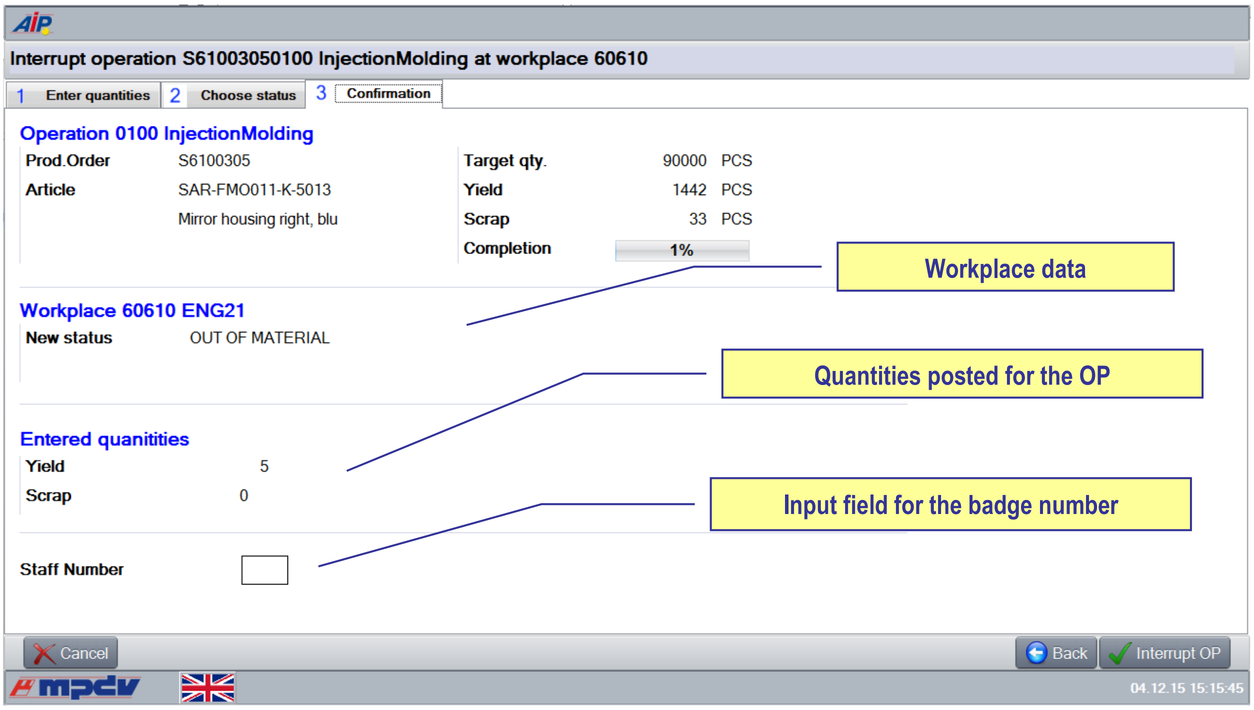
To open the next view (***Select status*** in the example), click the ***Next*** button or another tab (in our example: ***Select status*** or ***Confirmation).*** Please note in this context, that no view can be skipped when they are navigated upwards (view 1 → view 2 → view 3). This means: When you are in the first view *(enter quantities)* and you click the third view *(confirmation),* the second view *(select status)* will be displayed first.

Vice versa, when navigating downwards (e.g. from the *confirm* view to the *enter quantities* view), each view can directly be opened by clicking the required tab. In this case, views can actually be skipped. Using the ***Back*** button, views are opened one after the other (upwards).

As long as the dialog has not been confirmed, the data entered can be changed at any time by scrolling back and forth.



In the second view *Select status*, you select the workplace status that is set, when the operation has been interrupted. You can select the status from the status list displayed. This list can be restricted using the *Filter* field. Once the required values have been entered, the next view/sub-dialog can be opened by clicking ***Next*** (in our example it is the last view).



The sub-dialog *Confirmation* shows a summary of all values entered in the dialog. If the user agrees with the entered data, the ***Interrupt operation*** dialog can be confirmed, once the badge number has been entered. Then the dialog is sent to the server and posted.

If an input field of the dialog is not completed properly (e.g. a mandatory field is empty), the field is highlighted in red in the respective view and gets the focus. The user can then directly correct the value.

If a workflow dialog is opened, you can click the ESC key to directly exit the dialog. This exit is also possible, if the virtual keyboard is displayed. As a consequence, you cannot use the ESC key to close the virtual keyboard.

3 Main View with Tiles

With the AIP2, the user can switch between the tile design optimized for touchscreens and the list format. By default, the tile layout is shown, which is described in the sections that follow.

To ensure proper processing and posting, terminals with "MDE" operation mode must <u>not</u> be switched off during times without shift.

3.1 Main view – header and footer

Header

The *AIP* logo is displayed on the top left of the screen, which may be replaced with a customer logo after configuration.

Possible messages are displayed to the right of it (e.g. if a dialog is opened for more than five minutes).

A separate window opens to display error messages that occur during data collection (e.g. validity checks).

Main views

You can assign a maximum of 16 workplaces or machines to the *AIP2* terminal. The different workplaces are listed in the order that they were assigned to the terminal on the client. .

In the main view of the *AIP2,* you can use the button "< Overview“ to switch to the icon view of workplaces. In the terminal configuration of the client you can specify whether you want to use the icon view. The sections that follow describe the main view and the icon view.

Footer

The MPDV logo can be found at the bottom left of the AIP2 terminal. Double clicking the logo opens the info dialog where you can start further administration functions. This dialog closes automatically after approx. 5 seconds.

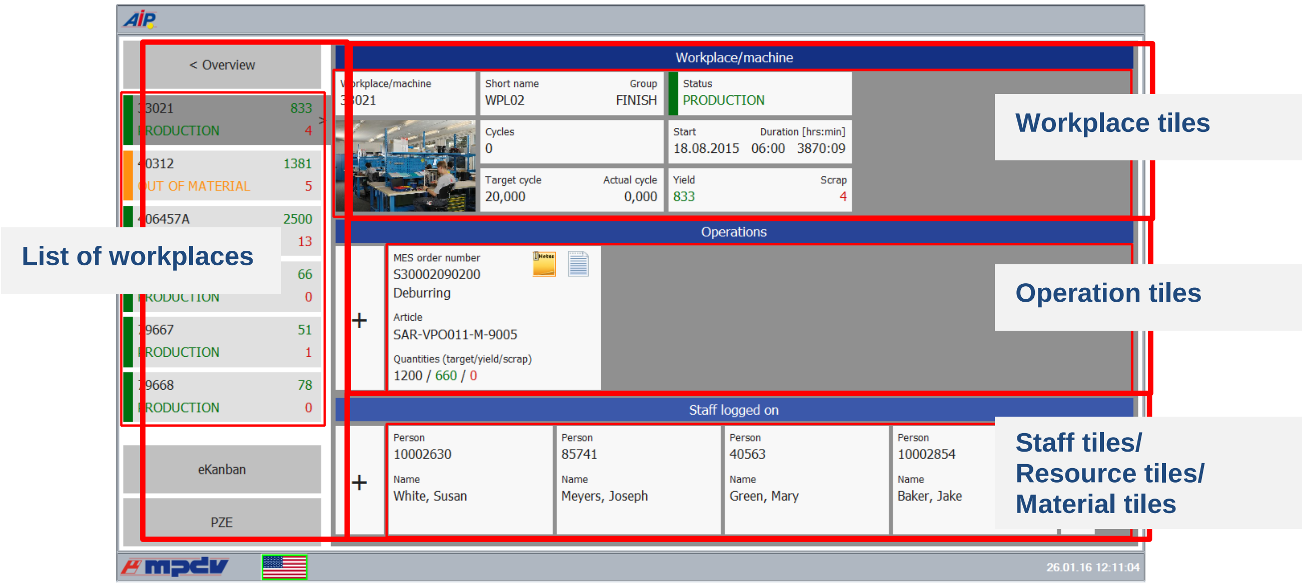
In the middle, further information is displayed: the current terminal status, AIP2 version number, date of the build, IP address of the server and the terminal number.

The current date and time are displayed to the right.

AIP2 statuses

	Network connection has been established. The terminal is ONLINE. Server communication is enabled. All saved data records have been transferred.
	The terminal is sending data to the server.
	No network connection or no connection to the server. The terminal is OFFLINE. Server communication is interrupted. Online functions, such as the display of information, are disabled. But you can still record certain postings. These postings are transferred to the server, once data connection has been re-established.
	Data is being received. The terminal reads files from the server or writes data to the server.
	The terminal is sending saved data records to the server.
	DEMO mode The terminal is in DEMO mode, i.e. server communication is disabled.

3.2 Main view with "tiles"



Please note: The actual display can be different to the above illustration.

Subject to the configurations made, the main view with tiles consists of two or three rows of tiles. While the first two rows of tiles (workplace and operation tiles) are always displayed, it is up to the user whether or not the third row of tiles is shown (optional display). In the configuration of workplaces you can configure for each workplace separately if you want to show the third row of tiles.

Additionally, there is the list of workplaces to the left. Here, you can select the workplaces for which details are displayed on the right-hand side.

List of workplaces

The workplace list shows all workplaces/machines assigned to the terminal. If many workplaces are assigned, swipe to get to the workplaces displayed further down.

This information is shown for the workplaces:

Machine/workplace number

Shows the machine and/or workplace number.

Status

The status is displayed for each machine in color on the left-hand side and also the status text is colored. Coloring is as follows:

- green: production
- yellow: assigned status
- red: not assigned

If the production lock is enabled, an exclamation mark is displayed in the same color as the status.

Quantities

On the right-hand side, the first figure shows the produced yield in green and the red figure shows the produced scrap.

If you have enabled the *Compensate manual quantities* option (e.g. set off scrap against yield) and the machine list also shows shift-related quantities, they will not be updated immediately. The application only updates the quantities, once the lists have been reloaded.

Unit for yield and scrap

If no operation is logged on, the primary quantity unit from the workplace/machine configuration is displayed as unit for yield and scrap. If an operation is logged on, the primary quantity unit of the operation is displayed.

Workplace tiles

The workplace tiles provide the following details:

Workplace/machine

No workplace or machine number is displayed.

Short name / group

The short name of the workplace or machine and the machine group are displayed.

Status

The status is displayed in color and as status text on the left-hand side. Coloring is as follows:
- green: production
- yellow: assigned status
- red: not assigned

If the production lock is enabled, an exclamation mark is displayed in the same color as the status.

Machine image

Shows the picture of the machine stored in the configuration of workplaces.

Clocks

Shows the recorded machine cycles of the current shift.

Start / Duration [hrs:min]

Point in time since the status has been available and the resulting duration at the current point in time.

With BDE workplaces[1], this point in time refers to the last manual status change. With MDE workplaces, it refers to the time when the last status change was identified (for machine connections). It also refers to the point in time when the status was last changed manually or to the time of the last shift change.

Target/actual cycle

Current target and actual cycle of the workplace.

The largest target cycle of all operations logged on to the workplace is shown. The largest target cycle is transferred to the MDE for monitoring.

If the target cycle is smaller than the minimum cycle time, the target cycle is still shown.

If an operation is logged off or interrupted, the largest target cycle of the remaining operations is identified and displayed. After logoff or interruption of the last operation at the workplace, the last target cycle set is still displayed.

If no operation is logged on, the target cycle specified in the machine list is displayed. Thus, even after a restart, the terminal can get the target cycle that last applied.

Yield / Scrap

Yield and scrap quantities of the current shift produced at the machine/workplace.

KPIs: OEE, utilization efficiency, scrap ratio

This function is only available if you enable the extension aipkpi.

Shows the KPIs OEE, utilization efficiency and scrap ratio. The application calculates the KPIs at cyclic intervals (scheduler job "MDE keyfigure calculation“). The KPIs always refer to the current shift. The application calculates the KPIs based on the formulas that are also used for the OEE report and/or the efficiency report on the MOC. The application shows the KPIs with two decimal places. To the right of the KPI, the AIP2 GUI highlights in color if limit values are exceeded or not reached. If you have not defined limit values, the application shows the KPI in gray, otherwise in the color you defined for exceeding/not reaching limit values. For further information on the configuration, refer to the document MDE_KPI_Configuration.pdf.

[1] An MDE workplace is a workplace that is assigned to a terminal, which runs in the “MDE” operation mode. Otherwise, it is a BDE workplace.

The AIP calculates and updates data at cyclic intervals. This may result in deviations between the collected values and the displayed KPIs.

Linked functions

If you click the workplace status, the dialog for changing the status opens. If you click one of the other tiles, the dialog opens where you can start the functions available for the selected workplace.

The buttons displayed depend on the selected workplace. The button *Lock production status*, for example, is only available for MDE machines.

List of operations logged on

The middle area on the right-hand side shows the logged on operations as tiles. The following data is shown:

MES order number

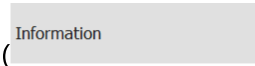
Order number and operation number of the operation logged on. The combination of these two numbers is the *MES order number.*

Article

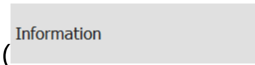
Article defined for the operation.

Quantities (target / yield / scrap)

Shows the target quantity defined for the operation, the produced yield and the scrap. The yield and scrap quantities integrate the counter readings of the available machine connections.

This icon is displayed if at least one note has been recorded for this operation in the graphic planning board of the client that must be shown on the terminal. To display the note(s), click the icon or click the operation and select the *Information* button (

).

This icon is displayed if at least one long text is stored for this operation. Click this icon to display the long texts or click the operation and select the button *Information* (

).

If you click an operation, a screen opens that shows the workplace/operation data. Via this screen, you can also select the operation-related functions.

Operation tiles

In addition to the fields already described, the operation tiles also show the following data:

Comments

This tile shows the user fields 53 and 54 (alphanumeric, 20 characters) of the operation. To edit these fields, you must store a respective user field key for the operation, which includes these two fields.

Completion in %

The bar shows the proportion of “yield”, which has been produced until now, compared to the “target quantity”.

Since logon (target / yield / deviation)

The production quantity to be expected since the OP has been logged on (depending on the cycle time, partitioning and the time when no production lock has been set for the machine). If the terminal program has been restarted after the OP logon, no value can be calculated.
Calculation:
Target Since Logon = Net Running Time[sec] * Partitioning/Target Cycle[sec/stroke]
Net Running Time: Time since logon while the production lock has not been set. This calculation does not integrate the breaks specified in the shift model or the status times posted to RPA 12 (resource performance account).

Deviation (in percent) between the calculated target quantity since logon and the quantity which has actually been produced “since logon”.
Calculation: Deviation[%] = 100% * (Yield Since Logon - Target Quantity Since Logon) / Target Quantity Since Logon.

As of CTAIP 8.2.1.32:

Up to now, the target quantity since OP logon was only calculated if the workplace/machine was assigned to a terminal with operation mode "MDE processing". As of CTAIP 8.2.1.32, the target quantity since OP logon is also calculated if the terminal is configured with operation mode "BDE processing".

If the terminal has been restarted after an operation logon, the target quantity cannot be calculated correctly. To improve transparency, an "*" (asterisk) is shown behind the target quantity (since OP logon) in this case. The asterisk indicates that the target quantity now displayed no longer refers to the time of the operation logon, but to the time of the terminal restart.

While the workplace/machine status 999 is displayed, the target quantity since OP logon is "---". If the status 999 is again changed within a "free shift", the target quantity since OP logon is calculated using the point in time of the OP logon, of the shift start or of the terminal start.

To disable the calculation of the target quantity since OP logon on the terminal, you can use the configuration CalcTargetYieldSinceLogon=0 in the hytnrcfg.ini. The quantity is then displayed using "---".

With workplaces/machines that are configured as "Machining centers", the calculation of the target quantity since OP logon is generally disabled because this calculation contradicts the principles of the machining center.

Planned duration

The field *Planned duration* displays the target processing time of the operation in format [h:mm].

Partitioning

Calculate the displayed partitioning as follows:

TLG_M	Partitioning of the workplace/machine (TLG in mnr.lst)
DIV_M	Pulse factor of workplace/machine (IMPFAKT in mnr.lst)
TLG_{AG}	Partitioning of the operation (TLG in anr.lst)

The application shows the calculated partitioning without decimal places, provided it is an integer value.

In case the partitioning or pulse factor of a machine or an order is 0, calculation is based on the value 1.

Displaying the "3rd list"

The third list is optional. You can configure the third list in the configuration of workplaces. The following lists can be displayed:

- List of staff logged on to the currently selected workplace (BDE)
- List of resources logged on to the currently selected workplace (WRM)
- List of materials/input batches (MPL/TRT) logged on to the currently selected workplace
- List of output batches produced in the currently selected operation (MPL/TRT)

In case you have enabled several lists, you can switch between these lists in the header line that is located above the third list. Activated lists can be selected one after the other.

Maintenance status

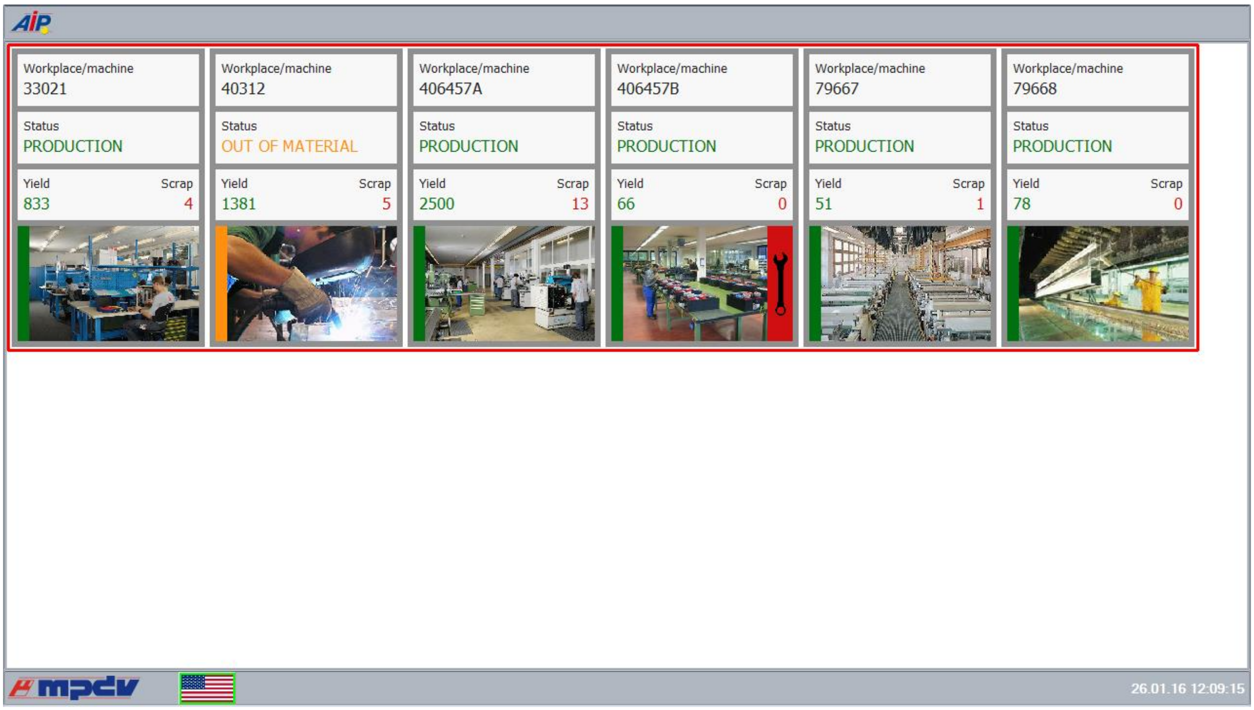
If you have purchased the license for the maintenance calendar, the maintenance status is displayed using a yellow or a red field showing a wrench. The color displayed depends on the required maintenance activity.

Calling functions

The functions available are assigned to the relevant objects. Example: The functions *Log person off* and *Log all staff off* are displayed if you click on a person logged on.

3.3 Icon view of workplaces

You can enable this view in the configuration of terminals via the client. Then open this view by clicking the button "< Overview“ in the main view. This view shows workplaces in a clear structure and with an image. It shows important information on the single workstations:



Please note: The actual display can be different to the above illustration.

A colored bar to the left of the image indicates the current status of workplaces/machines:

- green: production
- yellow: assigned status
- red: not assigned

Each tile includes the workplace/machine number, the status (text) of the workplace, the yield and scrap quantity and the image of the workplace.

A colored background with a caliper and/or wrench to the right of the image indicates if an inspection or maintenance is due for the workstation.

If you enable the extension aipkpi, the application also shows the KPIs OEE, utilization efficiency and scrap ratio.

The application calculates the KPIs at cyclic intervals (scheduler job "MDE keyfigure calculation“). The KPIs always refer to the current shift. The application calculates the KPIs based on the formulas that are also used for the OEE report and/or the efficiency report on the MOC. The application shows the KPIs with two decimal places. To the right of the KPI, the AIP2 GUI highlights in color if limit values are exceeded or not reached.

The AIP calculates and updates data at cyclic intervals. This may result in deviations between the collected values and the displayed KPIs.

If you click on a tile, the previously described main view is displayed and the workplace is automatically selected. From there, you can perform the postings for the selected workplace.

Use the option "< Overview" to exit the main view and to return to the icon view.

As part of the advanced configuration options, you can customize the layout of display lists, the displayed data fields and functions. For technical reasons, however, you cannot change the sort sequence of display lists in the main view of the terminal.

As of AIP 8.2.2.28, you can automatically change from the main view with tiles to the icon view of workplaces after a configured time. The configuration `AUTOMATIC-CHANGE-TO-START-DISPLAY` is described in the document AIP2_Configuration_hytnrcfg.pdf.

4 Basic Screen as List View

The new tile design can be disabled for the AIP2 terminal. This chapter describes the basic screen with disabled tiles.

In general, the AIP2 has been designed for entries to be made via touch screen. The corresponding functions can be started, selected or executed by touching the buttons or using the displayed virtual keyboard. Selection lists are provided in many cases, as an alternative to manual entries. Required entries can easily be selected from these lists.

Barcodes can be imported/entered in the current dialog using barcode readers, handheld scanners, or swipe card readers. Subject to the barcode prefix, certain data (e.g. operation data) can directly be assigned to the corresponding input field, without having to focus this input field explicitly.

It goes without saying that mouse and keyboard may also be used.

To ensure proper processing and posting, terminals with "MDE" operation mode must <u>not</u> be switched off during times without shift.

4.1 Basic screens – header and footer

Header

The *AIP* logo is displayed top left of the screen, which may be replaced with a customer logo after corresponding configuration.

Possible messages are displayed to the right of it (e.g. if a dialog is opened for more than five minutes).

A separate window opens to display error messages that occur during data collection (e.g. validity checks).

Basic screens

A maximum of 16 workplaces or machines can be assigned to the *AIP2* terminal. The single workplaces can be found within the list area in the order assigned to the terminal via the client. .

As regards the basic screen of the *AIP2* terminal, the user can choose between a tabular view, field-related view and an icon view. This can be configured via the configuration of terminals in the client. The single basic screens are described in the sections that follow.

Footer

The MPDV logo can be found at the bottom left of the AIP2 terminal. Double clicking the logo opens the info dialog where further administration functions can be started. This dialog closes automatically after approx. 5 seconds.

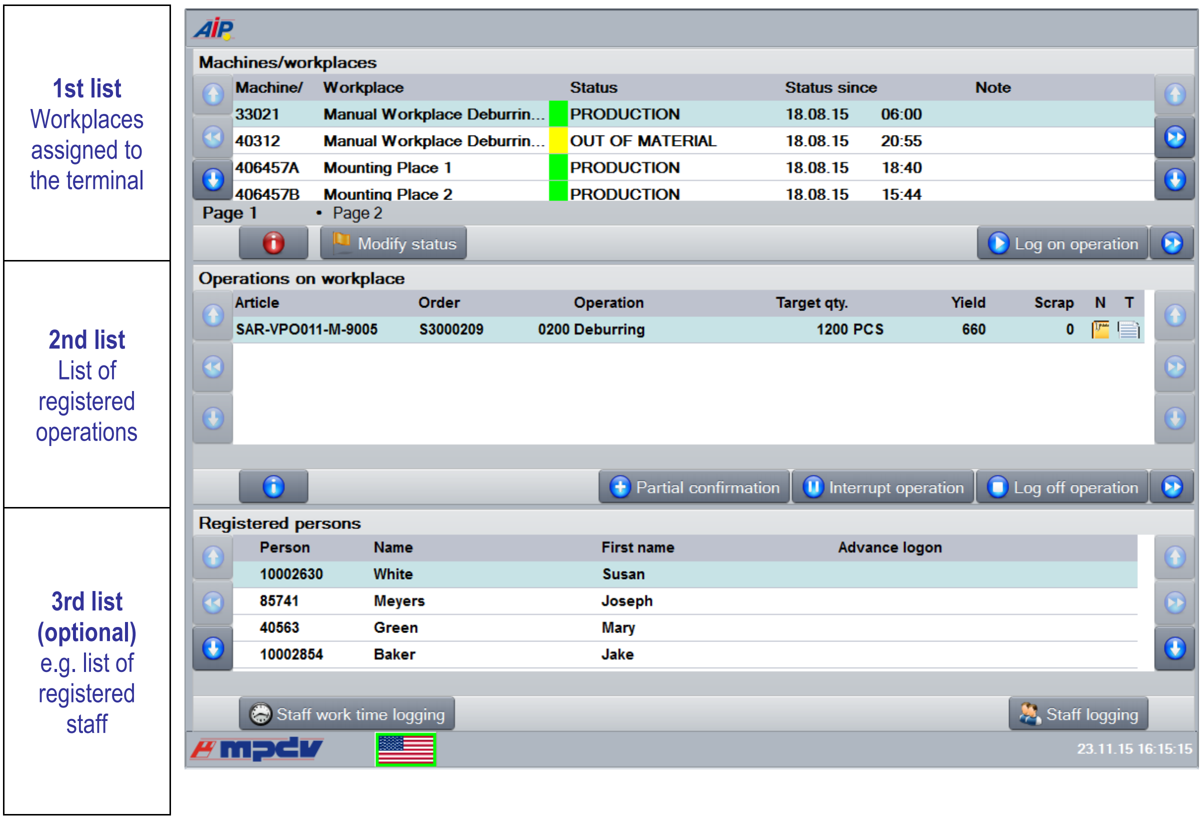
Further information is displayed in the center : the current terminal status, AIP2 version number, date of the build, IP address of the server as well as the terminal number.

The current date and time are displayed to the right.

Terminal status

	Network connection has been established The terminal is ONLINE. Server communication is enabled. All saved data records have been transferred.
	The terminal is sending data to the server.
	No network connection or no connection to the server. The terminal is OFFLINE. Server communication is interrupted. Online functions, such as the display of information, are disabled. But certain postings can be recorded anyway. These postings are transferred to the server, once data connection has been established.
	Data are being received. The terminal reads files from the server or writes data to the server.
	The terminal is sending stored data records to the server.
	DEMO mode The terminal is in the DEMO mode, i.e. server communication is disabled.

4.2 Basic screen “tabular view“



Subject to the configurations made, the tabular basic screen consists of two or three tables. While the first two tables are always displayed, it is up to the user whether or not the third table is shown (optional).

“Machines/workplaces" table

The upper table shows the workplaces assigned to the terminal. The following columns are displayed.

Machine/workplace

The machine or workplace number as well as a description are displayed.

Status

The status is highlighted in color and the status text is shown. Coloring is as follows:
- green: production
- yellow: assigned status
- red: not assigned

Status since

Point in time since the status is available.

For ADE workplaces[2] the point in time refers to the last manual status change. For MDE workplaces it refers to the time when the last status change was identified (for machine connections). It also refers to the point in time when the status was changed manually most recently or to the time of the last shift change.

Please note:

It is indicated here if the “lock production status” function is enabled for the machine/workplace.

Below the first list there is a row including the function buttons mainly relating to machines/workplaces. These functions are described in more detail in the sections that follow.

By way of “customizing” services it is possible to adapt the layout of the display lists, displayed data fields, sort sequences, etc. according to the customer’s requirements. For technical reasons, however, the sort sequence of display lists may not be changed in the basic screen of terminals. The software does not allow it.

Provided that the "compensate manual quantities" option (e.g. set off scrap against yield) is enabled and the machine list also shows shift-related quantities (no default setting), they will not be updated immediately. Quantities are only updated once the lists have been reloaded.

"Operations at workplace" table

The second table shows the operations currently logged on to the selected workplace. The following columns are displayed:

Article

Article defined for the operation

Order and operation

Order number and operation number of the registered operation. Together they build the *MES order number.*

Target quantity

Target quantity defined for the operation.

[2] We talk of an MDE workplace if this workplace is assigned to a terminal, which runs in the “MDE” operation mode. In any other case, it is an BDE workplace.

Yield

Yield already produced for this operation. The counters of possible machine connections are considered as well.

Scrap

Scrap quantity already produced for this operation. The counters of possible machine connections are taken into account as well.

N

It is indicated here if a note visible on the terminal has been recorded for this operation in the graphic planning board of the client. The note(s) is/are displayed by clicking the OP info button ().

T

If a long text is defined for this operation it is indicated here. The long text is displayed using the OP info dialog (button).

Below the second list there is a row that mainly includes function buttons relating to operations.

"3rd list" table

The third list is optional and may be configured. Information displayed in this list depends on the workplace configuration.

The following lists can be displayed:

- List of staff logged on to the currently selected workplace (BDE)
- List of resources logged on to the currently selected workplace (WRM)
- Materials/input batches logged on to the currently selected workplace (MPL/TRT)
- List of output batches produced in the currently selected operation (MPL/TRT)

The buttons below the third list (to the left) allow switching between these lists.

Please note

The staff logged on displayed in the third list is identical to the list displayed in the dialog “F5 staff logged on…”. Selecting a person in the third list does *not* affect the selection of the operation in the list of OPs running at the workplace. Therefore, it neither affects pre-assignment of the operation in the corresponding posting dialogs.

Toolbar in the basic screen

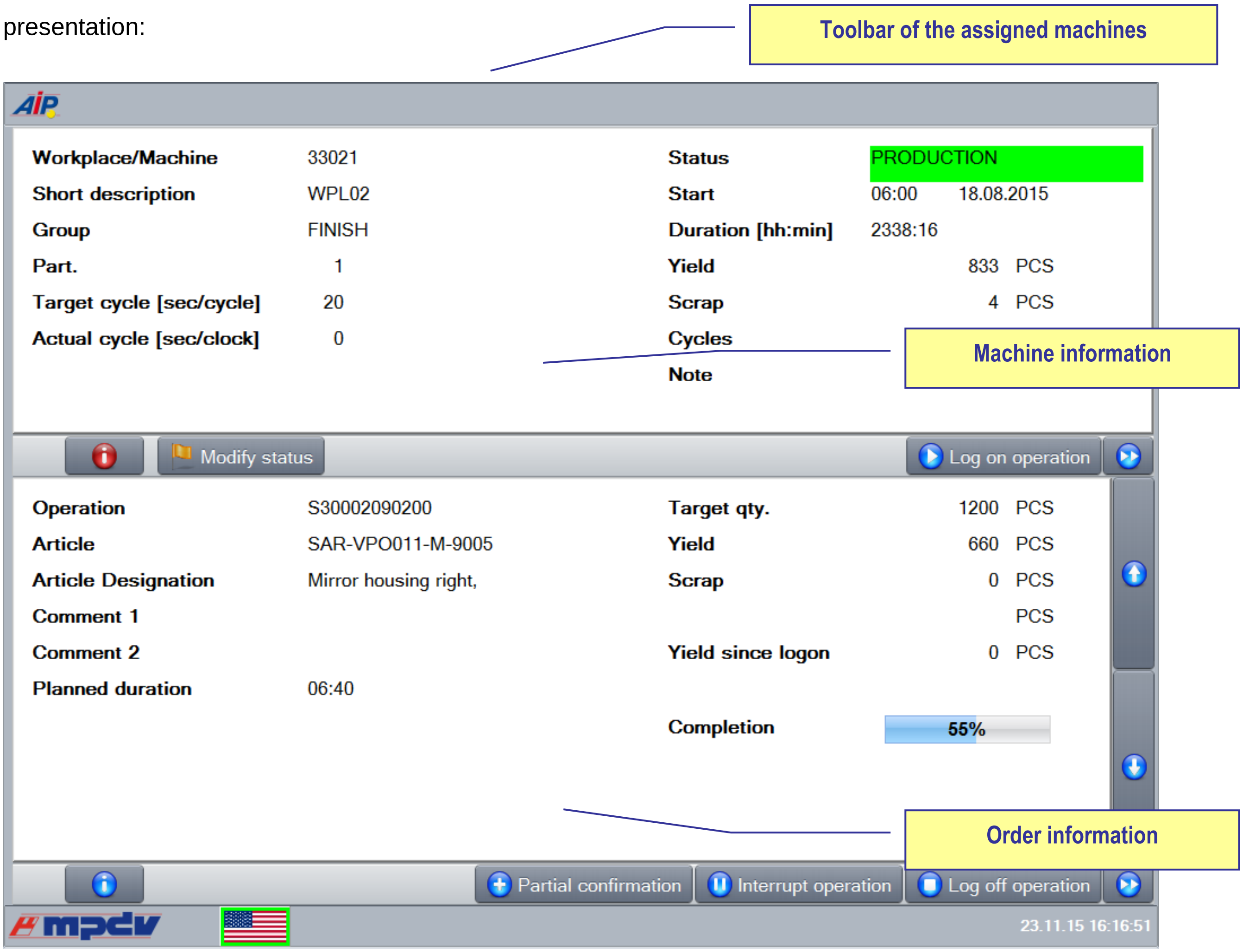
A toolbar, which may be customized, is assigned to each list included the basic screen. This makes the purpose of a function clear to the user. The “partial upload/confirmation” function can be found below the list of registered operations.

In fact, the toolbar may include several “tabs”, which can be made visible by scrolling to the right/left at the right/left end of the toolbar. A posting dialog (e.g. *change partitioning*) can be opened by clicking the corresponding button.

Please note
The displayed buttons depend on the context defined by the respectively selected workplace. Thus, the displayed buttons may vary when selecting another workplace/machine.

4.3 Basic screen "machine overview"

If the “change view” button is clicked in the basic screen, the view changes to the following presentation:



This presentation gives detailed information on a single machine, whereas the above toolbar still provides an overview of all assigned machines and workplaces.

The presentation consists of three sections:

Toolbar of the assigned machines

The color indicates the current status of all assigned workplaces. Coloring is as follows:
- green: production
- yellow: assigned status
- red: not assigned

It is possible to switch between machines by pressing a machine icon (requires a touch screen). Using the keyboard, the active machine can be selected by the arrow keys. To do this, the toolbar must be active.

Workplace/machine information

This display area shows information relating to workplaces/machines and shifts about the currently selected workplace.

Order information

This display area shows information on the registered order/OP. If several orders/OPs are logged on to the workplace, then extra arrow buttons are displayed. It is possible to switch between individual orders/OPs using these arrow buttons.

Notes on selected fields of the machine overview

Unit for yield and scrap

Provided that no operation is logged on, the primary quantity unit from the workplace/machine configuration is displayed as unit for yield and scrap. If an operation is logged on the primary quantity unit of the operation is displayed.

Partitioning

The displayed partitioning is calculated as follows:

$$\text{Partitioning} = \frac{TLG_M}{DIV_M} * \left[\frac{TLG_{OP1}}{DIV_{OP1}} + \frac{TLG_{OP2}}{DIV_{OP2}} + \cdots\right]$$

TLG_M	Partitioning of the machine (TLG in mnr.lst)
DIV_M	Pulse factor of the machine (IMPFAKT in mnr.lst)
TLG_{OPi}	Partitioning of the individual operation (TLG in anr.lst)

The resulting partitioning is displayed without decimal places, provided it is an integer value. Otherwise, 3 decimal places are shown.

In case partitioning or pulse factor of a machine or an order is 0, calculation is based on the value 1.

Having logged off all OPs, the machine continues working with the partitioning of the machine.

Target cycle

The largest target cycle of all operations running at the machine is always displayed in the machine overview of the terminal. If this OP is logged off the largest target cycle of the remaining OPs will be displayed.

In case no OP is logged on, the target cycle from the machine list is displayed. Thus, even after a restart, the terminal can get the target cycle that applied at last.

The largest target cycle is also transferred to MDE for monitoring.

Comment 1, comment 2

These two fields show the user fields 53 and 54 (alphanumeric with 20 characters) of the operation. To be able to edit these fields, a corresponding user field key containing these two fields must be defined for the operation.

Target since logon

The production quantity to be expected since the OP has been logged on (depending on the cycle time, partitioning and the time while the production status was not locked for the machine). No value can be calculated, in case the terminal program has been restarted since the OP was logged on.

Calculation:
TargetSinceLogon = NetRunningTime[sec] * Partitioning/TargetCycle[sec/stroke]
NetRunningTime: Time since logon while the production lock has not been set. This calculation does not take into account any breaks defined in the shift model or status times posted on RPA 12 (resource performance account).

Deviation [%]

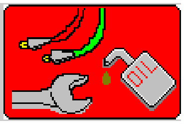
Deviation (in percent) between the expected target quantity since logon and the quantity which has actually been produced “since logon”.
Calculation: Deviation[%] = 100% * (YieldSinceLogon - TargetSinceLogon) / TargetSinceLogon

Completion

The bar represents the proportion of “yield”, which has been produced until now, compared to the “target quantity”.

Machine icon:

Provided that the WRM-WTK license has been purchased, the machine icon may be replaced with a picture showing a yellow or red oilcan. It all depends on the required maintenance activity:



4.4 “Machines as icons” basic display

This view can be configured as the default display in the configuration of terminals at the client It has the advantage that the user can tell from a distance whether or not all machines are in the “Production” status.

All MDE machines have their own buttons colored according to the corresponding status:

- green: Production
- yellow: Assigned status
- red: Not assigned

The button includes details on the workplace/machine number, the registered operation, the yield and scrap quantities as well as the status (text) of the workplace/machine.

If a button is touched , the “machine overview” basic display is shown for this workplace. From there, postings for the selected workplace can be performed using the standard buttons.

By clicking the “symbol” button (if configured) the view changes from the “machine overview” to the “icon view of machines”.

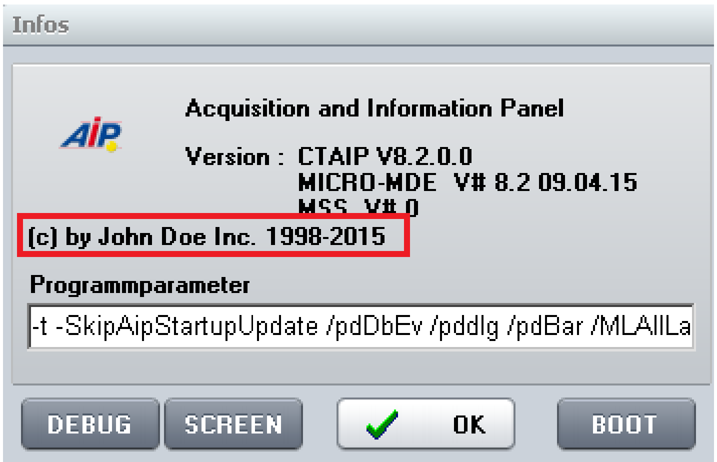
5 AIP2 -Local Configuration

5.1 Local Configuration ctaip.ini

The most important hardware and system settings are defined for each terminal in the CTAIP.INI file of the C:\MPDV\AIP2 directory.

Changes to the configuration file ctaip.ini are only enabled after rebooting the terminal software.

Layout configuration

Entry	Comment
Section [system]	
CompanyInfo=(c) by John Doe Inc. 1998-2015	Overwrites the copyright text in the "About" dialog 
parameters= … -t | +t	The virtual keyboard can be disabled/enabled, irrespective of the terminal type.
DEMO_ALL=ON	The information in the lower bar are normally hidden in demo mode. This option, however, causes all pieces of information to be displayed even in demo mode (version number and date, server, terminal number, online lights).
parameters= … -SkipDynDlgExtraction…	Internal option preventing the extraction of dialog fields/buttons when updating dialog data. (Only applies if WF directory ".\spool\$wf.$$$\" is available)
parameters= … -SkipAipStartupUpdate…	Internal option preventing the configuration files (*.ini,*.cfg,*.cfi) of the module DLLs (packets*.dll) and application DLLs (functions*.dll) from being synchronized when starting the terminal.
VirtScreenSize=640	All windows are started with the indicated resolution

Entry	Comment
VirtScreenRatio=16:9	The display ratio remains if the configuration VirtScreenSize is used to reduce the window. The width-to-height (aspect) ratio can be changed using VirtScreenRatio. Consequently, the width-to-height ratio 16:9 can be tested with a 4:3 monitor and vice versa. The value can be configured as "16:9" or as floating point value (e.g. 1,77777). Note: Information screens are specifically scaled if monitor screens in portrait format (e.g. 9:16) are used. This special scaling can be disabled in hytnrcfg.ini: [Tnr Konfiguration 0]->PortraitAlignment=off
Section [SKIN]	Configurations for the terminal's skin (see ctaiplay.ini) The online configuration of the terminal can be reached by the info dialog (click MPDV icon). ALT+F1 opens the control elements for the skin. Please note: Only in the design/GUI of AIP 8.1 and/or the dynamic dialogs
Saturation=0	Skin configuration / Default = 0
Hue=0	Skin configuration / Default = 0
Name=mpdv	Skin to be used / Default = mpdv
Active=false	Skins can only disabled in ctaip.ini! Default = true

Entry	Comment
Section [system]	
`Usr=21`	Distinct terminal number
`Hostname=192.9.200.24`	Internet address of the server
`Offlinetimeout=600`	The interval after which online access should be attempted the next time. The interval is specified in seconds
`Showcursor=on`	Show or hide mouse pointer in terminal application: on: mouse pointer active off: mouse pointer inactive
`Loadfile=` `ctnet\win\aip2.txt`	Configuration file to download the application from the server. The paths is relative to the installation directory of the system.

Entry	Comment
`Watchdog=on`	ON: Watchdog is activated OFF: Watchdog is not activated
`Demo=off`	'on': Offline demo mode; always off in the production environment!
`parameters=-t`	The –t parameter switches off the virtual keyboard.
`TMOUT_C=xxx`	Timeout for CONNECT to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
`TMOUT_S=xxx`	Timeout for SEND to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
`TMOUT_R=xxx`	Timeout for RECEIVE of the server If not specified, default = 120 seconds
`TMOUT_F=xxx`	Timeout for FILESERVER operations to the server If not specified, default = 10 seconds → Increase to 20 seconds for routing
Section [barcode]	**Configuration of customized barcode prefixes.**

`BarKenn90=MNR4`
`...`
`BarKenn99=ANR3`
	BarKenn90 > defines the prefix (here: 90); The ID from the dialog (= acronym) is assigned.

Section [comports]	

`com1=0`
`com2=MSS`
`com3=BAR`
`Com3=LEGIC`
`Com4=RFLESER`
	Assignment of serial interfaces to connected devices:
	MSS – machine interface
	BAR, LEGIC, RFLESER – various reading devices

Section [MSS-INIT]	
`MSS_DIALOG=10`	If the terminal is switched off longer than 15 minutes, a dialog is displayed on terminal restart. The user must then decide whether the counter pulses, which were recorded when the terminal was closed, are posted or discarded. The dialog closes automatically with "Yes" after an entered time has elapsed; in this case the counting impulses are posted. This value configures the time in seconds the dialog is open. If the terminal is switched off for less than 15 minutes, no dialog is opened; the counting pulses recorded in the switch-off phase are accepted and posted without confirmation. Please note: The value can also be configured in hytnrcfg.ini. Entries in the hytnrcfg.ini file take priority.

Entry	Comment
MSS_FILEAGE_MIN=5 MSS_FILEAGE_OVERTIME=delete	Additional configuration for **MSS_DIALOG** If the backup file for counting pulses of the terminal is older than 5 minutes, no dialog is opened and the back up file deleted. Quantities recorded at the time when the terminal was closed are not used/posted. Please note: The value can also be configured in hytnrcfg.ini. Entries in the hytnrcfg.ini file take priority.
Section [ext. software]	
Button=Editor WindowName=Editor SearchParts=On	Configuration of the button in the top line: A previously started program can be brought to the foreground at the push of a button. Button: button caption WindowName: Name of the program (e.g. from the taskbar). SearchParts=On: Parts of WindowName are sufficient SearchParts=Off: WindowName must be entered completely. The option "SearchParts=On" is recommended for programs such as MSWord that change the title bar subject to the document that is currently being loaded.
ProgFileName=c:\Programme\wincmd\Wincmd32.exe	The program that is started if the program mentioned above cannot be called to the foreground.
AutoStart=on	This option starts the program (ProgFileName) when starting the terminal program.
Section [DLL]	
BusDLL=PCC.EXE	PCC.EXE must be entered here if the terminal is configured with the option "operated as HYDRA-MDE terminal". If necessary, the AIP2 enters this value independently upon starting the program.
Section [PDV]	
PDVProtokoll=ON	Enables PDV logging (prot_pdv.txt) (by default=OFF)
PDVIOPDir=c:\IOPSim\	Path for DNC
Section [CAQ]	
;Supported barcodes	
FieldWNRBarcodeOnly=Y	If this entry is set the tool number may only be entered using a scanner.
FieldNestBarcodeOnly=Y	If this entry is set the cavity number may only be entered using a scanner.
FieldNummBarcodeOnly=Y	If this entry is set the number may only be entered using a scanner.

Entry	Comment
`FieldKNRBarcodeOnly=Y`	If this entry is set the badge number may only be entered using a scanner.
`BarcodeWNR=`	This field specifies which acronym is entered into the tool number field by the scanner.
`BarcodeNest=`	This field specifies which acronym is entered into the cavity number field by the scanner.
`BarcodeNumm=`	This field specifies which acronym is entered into the number field by the scanner.

5.2 PNG – Files / Bitmaps

The use of PNC files is recommended by MPDV. By default PNG files have a size of 24 x 24 px.

5.2.1 File pict.zip

The file "pict.zip“ is updated by the installation tool "inst32.exe“ while downloading and includes all default PNG files.

The default PNG files can be overwritten in the file pict_cust.zip. Several PNG files have the extension ".small.png" (e.g. aip.small.png). These PNG files are used with a screen resolution of 640x480.

5.2.2 File pict_cust.zip

The file "pict_cust.zip“ is loaded from the server directory (e.g. \<serverDir>\1\custom) when starting the program (as is the case for the hycust.mld).

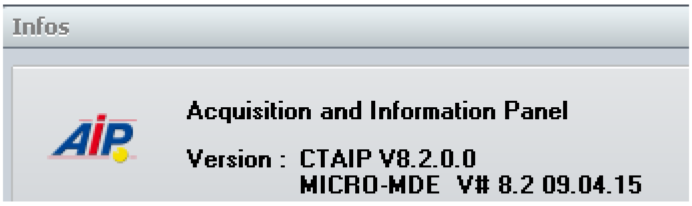
Customized PNG files may be stored in this file and loaded by the AIP2 terminal. Default PNG files may also be "overwritten".

Please note: file sizes are not adjusted.

Customize header

The AIP icon displayed in the header can be replaced by storing a separate AIP.png file in the pict_cust.zip file.

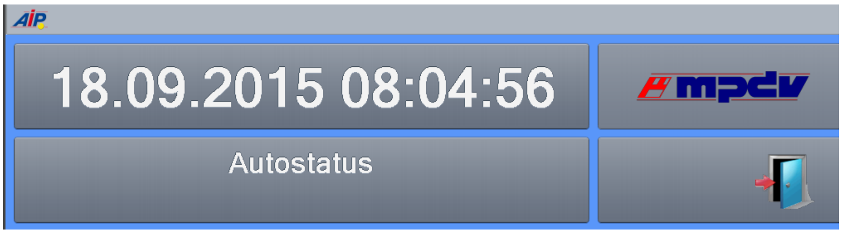
This AIP icon will also be replaced in the "About" dialog.



Customize footer

The MPDV icon displayed in the footer can be replaced by storing a separate company.png file in the pict_cust.zip file.

Customize PZE dialog



The MPDV icon displayed in the PZE dialog can be replaced by storing a separate pze_mpdv.png file in the pict_cust.zip file. In case the PZE terminal is operated with a screen resolution of 640x480, a customized pze_mpdv.small.png file has to be integrated in the pict_cust.zip file.

5.3 Multilingualism (*.mld files)

The below-mentioned files are required for the translation of the application. The table shows the priorities for the translation, ownership and the relevant storage locations.

Priority	File	Server directory	Owner	Description
1	ctaipkd.mld	./ctnet/win/aip2/custom	Customer	Customer-specific translations
2	hycust.mld	./1/custom/	MPDV	Customized translations by MPDV. The file is used by the AIP2, CTAIP and CTWIN terminal.
3	ctaip.mld	./ctnet/win/aip2	MPDV	Standard translation file

6 AIP2 - Central Configuration File hytnrcfg.ini

You can use this file as central place to store different configurations for all or for separate terminals.

For each section, a general version is available

[section 0].

The entries included in this section can be overwritten by entries in a terminal-specific section

[section <TNR-USER>]

<TNR-USER> = UserNo = terminal number + 2000 e.g. 2010,2101,..) for exactly one terminal/UserNo

The file hytnrcfg.ini is loaded from the server on every terminal start.

Section / Entry	Comment
	→→→
[Tnr Konfiguration 0]	
`FollowExternStatus=on`	Transfer of machine statuses when reloading machine list. Useful if status change is set by PDM or another terminal
[Terminal->Installation 0]	
`InstallFonts=on`	If set to "off", fonts are not installed during restart. ON=DEFAULT
`OnlyInstallFontsAfterDownload=false`	If "`InstallFonts=on`": If true, then fonts are only installed directly after a download. If false, then fonts are installed every time the terminal is restarted. (false = DEFAULT)
[Terminal->USR 0]	

Section / Entry	Comment
`AttachedApplication=First`	Displaying documents of OP info: With this configuration, the system first checks whether or not an application is linked in Windows that matches the file extension of the document. This application is then used to display the document. If no link is available, the viewers configured in ctaip.ini (→ [ext. software]) and internal viewers are used. If an extension is completely unknown, the system tries to display the document as text . Different settings are possible: **First** → search for linked application first **AfterUserViewer** → If a UserViewer is configured, this one overrides the linked application (also applies to ExcelViewer, WordViewer and PowerpointViewer) **Last** → Only if no ctaip.ini assignment is found for the file extension, then the system searches for a linked application (**default**). **Off** → The system does not search for a linked application.
`HTTPBrowser=standard`	Display of documents (via OP info): If documents are configured with a path of schema "http", the file is not downloaded to the terminal, but the link is transferred to a browser. The default browser for the terminal is htmview3.exe, as this one can be operated by touchscreen. If this entry is set, the default browser configured in Windows is used.
`SupressErrorMessage=70012`	Suppress message "material is not planned"
`MSS_DIALOG=10`	If the terminal is switched off longer than 15 minutes, a dialog is displayed on terminal restart. The user must then decide whether the counter pulses, which were recorded when the terminal was closed, are posted or discarded. After a configurable period of time, the dialog closes automatically with "Yes" (Yes, posting of pulses). This value configures the time in seconds the dialog is open.
`MSS_FILEAGE_MIN=5` `MSS_FILEAGE_OVERTIME=delete`	If the backup file for counter pulses on the terminal is older than 5 minutes, then no dialog is opened and the backup file deleted. Quantities recorded at the time when the terminal was closed are not used/posted.
[SignatureRecording->User 0]	
`ManualBadgeInput=true`	This configuration specifies whether or not the field *User* can be edited on the terminal (by default: no editing) **true** → activates keyboard input for field *User* on the terminal

Section / Entry	Comment
`Transparency=255`	The display of the signature dialog can also be transparent. **255** → Signature dialog is 0 % transparent (not transparent) **1** → Signature dialog is 99% transparent (maximum transparency) (Default = 155)
`ShowPosition=TR`	You can change the place of the signature dialog: **TL** Top – Left **TM** Top – Middle **TR** Top – Right **ML** Middle – Left **MM** Middle – Middle (Default) **MR** Middle – Right **BL** Bottom – Left **BM** Bottom – Middle **BR** Bottom – Right
`USE_SERVICE_ACCOUNT=1`	**0** (default) SSO: ServiceAccount is not used (requirement: the terminal must be started with the domain "user" (SSO)). **Note:** ServiceAccount=1 can only be used if all users are in the "root" domain. SubDomain users are not supported.
`SIGNATURE_1_USER_TYPE=REPORTING_USER_READONLY`	**REPORTING_USER_READONLY** The user identification using the Windows user is activated. The Windows user is then preassigned in field *User*. The *User* field is read-only. Requirement: The "SSO" option must be enabled for all reporting users. Otherwise, successful authentication is not possible. **REPORTING_USER_CHANGEABLE** The user identification using the Windows user is activated. The Windows user is then preassigned in field User. The *User* field can be edited. Requirement: The "SSO" option must be enabled for all reporting users. Otherwise, successful authentication is not possible.

<table>
<tr><th>Section / Entry</th><th>Comment</th></tr>
<tr><td><code>SIGNATURE_1_LOGON_TYPE=HYDRA</code></td><td>"" / Not set / "EMPTY"
There is also an alternative login procedure.

HYDRA
The user identification using the Windows user is locked. You must enter a user for identification.
Requirement: All reporting persons must have been created as users and the option "SSO" must not be enabled for the users. Otherwise, successful authentication is not possible.

ACTIVEDIRECTORY
The TAB for the user identification via User is locked. The Windows user must be used for identification purposes. Requirement: The "SSO" option must be enabled for all reporting users. Otherwise, successful authentication is not possible.

MIXED_BUT_UNIQUE
The setting of option "SSO" specifies whether the user login or the Windows login is used.
"SSO" enabled → Windows only
"SSO" disabled → user only</td></tr>
<tr><td><code>SIGNATURE_2_LOGON_TYPE=HYDRA</code></td><td><code>Identical to SIGNATURE_1_LOGON_TYPE (see above)</code></td></tr>
<tr><td><code>ExtendedSignatureRecording=true</code></td><td>Used for signatures on the terminal with quality data collection.</td></tr>
<tr><td>[MDE/Blade Configuration 0]</td><td></td></tr>
<tr><td><code>CONVERT-TO-ANSI-FILE=<list1|list2></code></td><td>Configuration of the files that are provided from the AIP to the MDEB2 blade in ANSI format when a combined operation is available.

The following lists are transferred by default if the entry is not available.
<code>counters.lst|schicht.lst|mnr.lst|mstat.lst|anr.lst|pnr.lst</code>

If you want to transfer further lists, you must specify the standard lists and the additional lists.</td></tr>
</table>

6.1 Layout configuration

Entry		Comment
Section and/or	**[terminal configuration 0] [terminal configuration 2XXX];**	**(general configuration) (2XXX terminal-specific configuration)**

Entry	Comment
AUTO-CONFIRM-UHR-ERROR-MESSAGE=TRUE	This setting specifies that in case of an error that occurred reading the clock (e.g. when activated after standby mode), the time is transferred without confirmation dialog and the terminal time is later synchronized with the server time via PDM command.
SUPPRESS-MAXIMUM-NUMBER-OF-MACHINES-WARNING=ON	Suppresses the warning after restart of terminal if more than 32 machines are assigned to the terminal (static/dynamic). (Default=OFF)
CalcTargetYieldSinceLogon=2	CalcTargetYieldSinceLogon=**1** The duration is calculated from the total runtime since login (all statuses) minus the configured shift breaks. CalcTargetYieldSinceLogon=**2** The duration is calculated from the total runtime since login (all statuses). Defined breaks are not used and are not deducted. (default value)
Section [QRD-PRINTER->TICKET 0] ;**(general configuration)** [QRD-PRINTER->TICKET 2xxx] ;**(2XXX configuration for a specific terminal)**	
COMPLETE-ABSENCE-OF-LOCAL-MNR-DATA-FOR-EVENT=< Events >	Reloads the machine row for the configured **<Events>**, if it is not available locally ➔ This configuration might be required for a group workplace without machine assignment.
COMPLETE-ABSENCE-OF-LOCAL-ANR-DATA-FOR-EVENT=< Events >	Reloads the order row for the configured **<Events>**, if it is not available locally ➔ This option has been implemented to access order data in the master data, e.g. when logging on orders.
COMPLETE-..-EVENT=< Events > COMPLETE-..-EVENT=#ALL# COMPLETE-..-EVENT=A_AN|A_P_AN	Explanation on the configuration of <Events> ➔ Using <#ALL#> the row (ANR/MNR) that is not available is reloaded for any event. ➔ <A_AN|A_P_AN> restricts reloading of information to specified events. The ID <DLGFAM> is preferred to the ID <DLG> in order to identify the <Event>.
Section **[AIP2 Initialization 0]**	
XML-GUI=OFF	Disables the new AIP2 design and uses the AIP 8.1 design.
CTWIN-STYLE=ON	Activates the GUI that is similar to CTWIN on the AIP2. The two button bars are shown below the two lists just like on the AIP 8.1.
CTWIN-BUTTON-LAYOUT=ON	If the option CTWIN-STYLE=ON is additionally set, the two button bars are displayed at the bottom of the screen.
AUTOMATIC-CHANGE-TO-START-DISPLAY=30	As of AIP 8.2.2.28: If this option is set, the display automatically changes to the main view after the configured time if no other interaction was performed in the meantime. The changing display is configured via the option *Show*

Entry	Comment

	machine/OP in the *Terminal configuration*, tab *MF functions*.
	• List:
	  ○ Change from the detail views or function menus (operation, person, resource, etc.)
	  ○ Change to the main view with "tiles"
	• Icons:
	  ○ Change from the detail views or function menus or from the main view with tiles
	  ○ Change to the icon view of workplaces
	The configuration is specified in seconds. Do not specify less than 10 seconds.
	An automatic change to the start screen is not made if an input dialog is open.

7 AIP2 - Local Configuration File ctaiplay.ini

The layout is configured for specific terminals in the file ctaiplay.ini stored in the terminal directory C:\MPDV\AIP2.

The layout is configured for specific terminals in the file ctaiplay.ini stored in the terminal directory C:\MPDV\AIP2.

This file is basically used for the configuration of grids in AIP2.

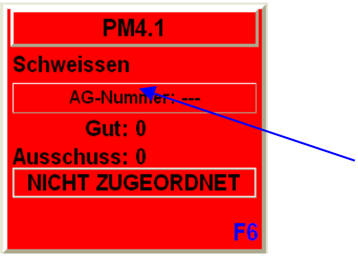
The complete standard INI files are located on the server directory \mip\ctnet\win\aip2 Any deviations from the standard are created in the customer-specific directories provided for this purpose, e.g. \mip\1\custom\aip2\tgrp_901.

Create the corresponding, empty file (e.g. ctaiplay.ini) in this directory. Modified sections are copied to this file. Make the respective configurations in this file.

After restarting the terminal, files from the main directory \mip\ctnet\win\aip2 are merged with files from the customized directory \mip\1\custom\aip2\tgrp_901. Then the merged file is transferred to the local terminal directory C:\MPDV\AIP2.

Changes to the configuration file ctaiplay.ini will not take effect until the terminal software has been restarted.

Entry	Comment
Section [OP info]	
Deaktiviert=AG_Bmk,AG_Fort	- indicated info pages are not shown - AG_Info (OP info) cannot be disabled - Entries affected by sorting are not disabled.
Sortierung=AG_TechInfo,*	- Order of info pages in the icon list - if the list ends with " ,* " the non-listed standard pages are added at the end. - Standard pages: AG_Info,AG_ZuInfo,AG_Bmk,AG_TechInfo,AG_Fort,AG_FertPap
Section [main]	
Nachkommastellen=0	Decimal places for quantities in the order/machine overview
Repaint_time=60	Cycle for updating the view (for machine list and machine info)
PopupSize->EmptyQueue=300	Empty popup window size for quick queue
PopupSize->ReloadPze=200	Reload popup window size for PZE configuration
SymbolSubstDesignation=MBEZK	The specified field replaces the machine number in the icon view.

Entry	Comment
SymbolAdditionalInfo=MBEZK	Display of any field from the machine list in the icon view between machine number and operation number:  The configured field replaces the machine number for lines and aggregates. Please note: Only in the design/GUI of AIP 8.1
MaxExpressions=50	For the configurations used in the list layouts for coloring rows or cells, 20 entries can be made by default. e.g.EXAMINE_CELLBKCOLOR20=.. The MaxExpressions setting can be used to increase the number of entries. → EXAMINE_CELLBKCOLOR50=.. This maximum index applies to all EXAMINE configurations in all grids. Internally, a corresponding amount of memory is always reserved for each grid, even if no EXAMINE configuration is used. (from AIP 8.2.1.12)
Sections for list layouts	
[Personenliste]	List displayed when staff is logged on
[Bedienposition]	Predefined list of "operator positions"
[Maschinenstatusliste]	Predefined list of "machine statuses"
[Ausschussgruende]	Predefined list of "scrap reasons"
[Abweichungsgruende]	Predefined list of "deviation reasons"
[Auftragsliste]	Lower list in the main view
[Vorgabeliste]	Order sequencing list
[Schichtinfo]	List of shift info
[Maschinenliste]	Upper list in the main view
[Eingangslosliste] input batch list	List of input batches, e.g. when "logging on the OP" in batch mode
[Ausgangslosliste]	List of preceding batches when "changing output batches"
Syntax of table formatting	
GRID_FONT=Arial	Font type
GRID_FONTSIZE=9	Font size
GRID_COLOR=clBlack	Font Color
GRID_BACKGROUND=clWhite	Background color
GRID_FIXCOLS=0	Number of fixed columns

<table>
<tr><th>Entry</th><th>Comment</th></tr>
<tr><td>GRID_ORDER=MSZEIB
GRID_ORDER=MSZEIB=-</td><td>Sorting in ascending order
Sorting in descending order

Sorting is executed according to the formatting of the column.

Examples:

ANR_DATB=C10,65,L, planned start
➢ Alphanumeric sorting (the date is provided in format MM/DD/YYYYY)

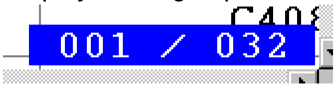
ANR_DATB=dd.mm.yyyy,65,L,planned start
➢ Sorting by date

If several criteria are indicated (separated by |) only the first criterion can be sorted in descending order. All other criteria are sorted in ascending order.
The following entry must be set in the configuration for the section so that the sorting is used in the display:
ORDER=#USE#INI#ITEM#

<u>Example for the section Sequencing List (Auto)</u>
[WF@ANR]
CMD=DLG=LIST;11|MOD=V|MNR=<MNR>|
…..
SECTION=Sequencing List (Auto)
…..
ORDER=#USE#INI#ITEM#</td></tr>
<tr><td>GRID_LIST_TYP=MNR
GRID_LIST_TYP=ANR</td><td>The list type of the section is indicated with this entry, if fields are displayed that need to be loaded additionally.
This entry also enables the search when starting.
The entry has to be entered above the IDs to be reloaded!!!
All identifiers to be reloaded can be found in the file headers.dat in the "spool" directory of the terminal. It consists of four lines:
1. 10|…: Fields that are always included in the machinery list
2. *10|…: Fields that can be reloaded for the machine list
3. 11|…: Fields that are always included in the order list
4. *11|…: Fields that can be reloaded for the order list</td></tr>
<tr><td>EXAMINE_CONTENTS_CHANGE=MGRP
EXAMINE_COLOR_C1=clWhite
EXAMINE_COLOR_C2=clSilver</td><td>The font color switches from clWhite to clSilver every time the MGRP value changes.
Up to 8 colors can be defined.</td></tr>
<tr><td>EXAMINE_SCANEXPR1=MGRP=71|72|73
EXAMINE_SCANEXPR2=MGRP=96|97|101
EXAMINE_SCANCOLOR1=ClGreen
EXAMINE_SCANCOLOR2=ClRed</td><td>The machine groups 71/72/73 are presented in green font color; the groups 96/97/101 are displayed in red font color.

Up to 8 colors each can be defined.</td></tr>
</table>

<table>
<tr><th>Entry</th><th>Comment</th></tr>
<tr><td>EXAMINE_SCANBKEXPR1=BATTRIB=1
EXAMINE_SCANBKEXPR2=BATTRIB=2
EXAMINE_SCANBKCOLOR1=clBlue
EXAMINE_SCANBKCOLOR2=clLime</td><td>All lines with BATTRIB=1 are shown in blue background color; rows with BATTRIB=2 are displayed in lime.
Up to 8 colors each can be defined.</td></tr>
<tr><td>EXAMINE_COLOR=TEXTCOLOR
EXAMINE_BKCOLOR=HGRCOLOR</td><td>Specification of a column that includes the color value for the row (e.g.: 0-Black; 255-Red, 16777215-White)</td></tr>
<tr><td>EXAMINE_CELLBKLEVEL2=EGR:AUSP,EGR:AUSP,<1*clLime|<=5*clYellow|>15*clRed</td><td>Setting of the background color depends on whether the field value reaches different threshold values.</td></tr>
<tr><td>EXAMINE_ROWBKLEVEL1=SGR:REST, SGR:REST,<=0*clRed|<=5*clLime|>15*clYellow</td><td>Setting of the background color depends on whether the field value reaches different threshold values. The entire row is colored because of the threshold value.</td></tr>
<tr><td>Syntax of column definitions</td><td></td></tr>
<tr><td>MNR=C8,80,R
MGRP=N6,60,R
AGR:AUS=N10.2,125,R
MSDATB=dd.mm.yy,70,L
MSZEIB=hh:mm,70,L
SKDATB=dd.mm.yyyy,90,L
SKZEIB=hh:mm:ss,80,L
MSDAUER=ddd.iii,60,R</td><td>Alpha-numeric, 8 characters, 80 pixels, right-aligned
Numeric, 6 characters, 60 pixels, right-aligned
Decimal, 10 digits, 2 decimal places
Displayed in the form "23.03.98", (left-aligned)
Displayed in the form "08:24"
Displayed in the form "23.03.1998"
Displayed in the form "08:24:39"
Displayed in industrial time unit " 22,982"</td></tr>
<tr><td>AGR:BMK11=hhh:mm:ss,80,R,TESTHEADER</td><td>TESTHEADER: new column caption</td></tr>
<tr><td>ALIAS KOPIE=MNR=N8,120,R,TITEL</td><td>ALIAS new name is being introduced
KOPIE= new identification
MNR= ID in data file
N8,120,R, Formatting
TITEL column caption in table</td></tr>
<tr><td>ALIAS AKA=MNR[1..3]=N8,120,R,ARRAY[1..3]</td><td>The first three characters from MNR are displayed.</td></tr>
<tr><td>ALIAS ATTR=MNR(2)=N8,120,R,PARAMETER(2)</td><td>The second part separated by „ ; “ is displayed.
Example: „ 12;20;130 “ ➔ „20“</td></tr>
<tr><td>ALIAS U=(R*6.2831)=N10.3,60,L,Scope</td><td>Conversion of a value
Syntax: see below</td></tr>
<tr><td>ALIAS SOLLZ={_INT((_DATETIME(|SKDATE|,|SKZEIE|)-_DATETIME(|SKDATB|,|SKZEIB|))*86400)}=hh:mm:ss,60,R,SOLLZ ;target time of shift</td><td>Complex calculations relating to several fields
Syntax: see below</td></tr>
<tr><td>GRID_BROWSEROW=0</td><td>only the active row is colored yellow</td></tr>
<tr><td>GRID_CELLPAINT=ON</td><td>Requirements for coloring rows column by column</td></tr>
<tr><td>GRID_REFRESH=5000</td><td>Cycle for updating the display [ms] → lists are not reloaded from the server!
Recommended if a constantly changing value is calculated using an ALIAS function.</td></tr>
</table>

Entry	Comment
GRID_POSITION=ON	Display of the grid position  Limited/no support when scrolling using scroll bars and page scrolling
GRID_CELLPAINT=ON EXAMINE_CELLBKCOLOR=WTK:STA,WTK:STA,0-clGreen|1-clBlue|2-clYellow|3-clRed can also be used with index: EXAMINE_CELLBKCOLOR1..8 EXAMINE_CELLBKCOLOR1..20	One single column is colored in every cell subject to a value. 1st value: ID of the column to be colored. 2nd value: ID of the reference column 3rd value: Configuration (color for possible values) Notes: – The reference column MUST be shown in the list, if required with length 0 – The values are converted into capital letters when being compared.
EXAMINE_CELLBKCOLOR=DMY,COLOR	Take over the color directly from the "color" column. The column <DMY> is shown in the color defined in the column <COLOR>
; Definition virt. column(1) EXAMINE_CELLVALUE1=CV1,REF1,S=DAT|P=ZEI|A=REST|N=INFO ; Definition virt. column(2) EXAMINE_CELLVALUE2=CV2,REF2,10=MGRP|20=MNR|30=COLOR|40=MSTTXT ... ; Layout/Position virt. column(1) CV1=CELLVALUE,150,Z,Data ... ; Layout/Position virt. column(2) CV2=CELLVALUE,150,Z,M/C/T	Filling of a virtual "Case" column with values from different columns subject to the value of a reference column. 1rd value: Identification of the virtual "case" column 2rd value: Identification of the reference column 3rd value: Configuration Reference value + ,=' + display column Please note: the virtual "case" column needs to be configured as follows <Identifier>=<Key word>,<Width>,<Alignment>,<Caption> e.g. CV1=CELLVALUE,150,Z,YST
GRID_RANDOMSORT=ON	This options randomly sorts the list. Please note: If this option is active, any configured sorting will be ignored.
GRID_CLIPBOARD=<BUTTON>@<SELECT>@<DATA>	This option copies data from a table/grid into the clipboard.
Special entries	
[Maschinenliste] ALIAS StkProMin=IZYSM= N8,48,R,Stk/min	Activation of calculation & display of the produced pieces per minute

Entry	Comment
[layout pze]	Configuration of the PZE terminal
KundenBitmap=kunde.bmp	**"Kundenbitmap=<File name>**“ file with customer logo When restarting the terminal, this file is copied from the server directory ".\ctnet\win\aip2\etc\“ into the application directory ".\etc\“.
DienstGangTaste=1,3	„**DienstGangTaste=1,3“** → Default [empty] By entering the function key numbers (1...4), a check specifying if the person is allowed to go on a business trip is performed during the posting.
StdSchrift=Arial StdDateSize=30 StdStatusSize=26 StdSpdBttnSize=16 InfoSchrift=Courier New InfoSchriftSize=20 SmallStatusFontSize=16 DateTimeLayout=dd.mm.yyy hh:mm:ss	Configuration of the used font types/font sizes as well as the layout of the date and time display.

7.1 Formulas used in grid layout

Simple conversion of a value

```
Syntax:      ALIAS <Alias>=(<formula>)=formatting
  <formula>: [1/]<ID>[<Operator><Value2>]
    <ID>: ID from list (The current value from the list is entered here
            in the formula)
<Operator>: + | - | * | / | ^
  <Value2>: 2nd Operand
```

Extensive formulas:

Formulas that can also relate to several table fields can be recognized by braces.

```
 Syntax: ALIAS <Alias>={<Formula>}
 <Formula>: (<Operand1>[<Operator><Operand2>])
 <Operand>: <Value> / <Function> / <Formula> / |KENN|
<Operator>: |+|-|*|/|^|
   <Value>: Constant ('0'..'9','e','E','.','-')
    |Kenn|: reads out a value from the table
<Function>: _<Fname>(<Operand>[,<Operand>[,...]])
           _DATETIME(<Date>,<Time>)
             <Date>: mm/dd/yyyy
```

```
 <Time>: ssss (Seconds of the day (0..86400))
 ==> Real value as TDateTime
_INT(<Operand>)
 returns a value without decimal places
_REAL(<Operand>)
 returns a value with decimal places
_ROUND(<Operand>)
 returns rounded value without decimal places
_MOD(<Operand1>,<Operand2>)
 ==> Operand1 mod Operand2
_ABS(<Operand>)
 absolute amount of a figure
_EXP(<Operand>)
 e raised to the power of X (e: basis of the natural logarithm)
_LN(<Operand>)
 natural logarithm (Ln(e) = 1)
_FRAC(<Operand>)
 proportion of decimal places
_LOG(<Operand1>,<Operand2>)
 LOG(N,X): logarithm to base N of X
_MAX(<Operand1>,<Operand2>)
 the greater value of two values
_MIN(<Operand1>,<Operand2>)
 the lesser value of two values
_SQRT(<Operand>)
 ==> Square root of Operand1
_PI()
 ==> 3.14151926535...
```

Examples:

Calculation of the target time of a shift (ZEISS):
*ALIAS SOLLZ={_INT((_DATETIME(|SKDATE|,|SKZEIE|)-_DATETIME(|SKDATB|,|SKZEIB|))*86400)}=hh:mm:ss,60,R,SOLLZ*

ALIAS TEST={_INT(|AGR:GUT|/|TLG|)},N3,30,R,Test

ALIAS test1={_LN(2.7182818)}=C8,80,L,Test1

New (V7.2.3.74): Utilization of intermediate variables in ALIAS functions:

ALIAS **U_Brutto**={|AGR:GUT|+|AGR:AUS|}=N8,40,Z,Brutto

ALIAS **U_BPMN**={_REAL(60000/|SZY|)*|TLG|}=N8,35,Z,BpmN

ALIAS TK_TEST={|***U_Brutto**|+|***U_BPMN**|}=N8,35,Z,TK*

<u>Setting of the background color depends on whether the field value reaches different threshold values.</u>

```
Syntax: EXAMINE_CELLBKLEVEL<i>=<Akro>,<Akro_ref>,
                              <Comp1><Val1>*<Col1>|
                              <Comp2><Val2>*<Col2>...
       <i>: Index 1..8
    <Akro>: Identification of the field to be colored
<Akro_ref>: Identification of the reference column
```

```
<Comp>: Limiting characters (<,>,<=,>=)
<Val1>: Limit value (integer or Real or ID
        of a reference value for comparison purposes)
        alternative: (Akro) - column of limit value
<Col1>: Color (Delphi name)
```

Threshold values are searched from the left to the right. If a "<" or a "<=" – criterion is met, the corresponding color is set and the evaluation/report is finished. If a ">" or ">=" criterion is met, it will first be checked whether or not the condition that follows is also met.

The direct comparison with "=" is not allowed. But the same function can be achieved by processing the comparisons relating to "<"..„<=" or „>"…">=".

An identification put in parentheses may also be indicated instead of the limit value. During the comparison, the current field content including the specified ID is read out from the same row as the limit value.

All three fields (field to be colored, reference field and limit value field, if required) must be configured as fields to be displayed. The field width can be set to zero if one of these fields should not be visible.

The color value clWhite may be entered to prevent sections from being colored.

The values are compared as they are displayed. The actual values 0.5 and 1 are considered being equal if displayed values are to be rounded to integer values.

Coloring of the field only works if the option "GRID_CELLPAINT=ON" is set.

The option "GRID_BROWSEROW=0" should also be set in order for the coloring to be recognized even if the row is selected.

```
Examples:
EXAMINE_CELLBKLEVEL1=MNR,MST,<=1*clLime|<=2*clYellow|>2*clRed
EXAMINE_CELLBKLEVEL2=FS,FS,<90*clLime|>=90*clYellow|>=100*clRed
EXAMINE_CELLBKLEVEL3=EGR:GUT,EGR:GUT,<(SGR:GUT)*clLime|>=(SGR:GUT)*clYellow
```

7.2 Translations in grid layout

Column contents can be configured to be translated and displayed by entering e.g. the configuration <XYZ=T10,100,L> instead of < XYZ=C10,100,L> in the configured grid columns. A <#> character must be prefixed for these "resource strings" to provide for better classification. This modification can be used in every INI file (hytnrcfg.ini,..) where grid layouts are configured.

Please note: The data do not include any translated values. In order for them to be displayed in e.g. dynamic dialog fields, an explicit translation must be performed using the VB script function < vbsTranslateDataValues(“<columns>“ , “<data row>“) >.

Column contents can be configured to be translated and displayed by entering e.g. the configuration <PSPERRE=U1,100,L> instead of <PSPERRE=C1,100,L> in the configured grid columns. The entry for the "resource string" that depends on the field has the following structure:

	„#<Acronym>#<Value>“		
e.g.	**„#PSPERRE#J“**	**"production lock enabled“**	
	„#PSPERRE#N“	**„ “**	**(blank character)**

This modification can be used in every INI file (hytnrcfg.ini,..) where grid layouts are configured.

Please note: The data do not include any translated values. In order for them to be displayed in e.g. dynamic dialog fields, an explicit translation must be performed using the VB script function vbsTranslateDataFields(“<columns>“ , “<data row>“) >.

7.3 Table of color values

Farbe	Name	Color value
	clWhite	$FFFFFF
	clBlack	$000000
	clBlue	$FF0000
	clLime	$00FF00
	clRed	$0000FF
	clYellow	$00FFFF
	clFuchsia	$FF00FF
	clAqua	$FFFF00
	clOrange	$0080FF
		$8000FF
		$FF8000
		$FF0080
		$80FF00
		$00FF80
		$808080

7.4 Modifications to GRID configuration / clipboard

The AIP2 provides for the configuration of copying values from the table into the clipboard.

Data can be copied into the clipboard using the shortcut "Ctrl + C", by right clicking with the mouse or an optionally configured button.

The copied values are transmitted as string in the internal format.

- Date columns as "MM/DD/YYYY“
- Time in "seconds after midnight"
- Durations in "seconds"
- Quantities with a dot as decimal separator

Data is copied including a header into the clipboard. The columns of the header and the corresponding values are separated by <TAB>. Lines are completed with <CR> <LF>.

The configuration is as follows:

```
GRID_CLIPBOARD=<BUTTON>@<SELECT>@<DATA>@<HEADER>
```

<BUTTON>	Optionally, using **"Y"** a button can be shown in the top right margin of the table. This button copies the selected data into the clipboard.
<SELECT>	Optional configuration of one or several selection criteria. Selection criteria are separated and/or linked with "|“. `GRID_CLIPBOARD=..@SELECT=X|*@..` The default selection criterion is "X“ (e.g. `@SELECT@` **becomes** `@SELECT=X@` **)**
<DATA>	The data to be copied into the clipboard can be configured here. - <ALL> All columns of the line - <VISIBLE> Visible columns (Pixel>0) - <COL1|COL2|COL3|…> configured columns For the configuration options <ALL> + <VISIBLE> the selection column is removed automatically from the columns to be copied if only one selection criterion is indicated. In case no selection criterion is stated, the selected line is copied into the clipboard according to configuration.
<HEADER>	As of CTAIP V# 2.0.3.35 **"N"** can be used to prevent the header from being displayed in the clipboard.

The following example shows the machine status list including multiple selection and copy button for the clipboard.

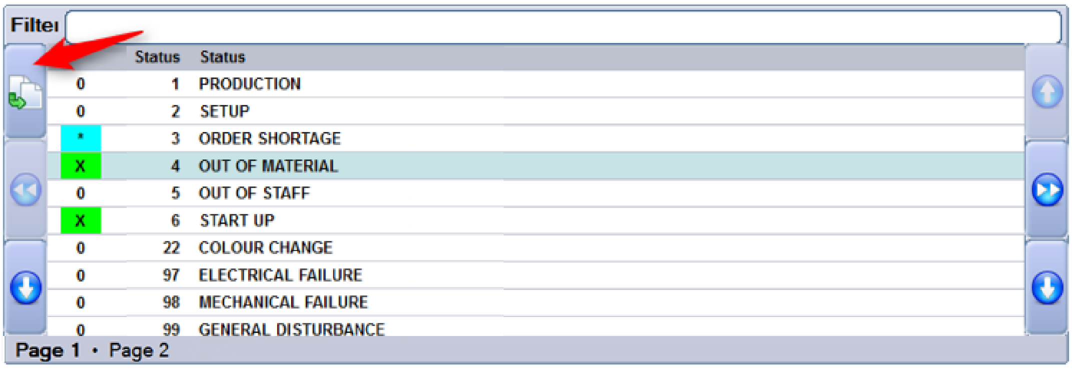
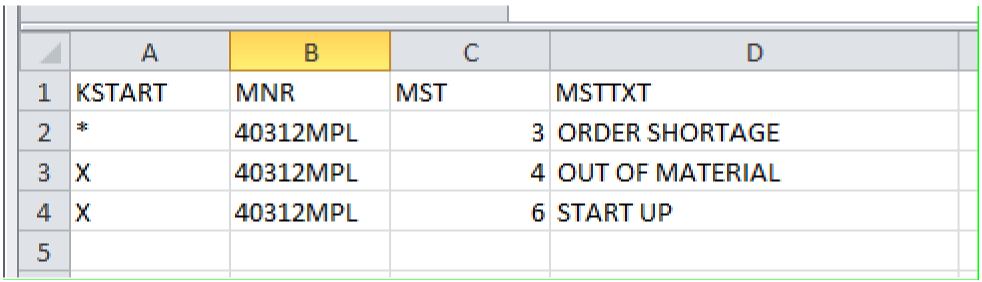


Fig. Configuration with button (red arrow) and multiple selection

```
[Maschinenstatusliste]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite
GRID_CLIPBOARD=Y@KSTART=X|*@KSTART|MNR|MST|MSTTXT
GRID_CELLPAINT=ON
EXAMINE_CELLBKCOLOR=KSTART,KSTART,X-clLime|*-clAqua

ALIAS LEER1=(DUMMY1)=C1,10,L
KSTART=C1,30,Z,*
MST=N8,60,R,
DUMMY=C3,10,R
MSTTXT=C70,150,L,Status
ALIAS LEER2=(DUMMY2)=C1,475,L
```

The data selected in the screenshot have been copied into Excel using the above-described configuration for the clipboard.



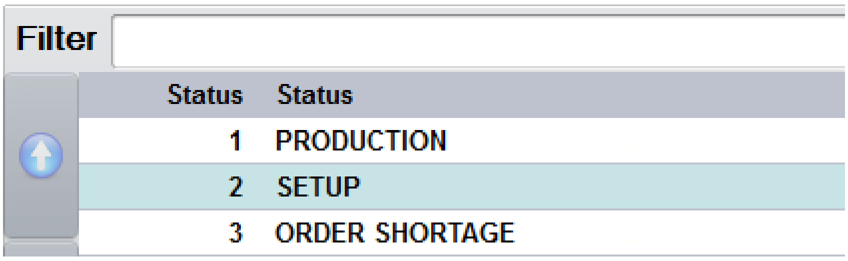
	A	B	C	D
1	KSTART	MNR	MST	MSTTXT
2	*	40312MPL	3	ORDER SHORTAGE
3	X	40312MPL	4	OUT OF MATERIAL
4	X	40312MPL	6	START UP
5				

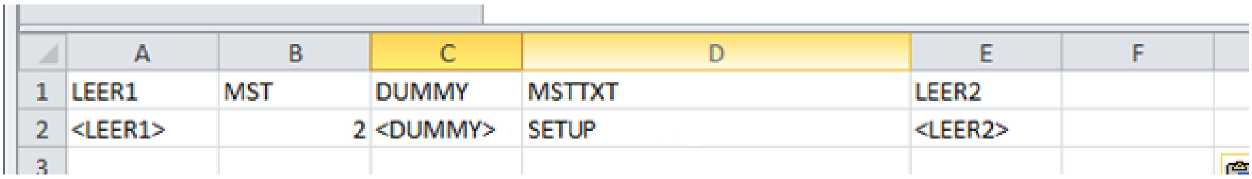
The modified configuration

```
[Maschinenstatusliste]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite
GRID_CLIPBOARD=@@<VISIBLE>

ALIAS LEER1=(DUMMY1)=C1,10,L
MST=N8,60,R,
DUMMY=C3,10,R
MSTTXT=C70,150,L,Status
ALIAS LEER2=(DUMMY2)=C1,475,L
```

copies data of visible columns (pixel > 0) of the selected line into the clipboard



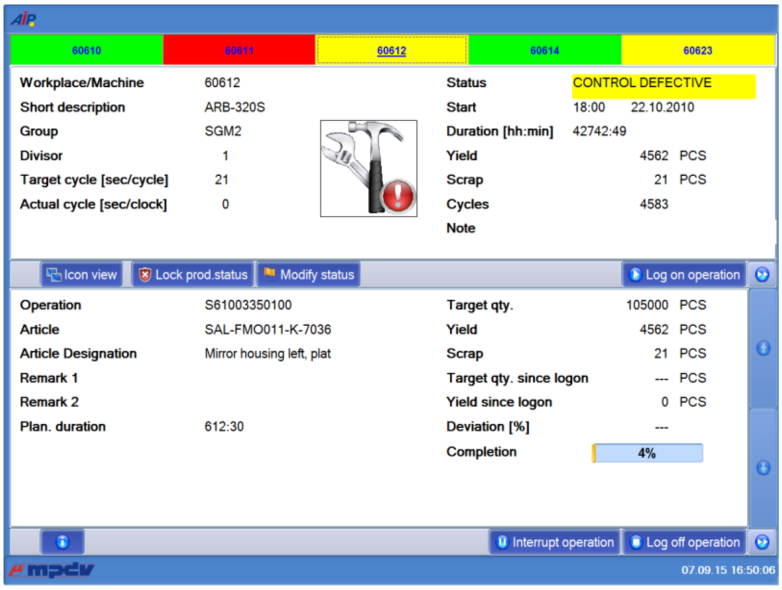


7.5 Configuration of basic screens

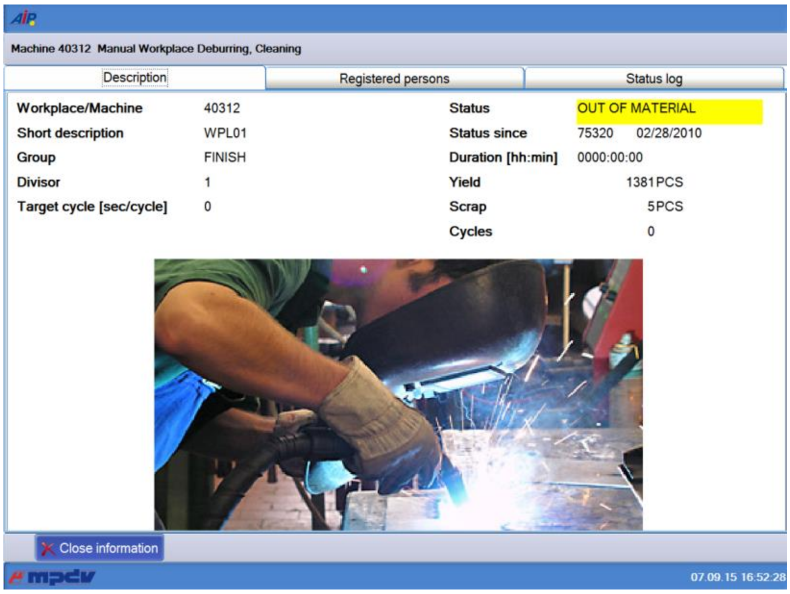
The dialogs/screens are configured using dynamic dialogs. For this reason, the following dialogs are always required:

MMINFO → Section referring to machines in the single machine view

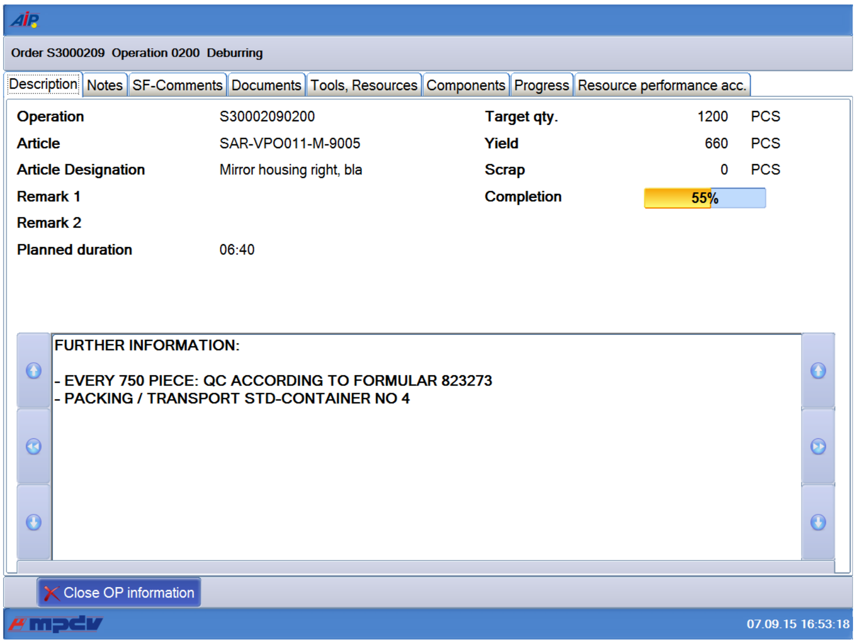
MAINFO → Section referring to orders in the single machine view



MINFO → Description of the machine information



AINFO → Description of the order information



The heights of the individual components of the basic screens and, as a result, the positions of the button bar are configured in the *ctaiplay.ini* file using the below-mentioned parameters:

Section [MainView1]	Configuration of the basic screen
MachineGridHeight=415 OrderGridHeight=500 ButtonBarHeight=50	Height configuration of components for the basic screen (machines, order grid, button bar) The configured heights are scaled to the current height. Consequently, the total sum of entered heights does not play a role.
Section [MainView2]	Configuration of the single machine view
MachineGridHeight=50 MachineInfoHeight=415 OrderInfoHeight=355 ButtonBarHeight=50	Single-row grid to select the machine Information on the machine Information on the order Height of both button bars The configured heights are scaled to the current height. Consequently, the total sum of entered heights does not play a role.

7.5.1 Available fields for the dialog configuration of basic screens

A script function completing the fields according to the customer's requirements is not available .

In general, the fields of the machine list and the order list are available. "MNR." or "ANR." must be prefixed for identification purposes.

Known quantity fields are formatted to match the configured number of decimal places.

Some fields are calculated. The following fields are additionally available:

Identification	Description
ANR.SOLL_SEIT	Target quantity since login The value is determined locally at the terminal. This is only useful for MDE machines. However, the order must be logged on locally after restarting the terminal.
ANR.ABWEICH	Deviation [%] Comparison of "target quantity since logon" and "actual quantity since logon“
MNR.SZY	Target cycle Field is transferred including "internal decimal places". The number of characters displayed is determined by the field of the dialog configuration.
MDE.IZY	Actual cycle The machine's current actual cycle - only if MDE processing is active for the machine at this terminal.
MNR.MSZEIB	Start time of the current status
MNR.MSDATB	Start time of the current status
MNR.MSDAUER	Duration of the current status
ANR.BEARBZ	Planned duration
MNR.MSTTXT	Status text
MNR.TLG	Partitioning Calculated based on the orders running at the machine.
ANR.FERTIG	Progress bar
TNRPSPERRE	Translated text for the production lock (corresponds to the configuration "TNRPSPERRE=U1,150,L,Hinweis“ in ctaiplay.ini) (the value J/N from the list can be found in MNR.TNRPSPERRE)

8 AIP2 - Local Configurations File ctaipbut.ini

Buttons are configured for specific terminals in the file ctaipbut.ini stored in the terminal directory c:\MPDV\AIP2.

The button pages of the main view and the OP info dialog may be configured in the configuration file ctaipbut.ini.

The buttons can only be configured like this in the main view if the new design of the AIP2 has been deactivated.

The server directory \<serverDir>\ctnet\win\aip2 contains the complete INI files of the standard. Deviations from this are created in the customer-specific directories provided for this purpose, e.g. \<serverDir>\1\custom\aip2\tgrp_901.

Create the corresponding, empty file (e.g.: ctaipbut.ini) in this directory. Copy all sections e.g. [ANR-ALL-Page1] to this file. The configuration is performed in this file.

After the terminal restart a merge (summary) of the files from the root directory \<serverDir>\ctnet\win\aip2 with the files of the custom directory \<serverDir>\1\custom\aip2\tgrp_901 takes place, which are transferred locally to the terminal in the directory c:\MPDV\AIP2.

All sections including the string "-Page" are imported.

Entry	Comment
Definition of sections [LST-MODUS-PageX.]	**General schematic structure of a button page** **Definition of a section** LST = List identifier of the button page (MNR, ANR, LIST3) MODUS = Mode of the machine LN = MPL – Mode DN = DLL – Mode LR = RF – Mode (reel-based manufacturing) LS = RS – Mode (cutting reels) LC = Handling unit (packing station) or XX = <MPL_MOD>[1] + <TYPE>[] YY = Value from the MNR.LST column <MNRBTN.MODUS> Otherwise if no applicable entry is found for the machine mode, the section ...ALL will be used (if available) X = Button page The definition specifying the mode a machine is running on is implemented in the AIP application program.
Sample configuration [MNR-...-Page1] 1=A_AN, L, log on OP 2=BLANK, L 3=$MPL-PAL$PAL_AN,L,log on pallet, 4=%BART:PZE=J%PZE,R,PZE,PZE.PNG Note: In one section numbering of entries <u>must</u> be consecutive from 1...n. A gap in numbering indicates the completion of a page!	<u>General structure:</u> x=<Function>,<Alignment>,<ButtonName>,<Icon> For example: 1=A_AN,L,log OP on,AGAN.PNG - Function A_AN - Alignment L or R (from the first "R“ on always "R“) - ButtonName Log on OP - Icon optional icon name (PNG, resolution 24x24 px) <u>Special functions:</u> $...$ (e.g. $MPL-PAL$) License check fails → Button is deleted %...% (e.g. %BART:PZE=J% .) Check field with value in (T)terminal (K) label ➔ only show if they match BLANK Insert distance between buttons

Entry	Comment
Configuration of functions using wildcards x=A_AN*,L, log on OP x=A_UN*,R, interrupt OP	The dialog to be opened is located as described below if buttons are configured using wildcards ID A_AN* - Calling dialog: A_AN - Identification of the machine type - Supplementing the dialog based on the machine type - Check whether or not the dialog is available if this is the case - calling dialog: A_AN_MPL - Evaluation of the posting type (only with A_AN) - Supplementing the dialog based on the *posting type* - Check whether or not the dialog is available if this is the case - calling dialog: A_*P*_AN_MPL - Calling up the located workflow or dialog If the function is <A_UN*> or <A_AB*>, it will be checked whether or not the OP to be logged off is a merged OP. - If this is the case, <A_UN*> is changed into <SA_UN> or <A_AB*> into <SA_AB> If the virtual column <MNRDLG.SUFFIX> includes a value, it will always be used (if available). ButtonFkt MNRDLG.SUFFIX Dialog e.g. A_AN* <XYZ> A(_*P*)_AN_XYZ A_TR* <ABC> A_TR_ABC

Further standard buttons:
`P_SPERRE`
`VIEW`
`ICON`
`PDV_ISTW`
`WF_BDE_KOM`
`SA_AN`
`DNC`
`MINIMIZE`
	Lock status "production"
	Switching of the basic view:
	List view ←→ presentation of individual machines
	Calling up icon view (only possible if configured in the machine configuration)
	Calling up the actual value view of PDV
	Input of BDE comments
	Log on merged operation
	Calling up the DNC startup screen
	Minimizing of the terminal program → Windows 7 requires the compatibility mode XP

`USER1…USER9`	User-defined buttons showing and starting external software The programs are configured in the section [ext. software] of the ctaip.ini file

Button IDs for the machine info
`[MNR-ALL-Page2]`
`1=M_INFO.INFO,L,show information`
`2=M_INFO.PERS,L,staff`
`3=M_INFO.MSPROT,L,machine status log`
	Consequently, the relevant info dialog including the selected page is opened in the foreground. Switching to other pages is allowed.
	M_INFO may be used to show the info page in the foreground:
	M_INFO=M_INFO.INFO

Entry	Comment

Button IDs for OP info:
`[ANR-ALL-Page3]`
`1=A_INFO.DOKU,L,documents`
`2=A_INFO.HILF,L,production resources and tools`
`3=A_INFO.KOMP,L,components`
`4=A_INFO.BMK,L,RPA`
`5=A_INFO.FORT,L,progress`
`6=A_INFO.NOTE,L,notes`
 	
`A_INFO.TEXT1,L,User text1`
`A_INFO.PICT1,L,User image1`
`A_INFO.SCRINF1,L,User scriptInfo1`
`A_INFO.DIALOG1,L,User dialogs`
	Consequently, the relevant info dialog including the selected page is opened in the foreground. Switching to other pages is allowed.
	A_INFO may be used to show the information page in the foreground:
	A_INFO= A_INFO.INFO
	
	Direct call of user-defined pages configured in the section [OP info] of the ctaiplay.ini file.
	Example:
	Dialog1=WF_BDE_KOM_LIST,BDE comments
	→ `A_INFO.DIALOG1,L,BDE comments`

`Configuration of a function with different modes`
`Just as it is the case for the configuration of the ANR/MNR info tabs, a function can be configured with different modes.`
`1=RES_WART.MNR,L,machine maintenance`
`2=RES_WART.RES,L,resource maintenance`
`3=RES_WART,L,other maintenance`
	In the configured examples, the dialog < RES_WART > is requested with the below-mentioned modes.
	1 < MNR >
	2 < RES >
	3 without mode
	
	The values can be read out as follows in the terminal script.
	VPar("BTN.FKT") Function + mode
	VPar("BTN.FUNC") Function
	VPar("BTN.MODE") Mode

Entry	Comment

Available button sections and buttons for pages of the OP info dialog:
 	
`[A_INFO-Page1]`
 	
`[A_INFO.DOKU-Page1]`
`3=AI_VIEW,R,open document`
`4=AI_VIEW_CLOSE,R,close document`
 	
`[A_INFO.HILF-Page1]`
 	
`[A_INFO.KOMP-Page1]`
 	
`[A_INFO.BMK-Page1]`
 	
`[A_INFO.FORT-Page1]`
 	
`[A_INFO.NOTIZ]`
 	
`[A_INFO.DEFAULT-Page1]`
 	
Recommended for all pages:
`1=AI_CLOSE,L,close OP information`
	Overview
	
	Document view
	
	Production resources and tools
	
	Components
	
	Resource Performance Accounts (RPA)
	
	Progress bar
	
	Notes
	
	Configuration of a default page (used if no section is defined for the tab).
	
	The IDs may also be used for the keys in the dynamic dialog (field "function").

Available button sections and buttons for pages of the machine info dialog:
 	
`[M_INFO-Page1]`
 	
`[M_INFO.PERS-Page1]`
`2=P_AN,R,log person on`
`3=P_AB,R,log person off`
`4=P_AAB,R,log everyone off`
 	
`[M_INFO.MSPROT-Page1]`
 	
`[M_INFO.DEFAULT-Page1]`
 	
For all pages:
`1=MI_CLOSE,L,close machine information`
	Overview
	
	Staff logged on
	
	Machine status log
	
	Configuration of a default page (used if no section is defined for the tab).

Entry	Comment
Definition of sections **[ButtonPanel]**	**General section for the configuration of global settings for all used button panels** → **2 in main view (MNR , ANR)** → **(W)ork(F)low**
functionkey_visible=on	Shows function keys (e.g. "F3") in button panels in order for the selection to be made using function keys (by default = off).
radiobuttonkey_visible=on	Presentation of function keys in radio group boxes of a workflow (by default = off).
functionkey_pze_visible=on	Display of function keys in PZE module (by default = off).
Definition of sections **[LIST3-ALL-Page1]** **as of CTAIP V# 2.0.2.33** **..=~<VISLIST-ID>~,L,,<PNG-File>** The characters "~" (or previously "§", should no longer be used) have been designed to identify third list buttons. Correct processing/updating (disabled/enabled) is only possible in the third grid list of the main screen.	**General section configuring functions of the configurable third list of the main screen.** INFO: The different types of the "3rd list" are configured in the machine label. The layout of a "3rd list" is defined in the "hytnrcfg.ini" file. ➔ All used lists have to be configured with their identifier „“ as follows. ➔ When changing machines, the "3rd list" is hidden/shown and buttons for "3rd lists" that are not configured are disabled.
1=~**M**~,L,,**PALETTE20x20.PNG**	Entry for "material list" ➔ "[VISLIST3(**M**)]“ from "hytnrcfg.ini“
2=~**P**~,L,,**PERSON20x20.PNG**	Entry for "list of persons“ ➔ "[VISLIST3(**P**)]“ from "hytnrcfg.ini“
3=~**R**~,L,,**RESS20x20.PNG**	Entry for "MNR_AMAT.LST“ ➔ "[VISLIST3(**R**)]“ from "hytnrcfg.ini“ with the configured Bitmap “ „
4=~**A**~,L,,**NUM.PNG**	Entry for "material list" ➔ "[VISLIST3(**A**)]“ from "hytnrcfg.ini“
5=~**G**~,L,,**PERSON20x20.PNG**	Entry for "list of persons GWP“ ➔ "[VISLIST3(**G**)]“ from "hytnrcfg.ini“

9 Barcode Input with Prefix

A barcode is interpreted as prefix barcode if the third character is a dot. In this case the first two characters identify the barcode type. The actual barcode starts with the fourth character.

ID+ Prefix	Example	Comment --> processing
--------	--------------------	---- **General** ---
00.	00.ABC123	Data not defined, ➔ 00. will be deleted and “ABC123” will be passed on to standard processing
01.	01.OK 01.ESC	Action barcode ➔ Dialog cancelled or ended with OK button or Esc button.
--------	--------------------	---- **HYDRA-ADE + HYDRA-LLE + HYDRA-MDE** ---
10.		(combined) Order/sequence/OP number → acronym <ANR>
11.		Order (header) → acronym <AUNR>
12.		Sequence → acronym <AFOLG>
13.		OP → acronym <AGNR>
14.		Sub-order number -> Acronym <UAGNR>
22.		Split no. → acronym <SPLNR>
15.		Upload/confirmation number → Acronym <RMNR>
16.	16.EXTRUDER-7 16,200	Machine → Acronym <MNR> ➔ Passed on to dialog with MNR=EXTRUDER-7 or MNR=200
17.	17.1 17.1001	Machine status → Acronym <MST> ➔ Passed on to dialog with MST=1 or MST=1001
18.	18.1 18.1001	Scrap reason → Acronym <EGG:AUS> ➔ Passed on to dialog with EGG:AUS =1 or EGG:AUS=1001
19.	19.1 19.1001	Deviation reason → Acronym < EGG:GUT > ➔ Passed on to dialog with EGG:GUT =1 or EGG:GUT=1001
20.	20.1 20.MF	Operator position → Acronym <BPOS> ➔ Passed on to dialog with BPOS =1 or BPOS = MF
21.		Wage and premium indicators → Acronym <LPKZ>
--------	--------------------	----**HYDRA-WRM + HYDRA-DNC + HYDRA-PDV + HYDRA-MPL** ---
40.	40,100 40.MONTAGE	Destination → Acronym <ZLO> ➔ Passed on to dialog with ZLO=100 or ZLO= MONTAGE
41.	41.KARTON 41.KISTE	Transport unit → Acronym <TPE> ➔ Passed on to dialog with TPE = KARTON or TPE = KISTE
42.		Batch number → Acronym <CNR>→➔
43.		Throughput batch number → Acronym <DLL>→➔
44.		Alternative batch number → Acronym <CNR:ALT1>→➔
45.		Alternative batch number → Acronym <CNR:ALT2>
46.		Alternative batch number → Acronym <CNR:ALT3>
47.		Alternative batch number → Acronym <CNR:ALT4>
48.		Alternative batch number → Acronym <CNR:ALT5>
49.		Alternative batch number → Acronym <CNR:ALT6>
--------	--------------------	---- **Mainly for the HYDRA-PZE module** ---

ID+ Prefix	Example	Comment --> processing
50.		Badge number → Acronym <KNR>
51.		Personnel number → Acronym <PNR>
52.	52.EDV 52.VERTRIEB	Cost center → Acronym <KST> ➔ Passed on to dialog with KST=EDV or KST=VERTRIEB
53.	53.1 53.1001	Absence reason → Acronym <FGR> ➔ Passed on to dialog with FGR=1 or FGR=1001
--------	**-------------------**	**---- Customer-specific barcodes ---**
90.		
…		

In case barcodes are required, which actually have a dot at the third place (e.g. if the machine/workplace number has a dot as third character), it is possible to define an alternative indicator for barcode prefixes in the HyTnrCfg.ini terminal configuration, e.g.

[Terminal->USR 0]
BarcodePrefixChar=$

If another prefix is actually required, the respective barcode font in use must be able to represent this prefix.

Examples for barcodes

Barcode printing: Font “Codedreineun” and prefix “.“

Prefix	Barcode	Raw data
10.	*10.123456780100* *10._ABCD12340100*	ANR = 123456780100 ANR=_ABCD12340100

Prefix	Barcode	Raw data
11.	*11.12345678*	AUNR = 12345678
12.	*12.01*	AFOLG = 01
13.	*13.0100*	AGNR = 0100
14.	*14.0000*	UAGNR = 0000
15.	*15.123456789012345*	RMNR = 123465789012345
22.	*22.02*	SPLNR = 02

Prefix	Barcode	Raw data
16.	*16.123456*	MNR = 123456
17.	*17.1122*	MST = 1122
18.	*18.1234*	EGG:AUS = 1234
19.	*19.123456789*	EGG:GUT = 132456789
20.	*20.13*	BPOS = 13
21.	*21.1221*	LPKZ = 1221

Prefix	Barcode	Raw data
50.	*50.1337*	KNR = 1337

9.1 Configuration of customized barcode prefixes

Section [barcode]	
BarKenn90=SAPCNR **BarKenn91=EGR:GUT**	The barcode prefixes 90...99 can be assigned here according to the customer's requirements. This means, if a barcode with the relevant prefix is used, it will be transferred to the dialog along with the assigned ID. Then the barcode has the following structure: <Prefix>.<Net barcode> e.g.: "90.12345" → SAPCNR=12345

Firmly assigned barcode prefixes:
10:ANR
11:AUNR
12:AFOLG
13:AGNR
14:UAGNR
15:RMNR
16:MNR

17:MST
18:AUSGRD
19:AGGGRD
20:BPOS
21:LPKZ
22:SPLNR
40:ZLO
41:TPE
42:CNR
43:DLL
44:CNR:ALT1
45:CNR:ALT2
46:CNR:ALT3
47:CNR:ALT4
48:CNR:ALT5
49:CNR:ALT6
50:KNR
51:PNR
52:KST
53:FGR

10 AIP2 - GUI Configuration

10.1 Overview

The layout for the new GUI of the AIP2 terminal (tile design) is stored in XML files. XML files can be edited using a standard text editor. Microsoft's *XML Notepad 2007* can also be used. This XML editor provides a clearer presentation, a user-friendly copy function for entire objects and the possibility to move complete objects. *XML Notepad 2007* was used for the generation of screenshots included in this document.

In the configuration, font sizes and positions are given in points in relation to a screen resolution of 600 points in height. Values must be scaled and then rounded to whole dots when using a screen with a higher resolution. Therefore, the proportions can slightly vary depending on the screen resolution.

Colors are specified in XML files in a reversed RGB notation (**B**lue/**G**reen/**R**ed instead of **R**ed/**G**reen/**B**lue). If the entry is in the hexadecimal format, the two places behind the symbol $ define the color blue, the next two places the color green and red is defined by the last two places.

10.2 Filing XML files in the server

Like the INI files ctaiplay.ini and ctaipbut.ini, the XML files that define the GUI are located on the server in the sub directory ctnet\win\aip2. XML files are filed in the sub directory *gui.*

10.2.1 Scope Concept

The INI files in the subdirectory <SystemNo>\custom\aip2 and the XML files in the subdirectory <SystemNo>\custom\aip2\gui can be overridden customer-specifically in deviation from the standard. Various *scopes* are provided in order that the different changes do not overwrite each other.

Scope	Directory	Examples
Standard	ctnet\win\aip2\<x>.<y>	ctnet\win\aip2\ctaiplay.ini ctnet\win\aip2\gui\l_anr.xml
Standard scope	<SystemNo>\custom\aip2\<x>.<y>	1\custom\aip2\ctaiplay.ini 1\custom\aip2\gui\l_anr.xml
Custom Scope	<SystemNo>\custom\aip2\<x>**@custom**.<y>	1\custom\aip2\ctaiplay@custom.ini 1\custom\aip2\gui\l_anr@custom.xml
VAR scope	<SystemNo>\custom\aip2\<x>**@var**.<y>	1\custom\aip2\ctaiplay@var.ini 1\custom\aip2\gui\l_anr@var.xml
Local scope	<SystemNo>\custom\aip2\<x>**@local**.<y>	1\custom\aip2\ctaiplay@local.ini 1\custom\aip2\gui\l_anr@local.xml

Standard scope

The Standard Scope is used to create a copy of the standard. This ensures that changes to the standard, sometimes supplied with a service pack, do not interfere with the collection terminal and therefore, the collection software remains unchanged. After generating the copy, the required changes to the standard must be either copied again or synchronized. The changes can then be integrated into the Standard Scope.

Custom Scope

The Custom Scope is reserved for MPDV to file customer specific configurations. A file is stored in the Custom Scope if the file name includes **@custom** before the extension.

VAR scope

The VAR Scope is reserved for partners (Value Added Reseller) to store changes for customers of partners. A file is stored in the VAR scope if **@var** is inserted before the extension.

Local scope

The Local Scope is reserved for customers to store their own customized files. A file is stored in the Local Scope if **@local** is inserted before the extension.

The priority of the different scopes is ascending from the standard scope to the local scope. A file in the local scope takes priority over a file included in the standard scope.

INI and CFG files are processed differently to XMLfiles:

INI and CFG files are merged per section, that means sections in the standard are totally replaced by potentially existent sections deriving from individual scopes. Settings in the Local Scope have highest priority as they are processed at last and therefore overwrite settings from the scope located above.

XML files are not merged but accepted. That means only files are processed located in the list of scopes at the bottom.

The only exception is the file *globaldefines.xml.* The content of that file is merged with the settings of the individual scopes. It is therefore possible to overwrite individual settings (i.e. font size or color) without copying the complete file. If you would like to overwrite a certain element of the file *globladefines.xml,* please copy the file, delete all elements to be accepted from the standard file and then store the file in the relevant Scope.

10.2.2 Specific layouts of terminals or terminal groups

Like the INI files ctaiplay.ini and ctaipbut.ini, the XML files that define the GUI are stored on the HYDRA server in the subdirectory <SystemNo>\custom\aip2. Standard XML files are filed in the sub directory *gui.*

If different GUI layouts are required for specific terminals or terminal groups, the non-standard XML files are stored in the following subdirectories:

Reference	Subdirectory	Example
Terminal group	tgrp_<*Terminal group*>\gui	tgrp_900\gui
Terminal	tnr_<*Terminal number*>\gui	tnr_100\gui

XML files stored in these sub directories replace standard files with the exception of the *globaldefines.xml* file whose content is merged with the relevant standard files.

10.2.3 Loading configuration files during restart

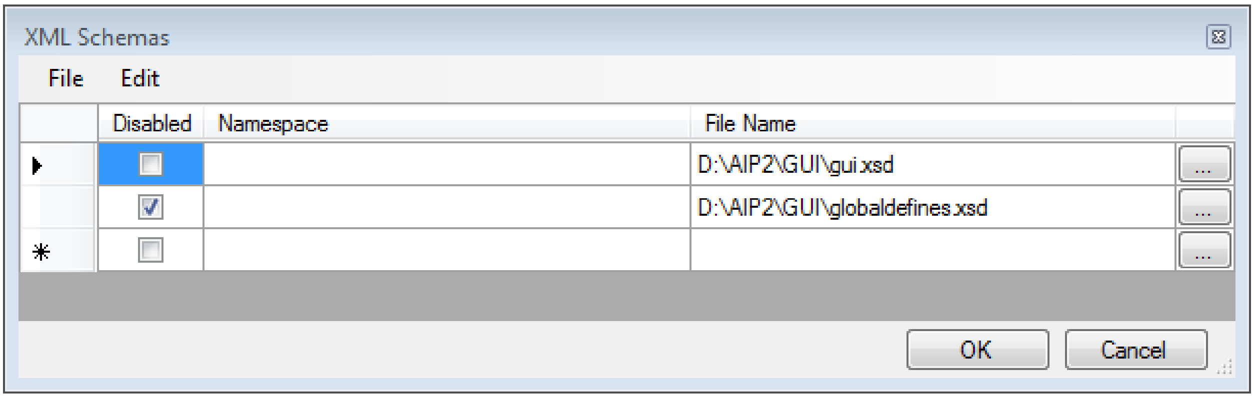
Every time the AIP2 is started, INI, CFG and XML files are updated from the server and automatically activated in the terminal.

When changing the layout, please note that the changed layouts are not overwritten when the AIP is started. Updating of configuration files can be deactivated in the file ctaip.ini by adding the parameter *SkipAipStartupUpdate* to the entry *parameters=* in the section *[system].*

10.2.4 Syntax check via XML Schema Definition (XSD)

An XML Schema Definition (XSD) defines the structure of an XML file. Depending on the editor used, a syntax check of the edited XML file is performed. In addition, you can select the value of specific fields via a selection list.

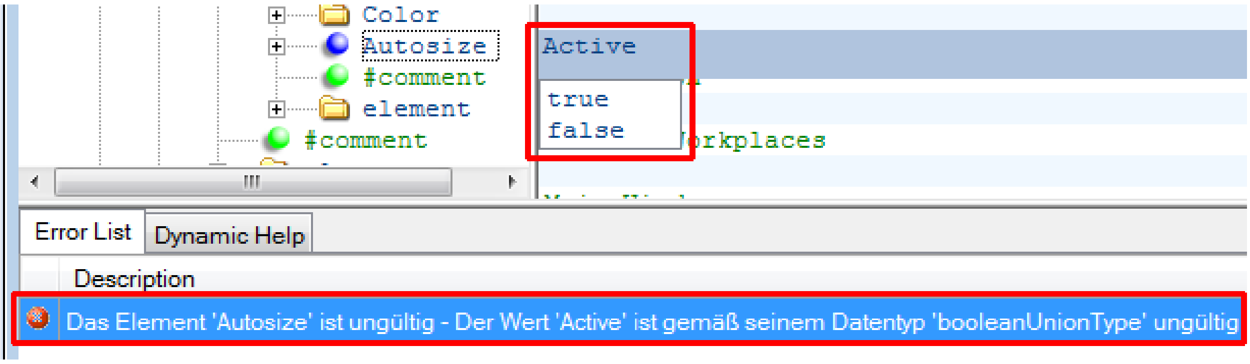
In the *XML Notepad 2007* of *Microsoft,* you can enter XML Schema Definitions via the menu item *View – Schemas…* and enable or disable them:



The file *globaldefines.xsd* includes the schema for the file *globaldefines.xml*. The file *gui.xsd* includes the schema for the other XML files used for the GUI configuration. The two XSD files are located in the HYDRA server in the same directory as the corresponding XML files.

The two XSD files *globaldefines.xsd* and *gui.xsd* are not compatible. For this reason, one of these files must always be *disabled*.

The following example shows a syntax check and a selection list:



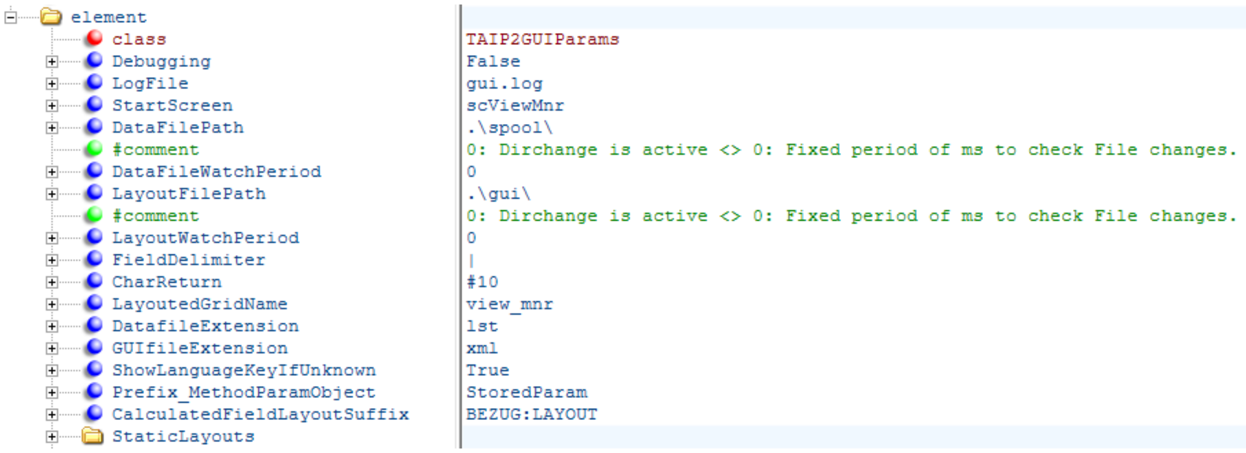
10.3 Settings

The file *globaldefines.xml* includes general settings, constants, data sources, calculated fields and functions.

Changes in the file *globaldefines.xml* are only active after restart of the AIP2.

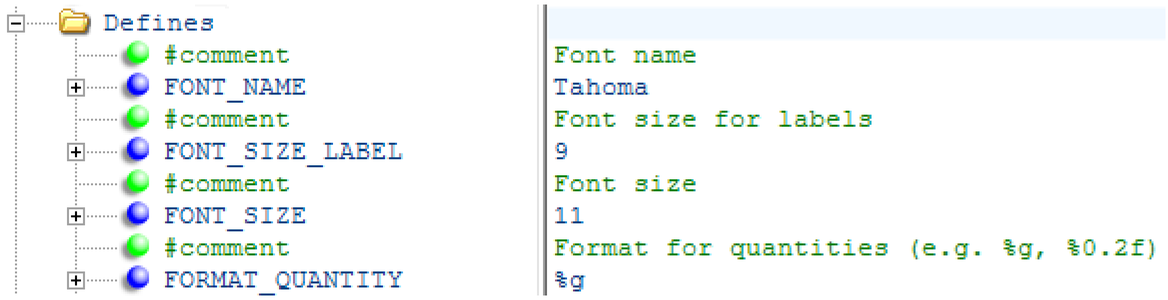
10.3.1 General settings

Standard settings control program processing and may not be changed.



10.3.2 Constants (Defines)

The section *Defines* specifies the Constants used for different layout configurations. For example, you can change color or font size at a central location using these constants.



The following constants can be set:

FONT_NAME

This constant sets the font of the GUI. The standard setting is „Tahoma".

FONT_SIZE_LABEL

This constant sets the font size for the GUI labeling. The standard setting is 8.

FONT_SIZE

The constant FONT_SIZE sets the font size for the displayed data in the GUI. The standard setting is 10.

FONT_SIZE_HEADING

FONT_SIZE_HEADING defines the font size for the headings on the right hand side. The standard setting is 10.

FORMAT_QUANTITY

This constant sets the format for the display of quantities.

Standard setting is „%g".

"%g": Automatically formats the shortest display: If the quantity is an integer, then without decimal places and without decimal separators. If the quantity is not an integer, the existing decimal places are output after a decimal separator (maximum 15 decimal places). No thousand separator is output.

"%0.0f": No decimal places, without thousands separator.

„%0.2f": Always two decimal places, without thousands separator.

„%0.2n": The format n is the same as the format f, but the resulting string contains thousands separators, if a thousands separator is configured in the regional settings.

The set format only affects configured layouts and not dynamic dialogs. A delimiter for thousands is not available for dynamic dialogs.

FORMAT_CYCLE

FORMAT_CYCLE defines the output format for cycle times (target cycle and actual cycle) if they are not output as durations (as in the standard system) but in seconds. Formatting is issued like in the previous field. Standard setting is "%0.3f". Please refer to section "1.2.4 CalculatedFields" for further information.

The set format only affects configured layouts and not dynamic dialogs.

COLOR_MENU_BUTTON

This constant defines the background color for specific buttons on the left hand side, like the button "<back" and "PZE". For example the button "< Back" and the button "PZE" on the start page belong here. Standard setting is "$C0C0C0" (light gray).

COLOR_MENU

This constant controls the background color of the buttons used for selecting workplaces in the "Home"-page and for calling functions if an object was selected. Standard setting is "$E0E0E0" (lighter gray).

COLOR_MENU_ACTIVE

This constant can set a background color for the selected workplace. Standard setting is "$909090" (gray).

COLOR_BACKGROUND

This constant can set a background color to display data on the right hand side. The color should correspond with the COLOR_MENU_ACTIVE. Standard setting is "$909090" (gray).

COLOR_MARGINS

This constant sets the color for the borders of the layout. Standard setting is "$FFFFFF" (white).

COLOR_HEADING

This constant defines the background color for the upper headings on the right hand side. Standard setting "$833014" (dark blue).

COLOR_HEADING_2

This constant defines the background color for the headings in the middle on the right hand side. Standard setting is "$974428" (blue).

COLOR_HEADING_3

This constant defines the background color for the lower headings on the right hand side. Standard setting is "$AA583B" (light blue).

COLOR_FONT_HEADING

This constant defines the font color of the headings on the right hand side. Standard setting is "$FFFFFF" (white).

COLOR_TILE

This constant defines the background color of the individual tiles on the right hand side. Standard setting is "$F8F8F8" (very light gray).

COLOR_FONT

COLOR_FONT sets the standard font color. Standard setting is "$202020" (dark gray).

COLOR_STATUS_PRODUCTION

This constant defines the color for the status *production.* Standard setting is "1077248" (dark green, $107000). With this constant, the color must be entered as a decimal value to avoid the display in a different color on specific screens.

COLOR_STATUS_NO_PRODUCTION

This constant controls the color for the all statuses except *production.* Standard setting is "$1090FF" (dark yellow).

COLOR_STATUS_NOT_ASSIGNED

This constant defines the color for the status "Not assigned". Standard setting is "$1010D0" (dark red).

COLOR_YIELD

This constant defines the color to display yields. Standard setting is "1077248" (dark green, $107000). With this constant, the color must be entered as a decimal value to avoid the display in a different color on specific screens.

COLOR_SCRAP

COLOR_SCRAP controls the color to display scrap. Standard setting is "$1010D0" (dark red).

COLOR_INSPECTION_DUE

This constant can set a background color to display a CAQ inspection due. Standard setting is "$1090FF" (yellow).

COLOR_INSPECTION_DONE

This background color indicates if a minimum inspection scope was reached. Standard setting is "1077248" (dark green, $107000). With this constant, the color must be entered as a decimal value to avoid the display in a different color on specific screens.

COLOR_INSPECTION_ERROR

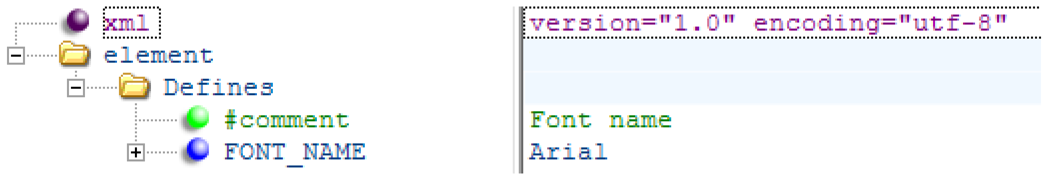
If an error occurs in the inspection planning, the relevant area is shown in the color set. Standard setting is "$1010D0" (dark red).

COLOR_MAINTENANCE_STATUS_0, …, COLOR_MAINTENANCE_STATUS_3

Color to display maintenance status of resources. Standard settings are "$FFFFFF" (white), "$873418" (blue), "$1090FF" (yellow) and "$1010D0" (red).

The name of customer specific constants must include the prefix "U". This way, they cannot be mixed up with constants of the standard.

The following example shows how to override a constant in a scope so that it is merged with the settings in the standard scope.



10.3.3 Data sources (ProviderDefinition)

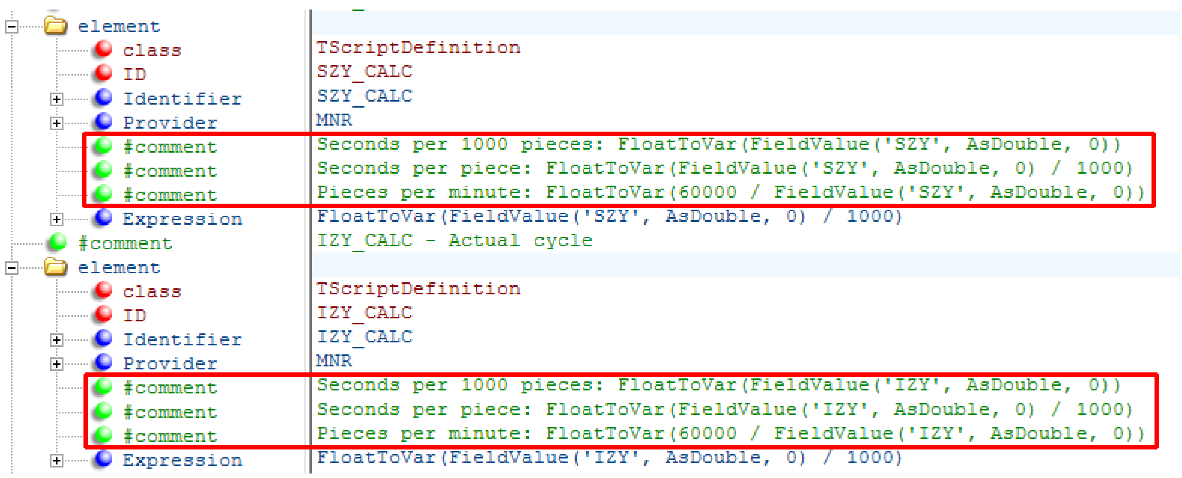
The settings define the correlation between the individual data sources and may not be changed.

10.3.4 Calculated fields

Calculated fields configure the display of the target and actual cycle. Both fields can be provided either as "time for 1000 pieces", "time for one piece" or "pieces per minute". The setting is done using the calculated fields *SZY_CALC* and *IZY_CALC*. The following formulas can be stored in the attribute *Expression* :

Field	Presentation	Formula
Target cycle	Time for 1000 pieces	FloatToVar(FieldValue('**SZY**', AsDouble, 0))
	Time for one piece	FloatToVar(FieldValue('**SZY**', AsDouble, 0) / 1000)
	Pieces per minute	FloatToVar(60000 / FieldValue('**SZY**', AsDouble, 0))
Actual cycle	Time for 1000 pieces	FloatToVar(FieldValue('**IZY**', AsDouble, 0))
	Time for one piece	FloatToVar(FieldValue('**IZY**', AsDouble, 0) / 1000)
	Pieces per minute	FloatToVar(60000 / FieldValue('**IZY**', AsDouble, 0))

Three options can be used as comments in the file globaldefines.xml:

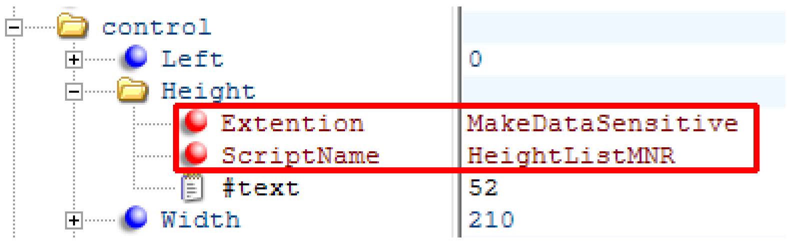


The formatting of both fields is configured using the constant FORMAT_CYCLE. Please refer to section "1.2.2 Constants (Defines)" for further information.

10.3.5 Functions (ScriptDefinitions)

The settings may not be changed during configuration of the GUI.

Fields that start a function are identified by the attributes *Extention* and *ScriptName*. The following example controls the height of the entries in the list of workplaces in the main view (a_list_mnr.xml):



The entry in the field *#text* is not active in this case. It is overwritten by the result of the previously entered function. If you want to change the height entered in this field, you must delete the two attributes *Extention* and *ScriptName.* Please note that in this case the data dependent identification of the height is disabled.

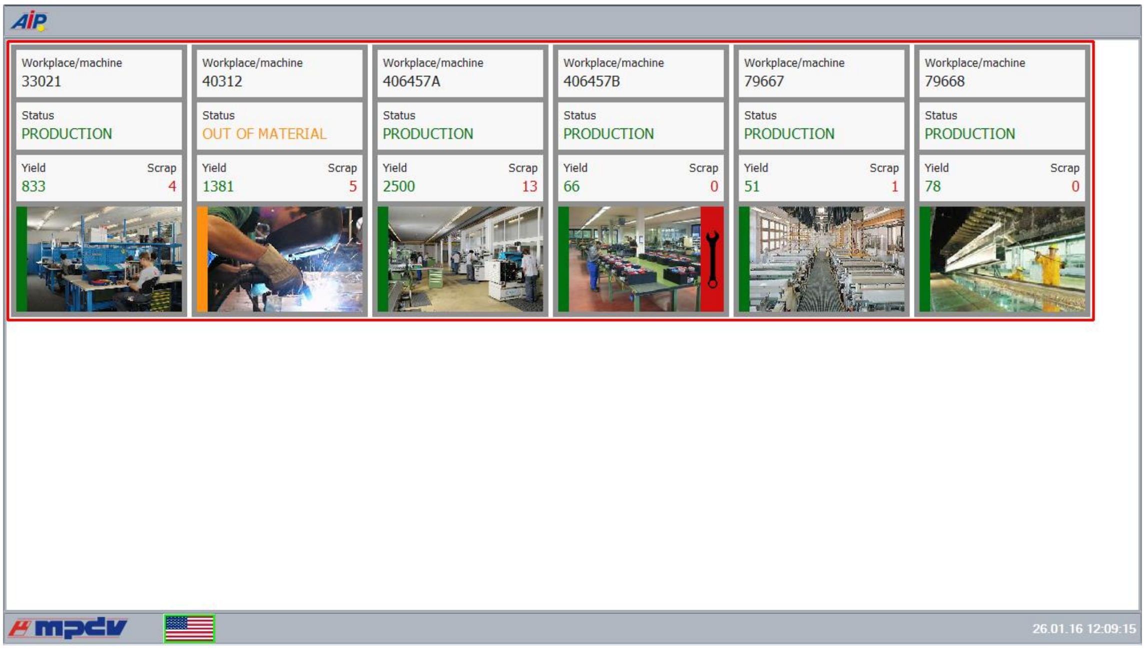
10.4 Layout definition

The definition of the layout is separated into layout files beginning with „l_“ and areas consisting of file names beginning with "a_".

There are the following layouts and areas in the standard:

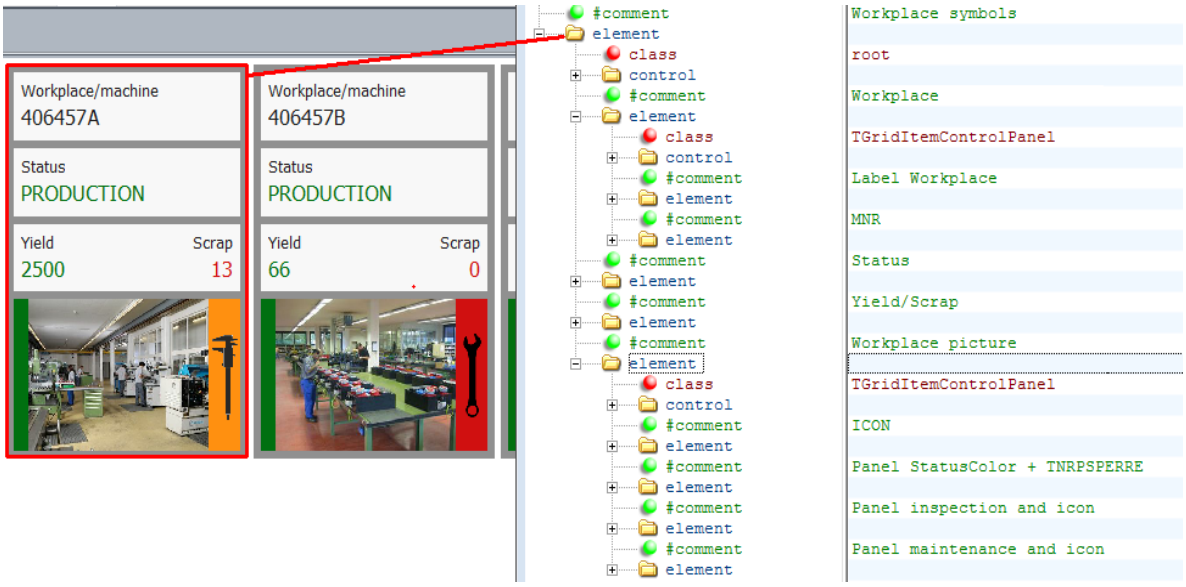
l_view_mnr.xml

The tile view shows workplaces assigned to the terminal:

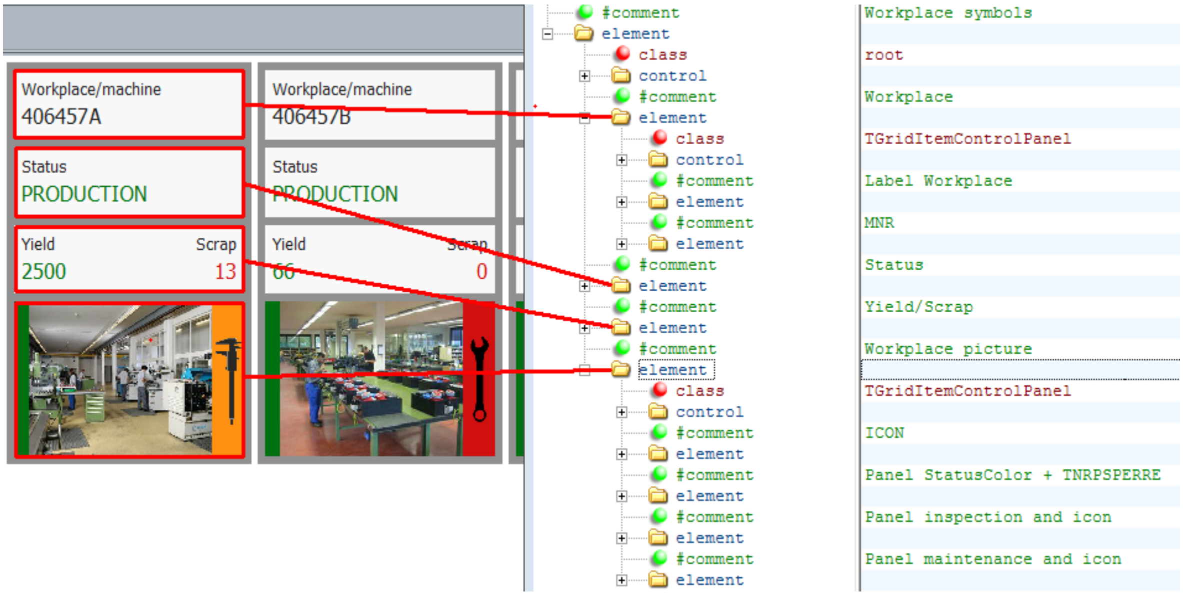


The structure for the individual workplaces is stored in the file **a_view_mnr.xml.**

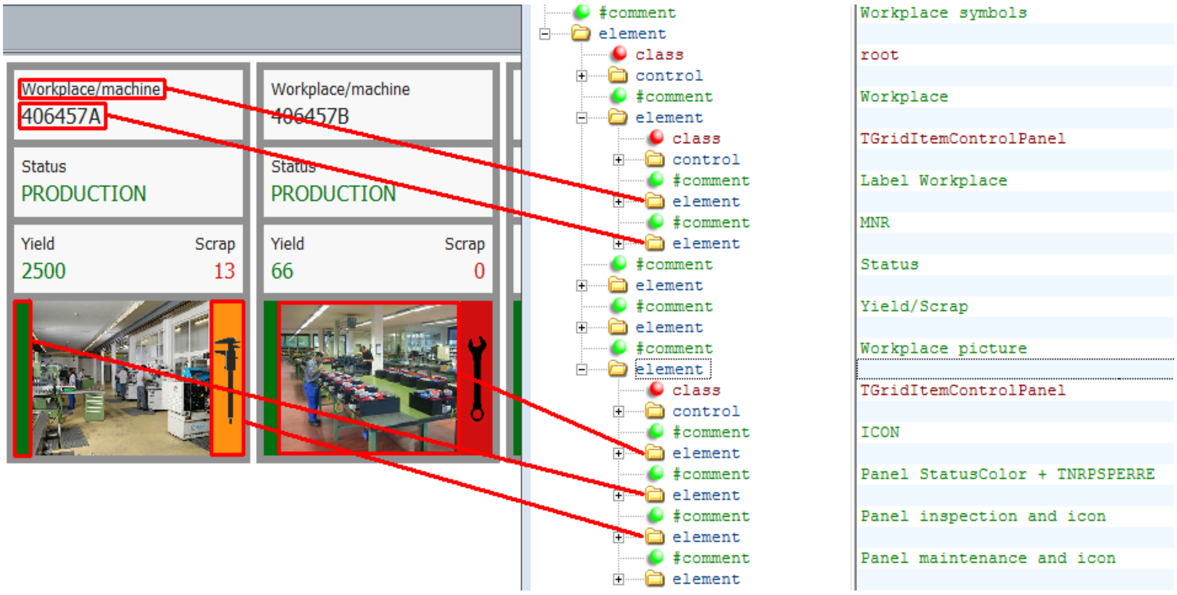
The following screenshots show the assignment of the elements in the file to the objects in the GUI. The element on the highest level of the tree structure is the big outer tile with gray frame:



The elements in the next level of the tree structure are the 3 tiles in light gray and the tile including the image:



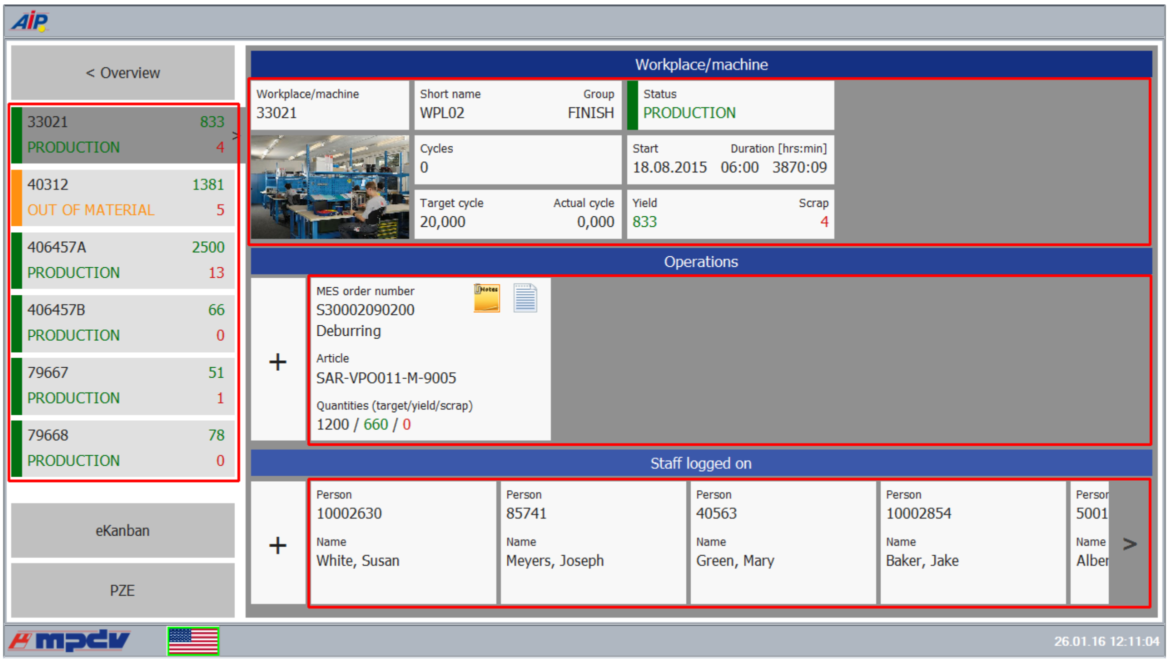
The elements in the next level of the tree structure are assigned as shown in the example of the light gray tile on top and the tile including the image at the bottom:



The lowest element in the list defines the layout of the tile including the screwdriver.

l_main.xml

Select a workplace to reach the main view:



The file **a_list_mnr.xml** contains the structure of the view of a workplace located in the list on the left hand side.

The file **a_data_mnr.xml** defines the upper area on the right hand side to display data for the workplace. This area is used for all layouts that follow.

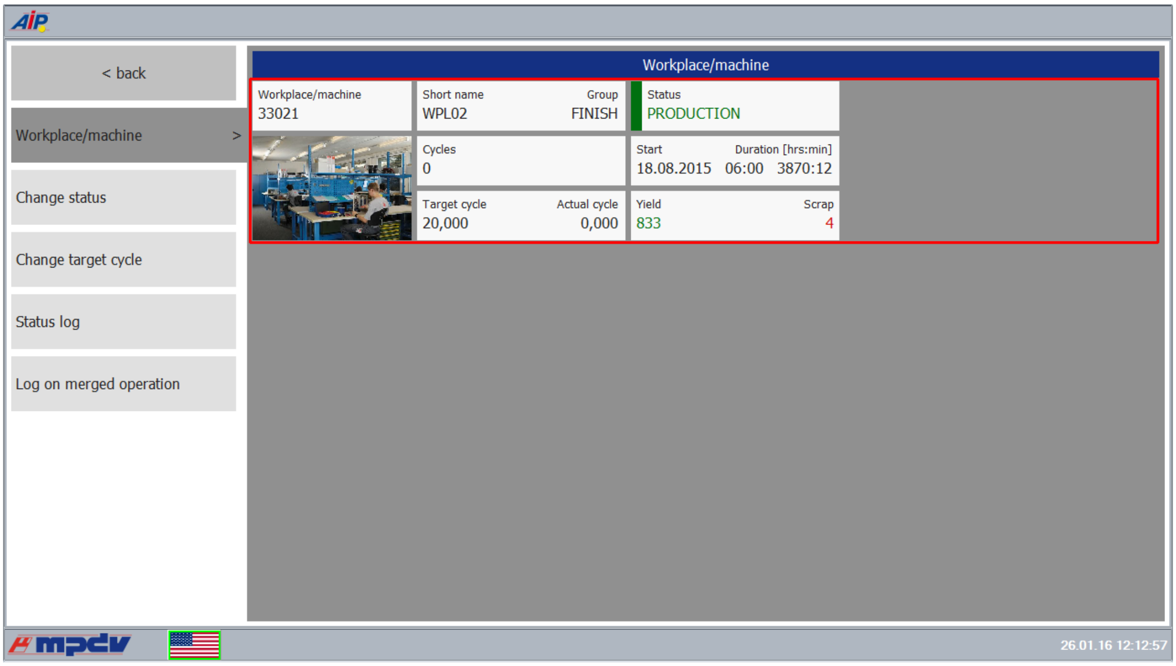
The file **a_list_anr.xml** stores the layout of an operation in the middle of the screen on the right hand side. The button to the left, which is used to log on an operation, as well as all other buttons outside a red frame are located directly in the **l_main.xml** layout.

Various lists can be displayed at the bottom on the right hand side. You can set in the workplace configuration which of the 3 lists are available at a workplace. The displayed data are defined in the following files:

- **a_list_pnr.xml**: Persons logged on
- **a_list_pnrg.xml**: Persons logged on at a group workplace
- **a_list_res.xml**: Resources logged on
- **a_list_emat.xml**: Logged on input material
- **a_list_amat.xml**: Produced output batches

l_mnr.xml

If you click on the area showing the data for the selected workplace, you will get to the workplace layout:

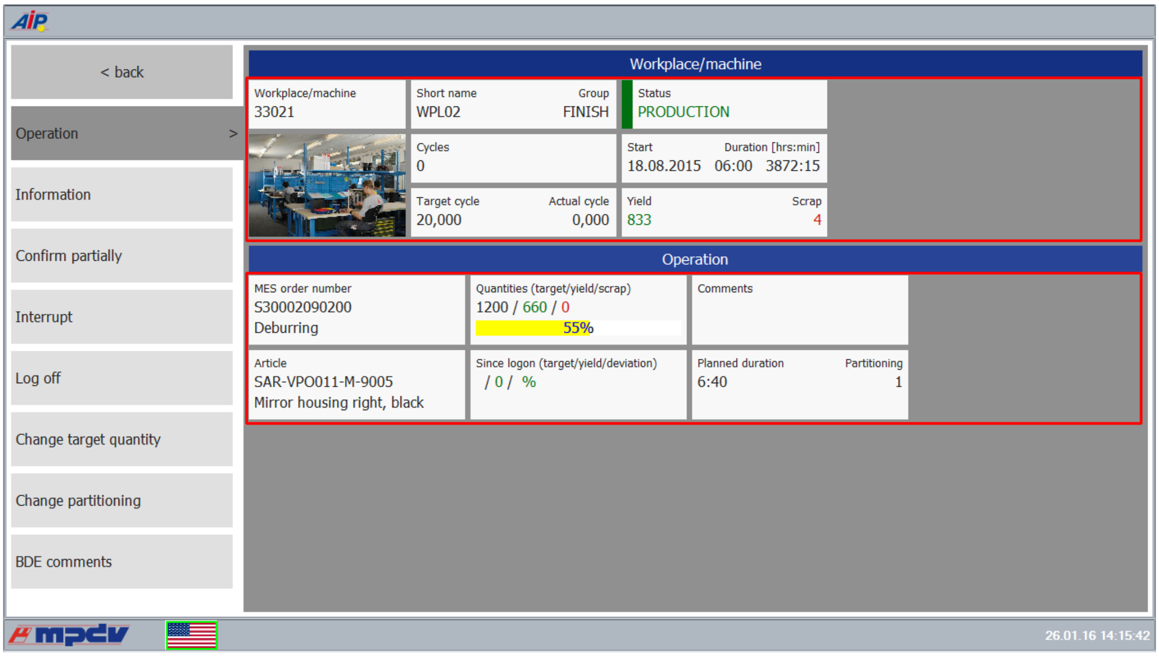


As described, the file **a_data_mnr.xml** defines the upper area on the right hand side to display data for the workplace.

The buttons on the left side are located in the layout **l_mnr.xml**.

l_anr.xml

If you click an operation in the main view, the operation layout appears:



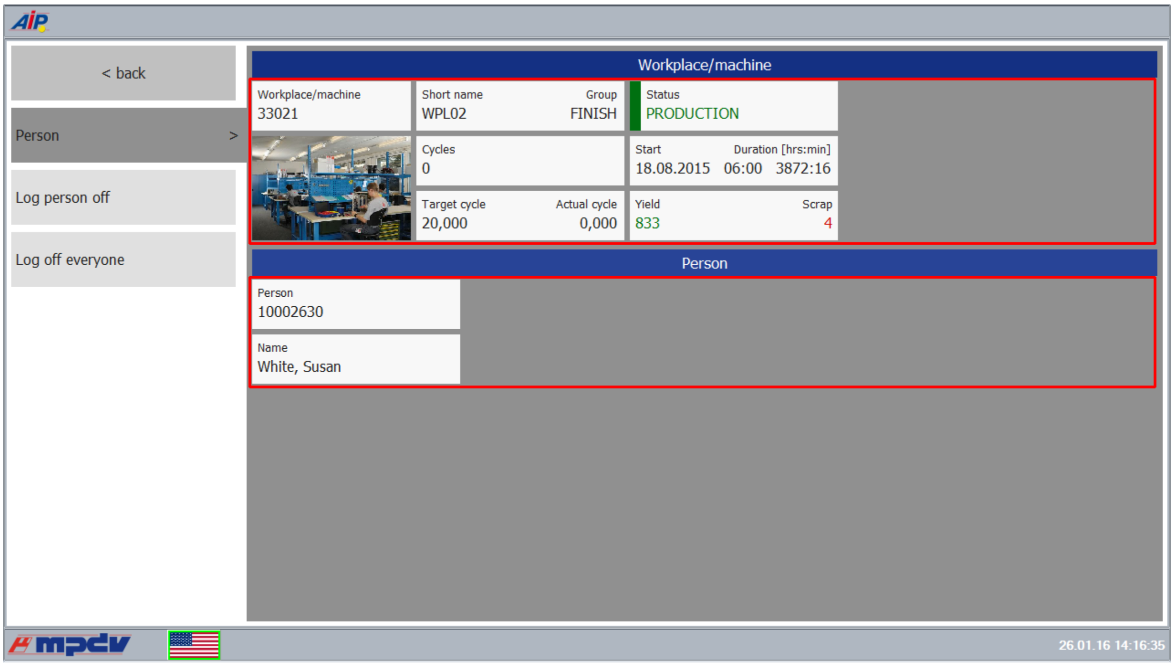
As described, the file **a_data_mnr.xml** defines the upper area on the right hand side to display data for the workplace.

The file **a_data_anr.xml** contains the definition for the area in the middle of the layout showing data for the operation.

The buttons on the left side are located in the layout **l_anr.xml**.

l_pnr.xml

The layout for staff appears if you select a person in the main view



As described, the file **a_data_mnr.xml** defines the upper area on the right hand side to display data for the workplace.

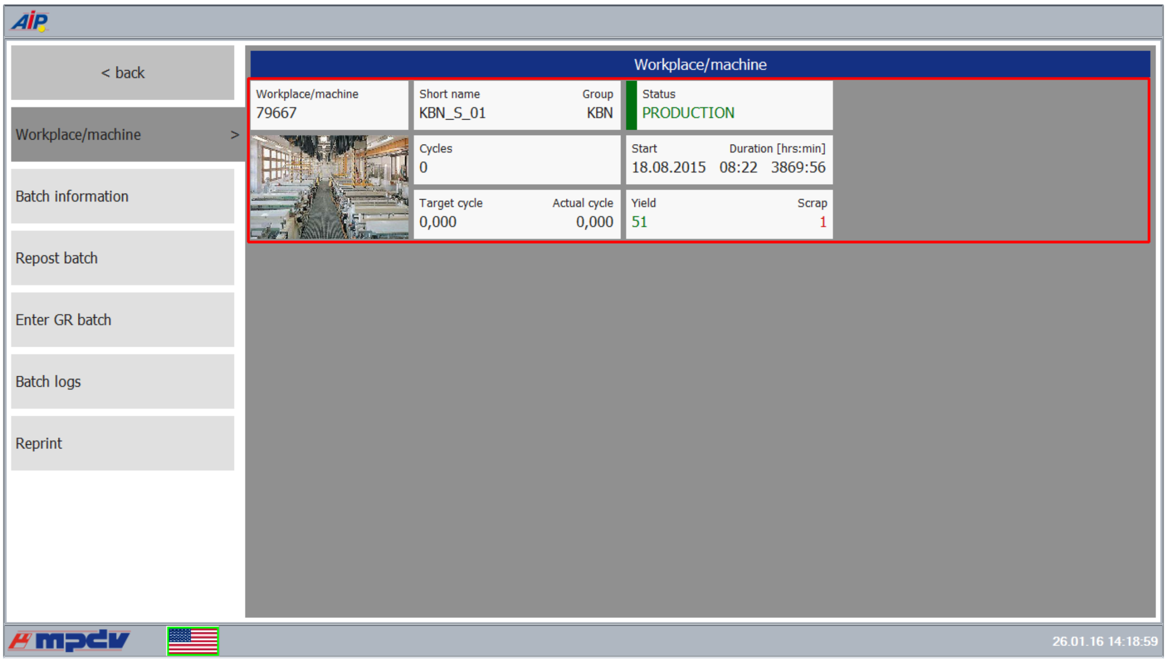
The file **a_data_pnr.xml** contains the definition for data relating to staff in the middle of the layout.

The button on the left side is located in the layout **l_pnr.xml**.

This layout is also used to display data and to request functions for staff logged on to a group workplace. When requesting the functions, an error message appears as staff and operations are logged on together to a group workplace.

l_res.xml

The resource layout opens if in the main view the third list "Resources logged on" is displayed and you click a resource:



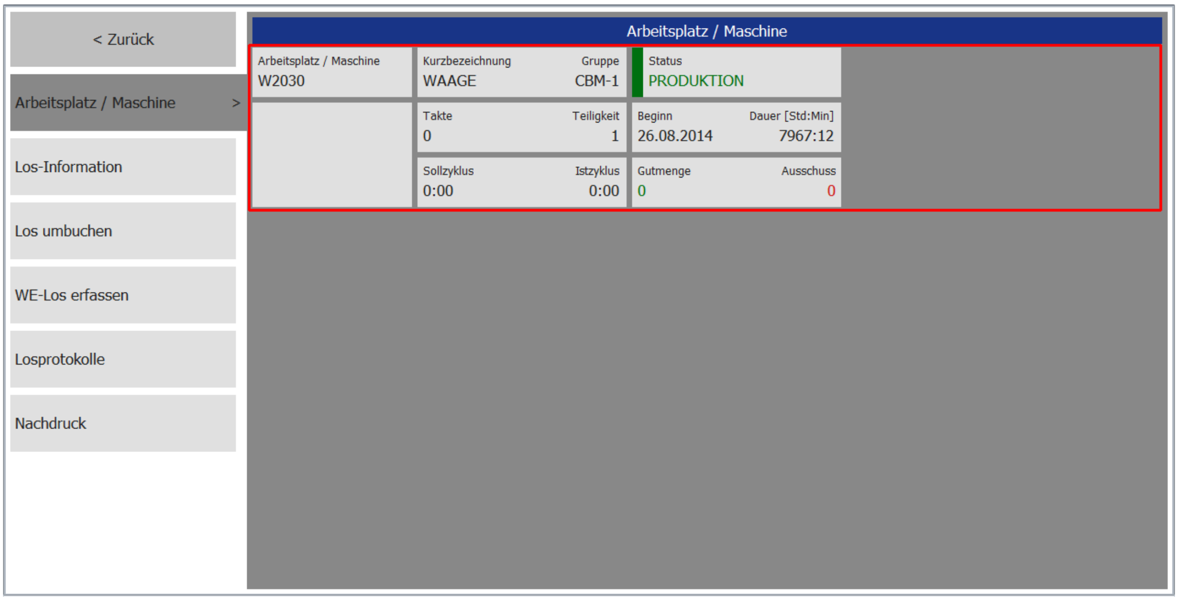
As described, the file **a_data_mnr.xml** defines the upper area on the right hand side to display data for the workplace.

The file **a_data_res.xml** contains the definition for data relating to a resource in the middle of the layout.

The buttons on the left side are located in the layout **l_res.xml**.

l_mat.xml

To request material layout, go to the main view. Click the button containing three dots to the left of the 3. lists "Input material logged on" and "Produced output batches":



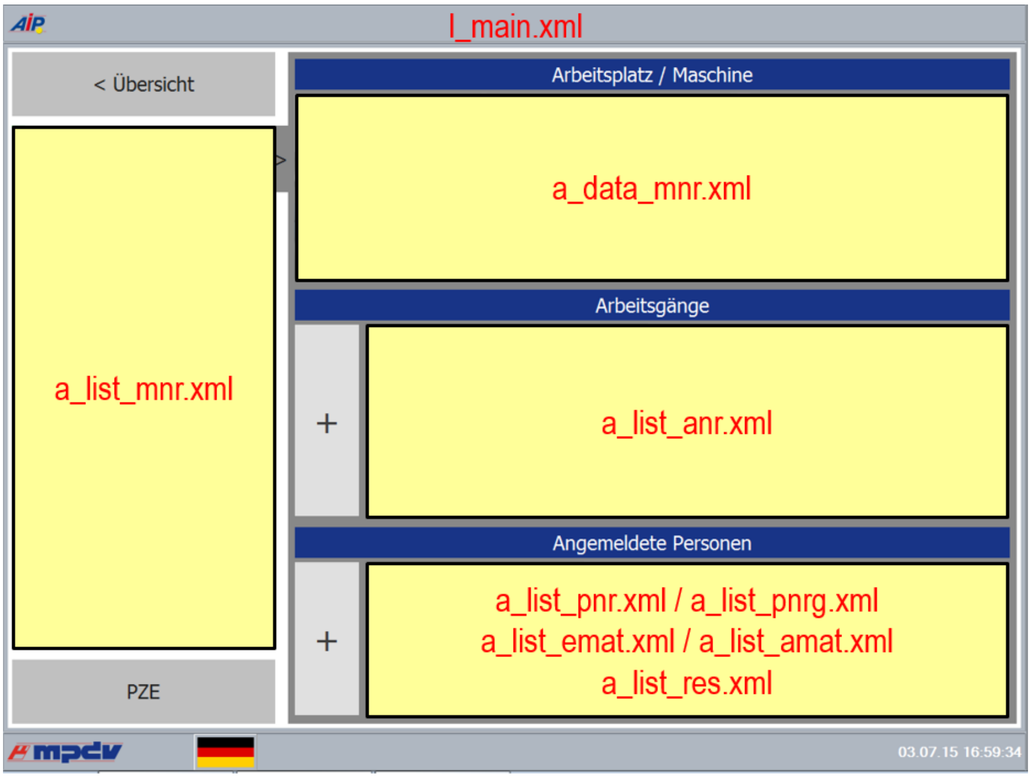
As described, the file **a_data_mnr.xml** defines the upper area on the right hand side to display data for the workplace.

The buttons on the left side are located in the layout **l_mat.xml**.

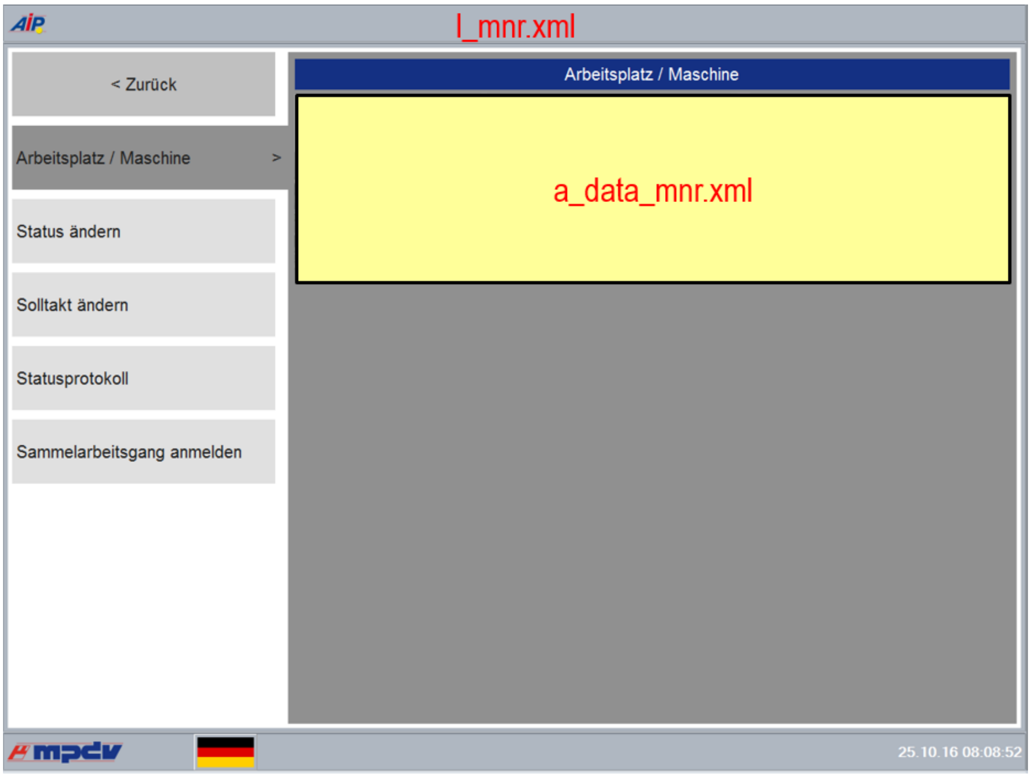
10.4.1 Overview XML files

Icon view:

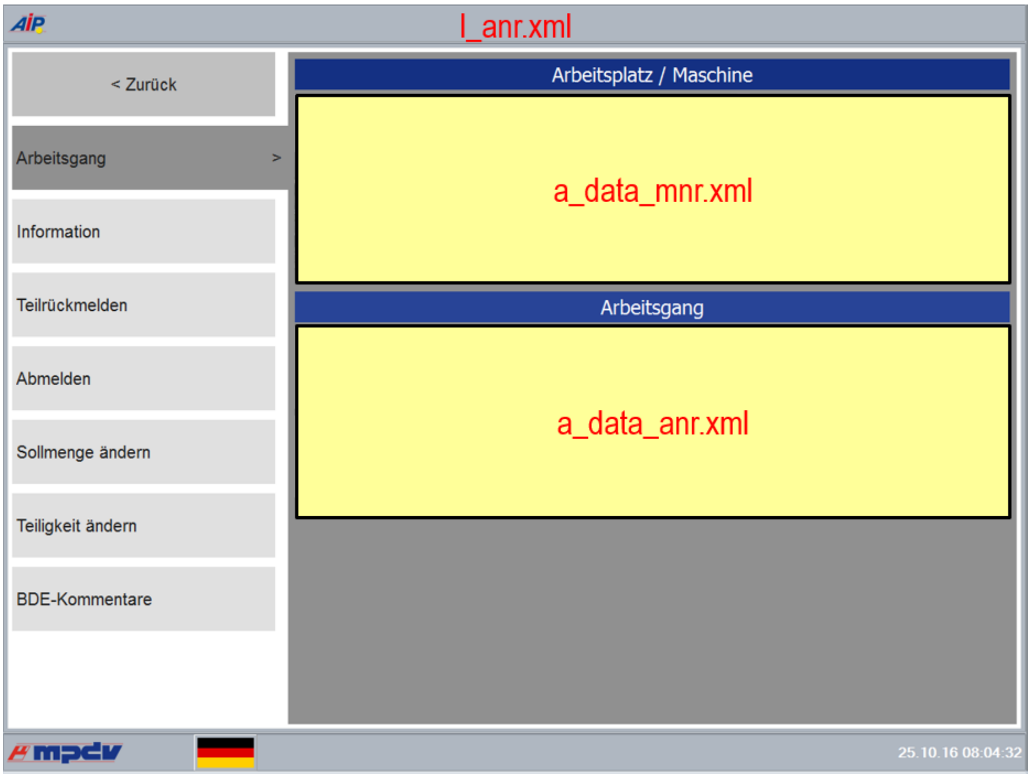
Main view:



Workplace layout:



Operation layout:



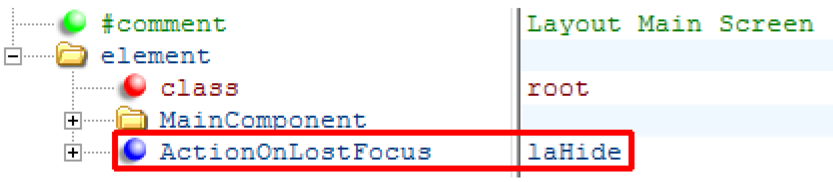
10.4.2 Layouts depending on the worplace type

You can override the layouts requested in the main view depending on the *batch management* and the *workplace type*. Both settings are stored in the dialog *Workplace and resource configuration* in the tab *Workplace configuration* and consist of one letter. If you copy a layout and both letters are written as lower case letters, are separated by an underscore ("_") and added to the file name, then this layout is used for all workplaces including batch management and used with the relevant workplace type.

For example, buttons should be made available with other dynamic dialogs for order postings at a packing station (letter "C") without batch management (letter "N"). To do so, copy the layout **l_anr.xml** onto the file name **l_anr_nc.xml**. You can then modify the buttons for the packing station in this layout.

10.4.3 Taking over changes in the layout configuration

Using the attribute „ActionOnLostFocus“, you can make the following settings per layout:



laFree

Using this setting, the displayed layout is discarded on changing to another layout and loaded anew from the XML files on the next start of this layout. Changes in the configuration of this layout, which were made in the meantime, are taken over.

laHide

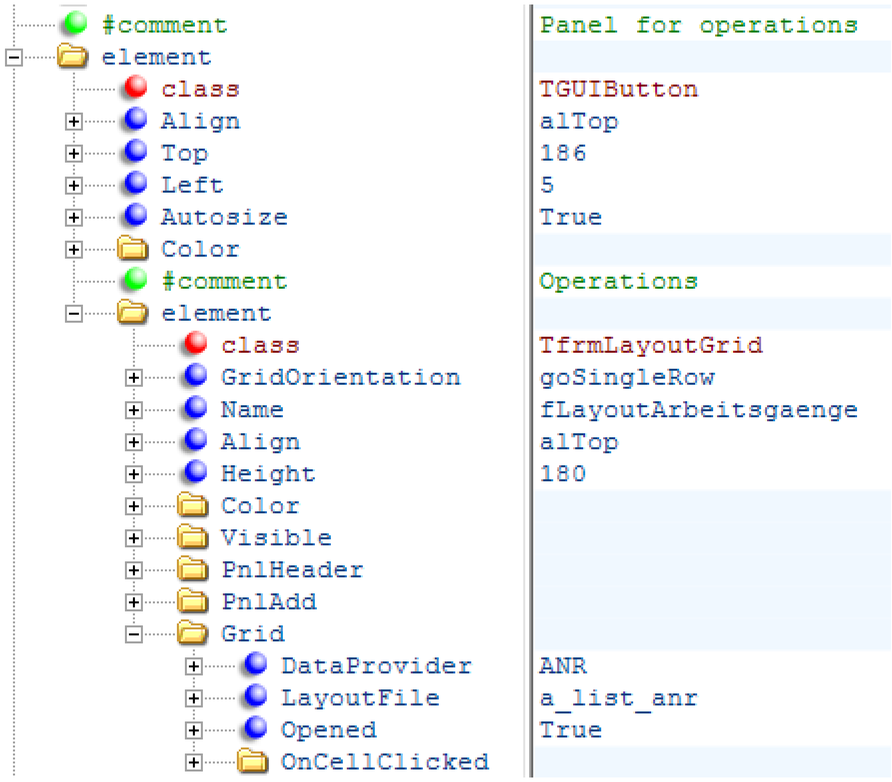
With this setting, the layout is not discarded, but stays in the background when you change to another layout. Changes of the layout configuration, that were saved after the first layout display, do not have an effect as the layout is not loaded anew from the XML files.

The setting *laHide* is applied by default in the layout of the main view (**l_main.xml**) to keep the scroll position in the lists on the right hand side when you return to this layout from another layout.

When you change the language during runtime using the flag in the status bar, the currently displayed layout is loaded. Changes of the layout in the main view are then taken over.

10.5 Configuration of lists

The configuration of lists is explained using the list of the logged on operations in the layout **l_main.xml:**



Lists of operations have their own panel in order to keep their position if operations of aggregates (a line is separated) are hidden.

The class *TfrmLayoutGrid* is responsible for the list display.

The settings below *PnlHeader* specify if and how a heading is displayed above a list.

The settings below *PnlAdd* define the button to create a new entry. In the above example, it's the button containing a "+"-symbol .

In the *Grid* area the data source (*DataProvider*) is set for the list. *LayoutFile* specifies which file defines the display of an element in the list. Below *OnCellClicked* you control what happens if you click an element in the list.

10.5.1 Filtering the displayed elements in the user interface (as of version 8.2.1.1)

The displayed lists can be filtered in the user interface. A text field must be included in the header of the list of type *TFrmLayoutGrid.*

The search syntax is equal to the search syntax of the lists in the dynamic dialogs.

A search field is integrate to a list of the type *TFrmLayoutGrid* eingebunden:

```
<element class="TfrmLayoutGrid">
	<SearchPanel>
		<Settings>
			<EnableSearch>true</EnableSearch>
			<SearchType></SearchType>
			<ExecuteEvent>return</ExecuteEvent>
			<SearchFields>ANR|AGNR</SearchFields>
			<SearchControlWidth>50</SearchControlWidth>
		</Settings>
	</SearchPanel>
	<SearchPanelPosition>header</SearchPanelPosition>
…
</element>
```

Details on the properties

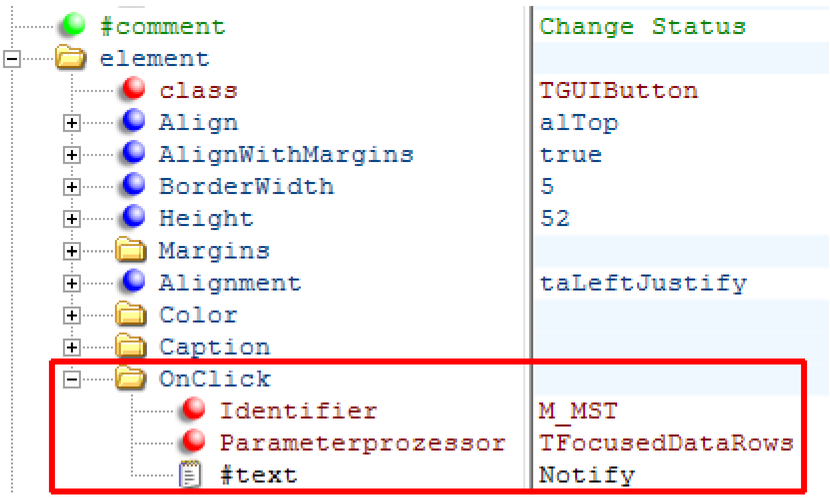
Properties	Description
Settings.EnableSearch	Display search panel
Settings.SearchType	Currently, only text is possible. Describes the visual search component.
Settings.ExecuteEvent	Event that should trigger the search. The following configurations are possible: 1. **KeyDown** The search is performed immediately on pressing the key. Compared to the following configurations, the above configuration costs a lot of time and shoud only be used for small data quantities. 2. **Button** The search is performed by clicking an additionally displayed button. 3. **Return** (Extension of "Button") The search can also be performed by pressing the return key.
Settings.SearchFields	Fields in the data source where you want to find the text you entered.

Settings.SearchControlWidth	Width of the input text box
TfrmLayoutGrid.SearchPanelPosition	Search panel position **row**: A row above the cells **header**: In the header of the TFrmLayoutGrids

If the search line is displayed, the list of type *TFrmLayoutGrid* requires more space in height. This space is at the expense of the tiles showing the data in the view. It is therefore recommended that you also change the height of the tiles with your data when using this functionality.

10.6 Request dynamic dialogs

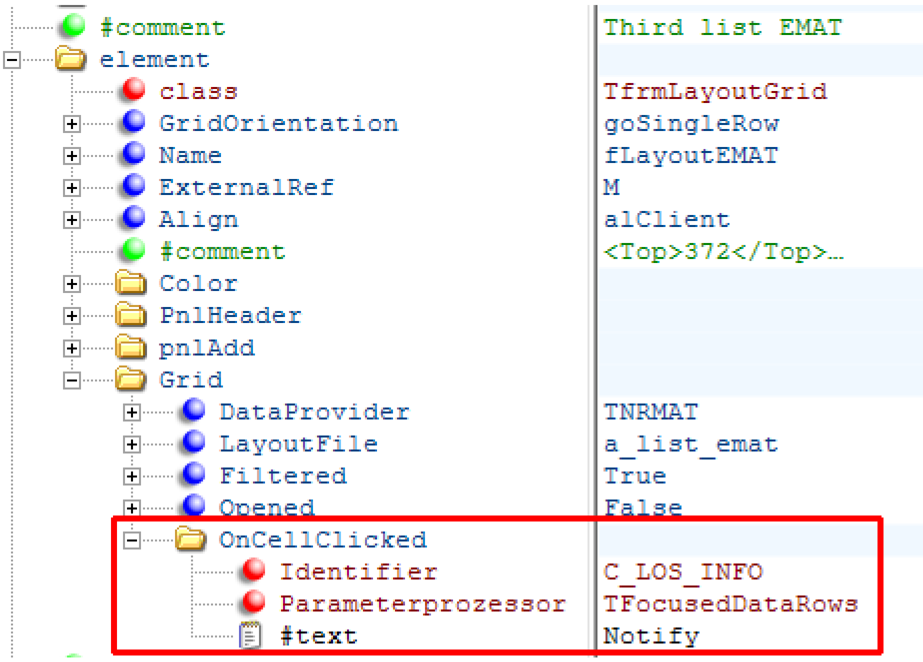
You can store an action in the individual fields and elements in order to perhaps request a dynamic dialog. In case of a button which is included on the left hand side in many layouts, you can enable this by using the entry *OnClick*:



This example shows the button "Change status" in the layout **l_mnr.xml.**

The entry *Identifier* defines the dynamic dialog to be requested. Both other entries are constant.

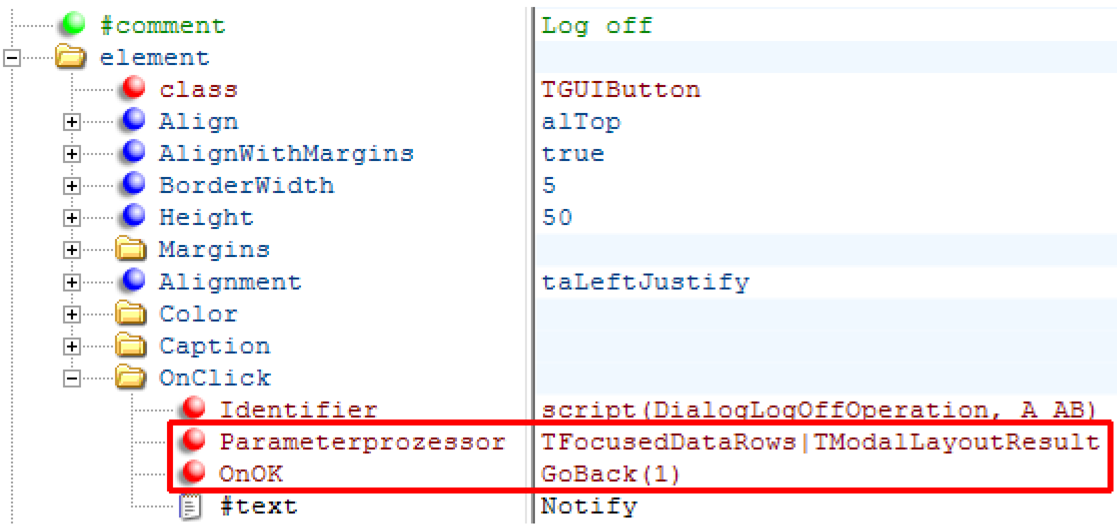
In a list of the class *TfrmLayoutGrid* containing several objects, the entry is called *OnCellClicked* and affects the elements below:



This example shows the request "MES Batch information" when selecting the input material in the layout **l_main.xml.**

10.6.1 Return after a dynamic dialog

After the execution of a dynamic dialog, you return to the layout where the dialog has been requested from. Alternatively, you can leave this layout and return to the previous layout which is normally the main view. Once an operation is logged off, it makes more sense to return to the main view than still displaying the data of a logged off operation.



A script request is stored in the setting *Identifier*. Depending on the workplace settings, the script request controls which dynamic dialog is requested. The first parameter specifies the script to be run and the second parameter the default value which is used if the script does not exist.

You then return to the previous view no matter if the dialog was completed or not, if errors occurred or if the dialog was interrupted without a change.

10.7 Positioning

The individual elements in the layout are arranged in a tree structure. The positioning of a subordinate item is always done in relation to the superior one (folder).

If a field shows a description and a corresponding value, the position of the description and the value refers to the top left corner of the field.

There are two ways to specify the position of the information. They are described in the following chapters.

10.7.1 Fixed positioning

Here, the position and the size of an element is specified with the following properties:

Top

Distance from the top

Left

Distance from the left

Height

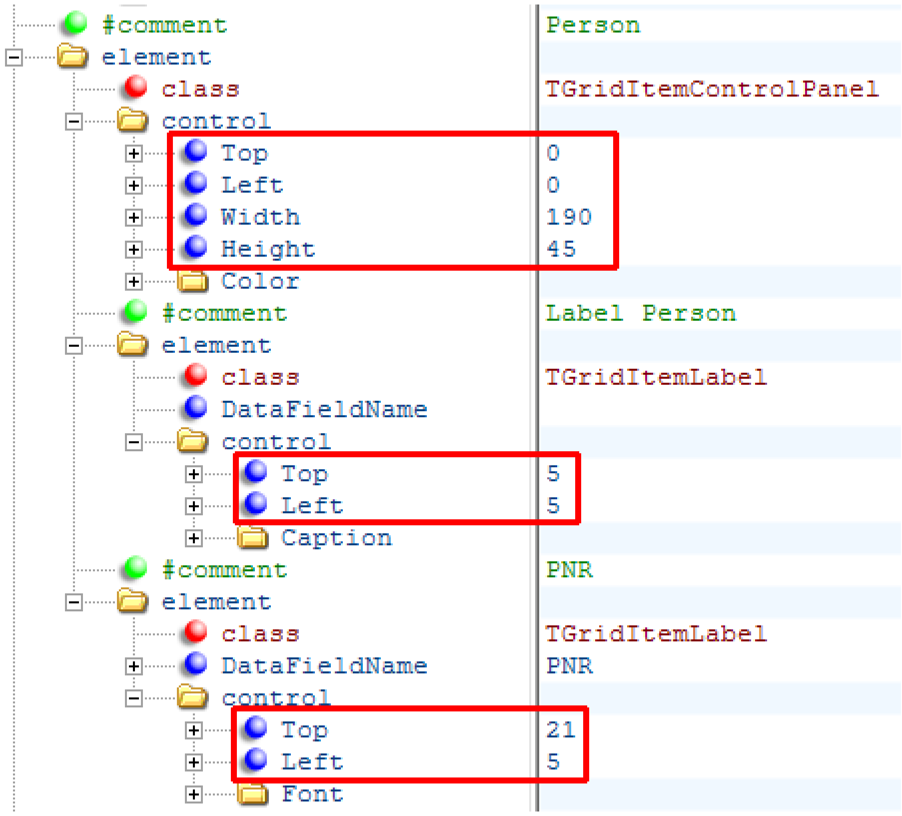
Height of the element

Width

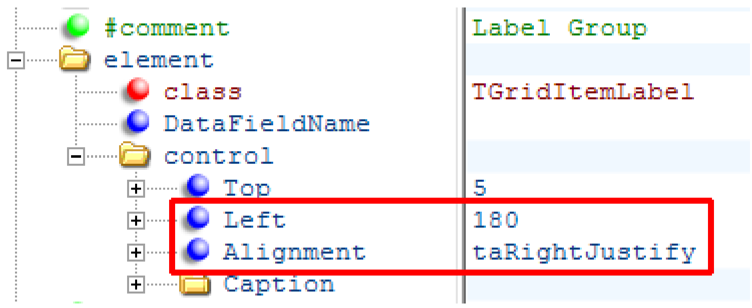
Width of the element

Properties can be found below the entry *control*.

The following example shows a logged on person for **a_data_pnr.xml**:



The property *Alignment* also specifies if the element is positioned towards the left (*taLeftJustify*), or the right (*taRightJustify*) or towards the center (*taCenter*). If no other property is explicitly set, the standard setting is left-aligned. The following example shows a position towards the right of the label *Group* of workplace data (**a_data_mnr.xml**):



10.7.2 Dynamic positioning:

If the positioning is done dynamically then the elements adapt their position and size to the one's above or next to it. The property *Align* can set the following:

Align=alTop / Align=alBottom

This element takes over the upper or lower limit and the width of the superior element. The property *Height* specifies the height.

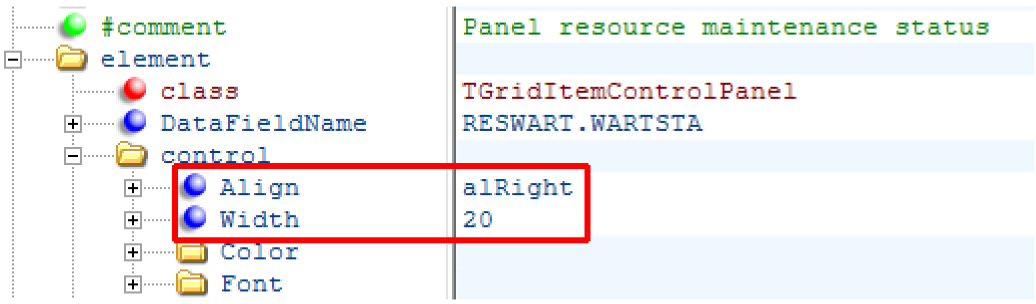
Align=alLeft / Align=alRight

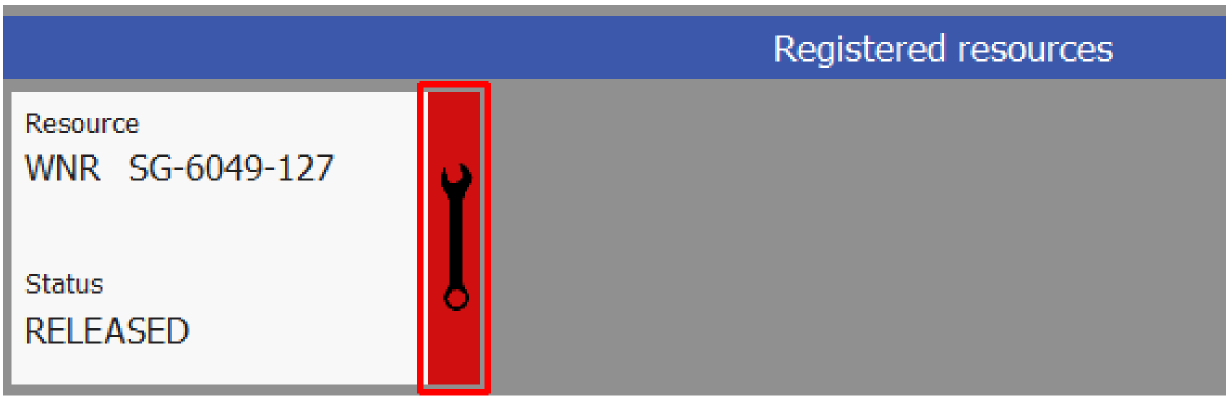
This element takes over the right or left limit and the height of the superior element. The property *Width* specifies the width.

Align=alClient

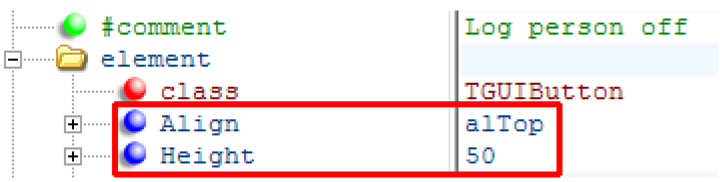
Aligns itself to the complete space of the superior element.

The following example defines the area for the color display of the maintenance status in the list of resources (**a_list_res.xml**):



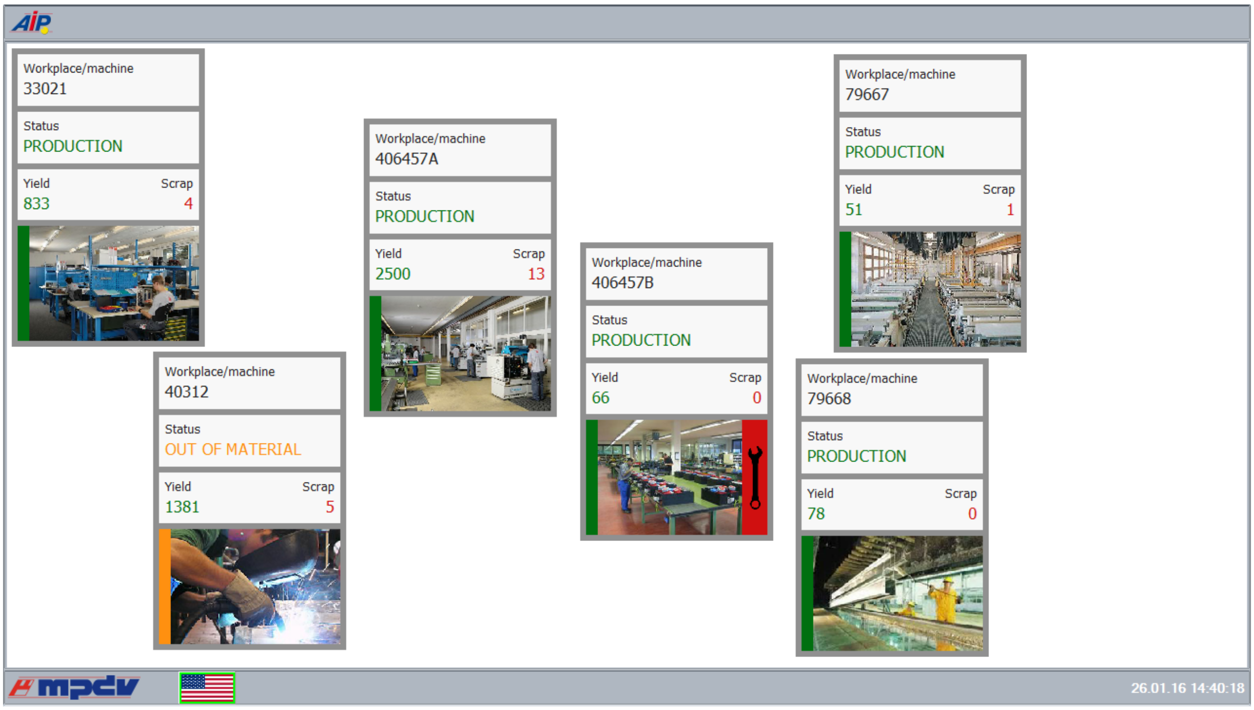


If neighboring elements have the same entry in the property *Align,* then they are displayed below or next to each other. This functionality is used in the button bars to request individual functions and ensures that there are no gaps if a function is hidden:



10.7.3 Positioning of workplaces in the icon view

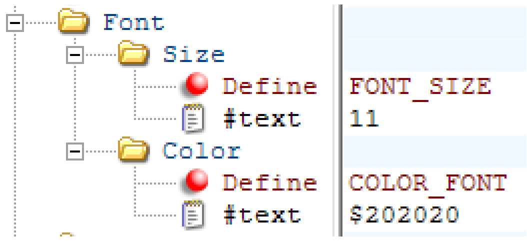
Positioning of individual workplaces in the icon view can be changed during runtime. You can start the design mode (password protected "mos6050") by double click the AIP icon in the top left corner. You can then position the workplaces by Drag&Drop. Double click the AIP icon to finish the design mode. The positioning of the workplaces is stored in the file *gui\p_view_mnr.xml.*



To reset the positioning, please delete the file *gui\p_view_mnr.xml.*

10.8 Text formatting

Text formatting in the GUI is performed using the entries below the field Font:



You can make the following settings:

Size

Set the font size

Color

Set the font color in reversed RGB notation

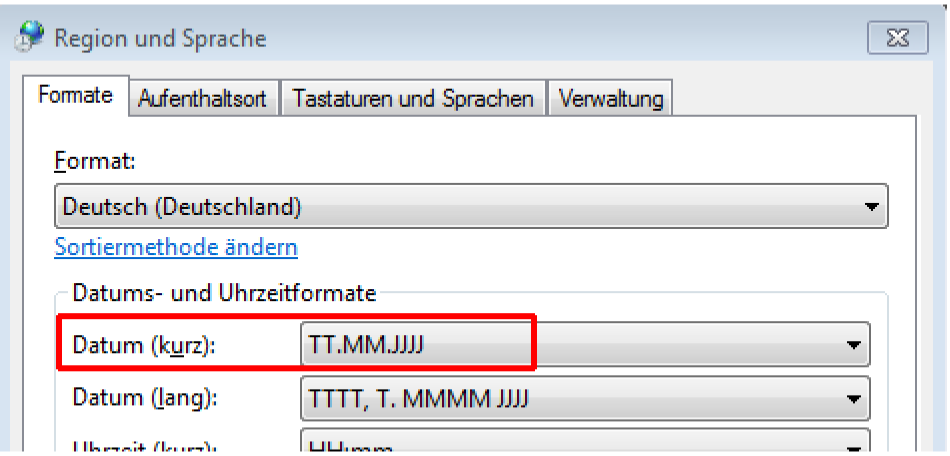
If the attribute *Define* is applied, the value entered in the field *#text* is not used. Instead the content of the entered constant is used (with both settings).

10.9 Formatting functions

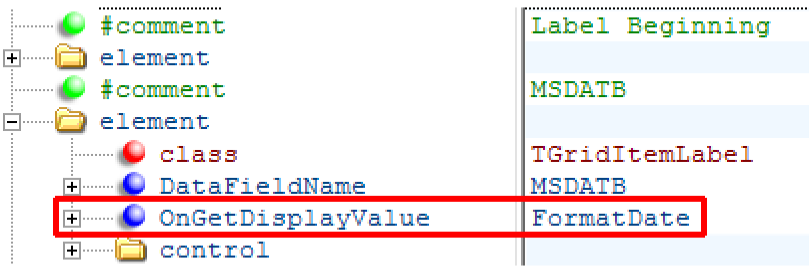
You can display data with the aid of different formatting functions:

FormatDate

This function sets a date depending on the date format *date (short)* set in the operating system:

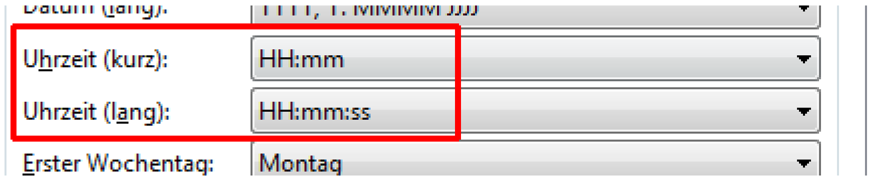


There is an example located in the workplace data (**a_data_mnr.xml**) for the start date of the current status



FormatTime / FormatTimeLong

The function *FormatTime* sets the time depending on the time format *Time (short)* set in the operating system:



FormatTimeLong uses the format *Time (long)*.

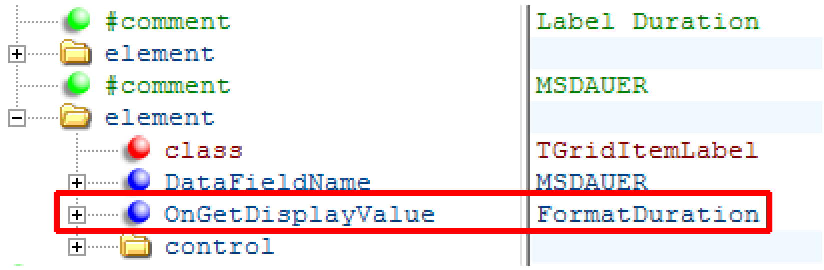
FormatDuration / …

There are various functions to display *durations* in different output formats.

Function	Format
FormatDuration	Hours:Minutes
FormatDurationMMSS	Minutes:Seconds

FormatDurationHHMMSS	Hours:Minutes:Seconds
FormatDurationHHII	Hours,decimal hours (Industrial minutes)
FormatDurationHHIII	Hours, decimal hours (3 decimal places)
FormatDurationMMII	Minutes,decimal minutes

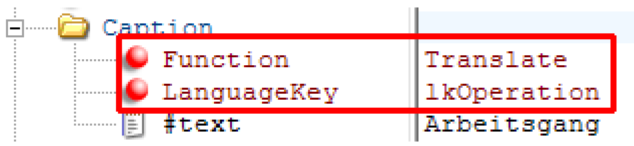
You can find an example for the output of a duration in workplace data(**a_data_mnr.xml**). It states the duration of the current status :



10.10 Multilingualism

AIP2 uses just like the AIP the Multilizer to translate text into another language. Language keys with the prefix "lk" are used for the new GUI. German texts without the prefix "lk" are also processed if they are included in the mld file.

Text for translation can be added using the function "Translate" and the entry "LanguageKey" (in accordance with the language set).



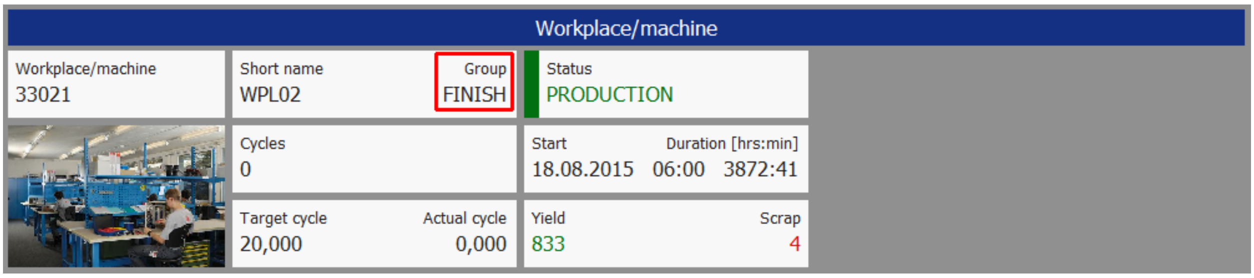
This example contains a German text "Arbeitsgang" (operation) which does not affect the processing. The text is replaced by the translated text using the language key during runtime.

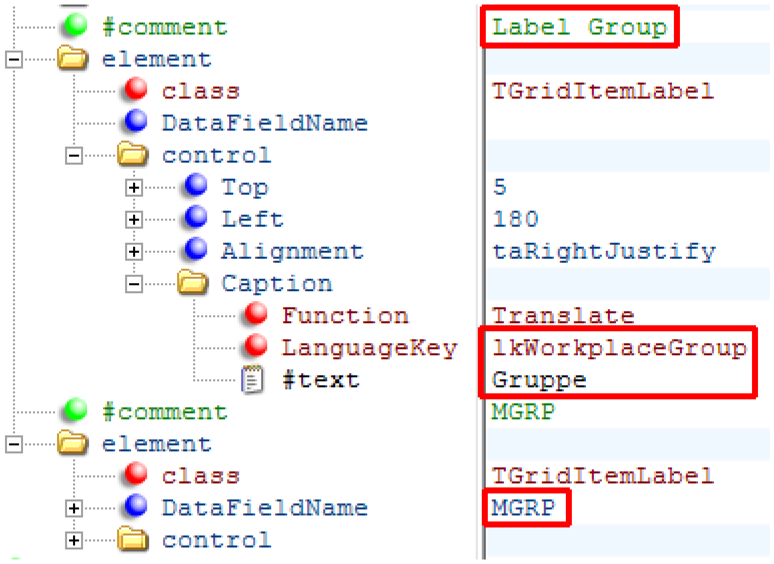
10.11 Examples / exercises

This chapter shows customization options of the layout using examples.

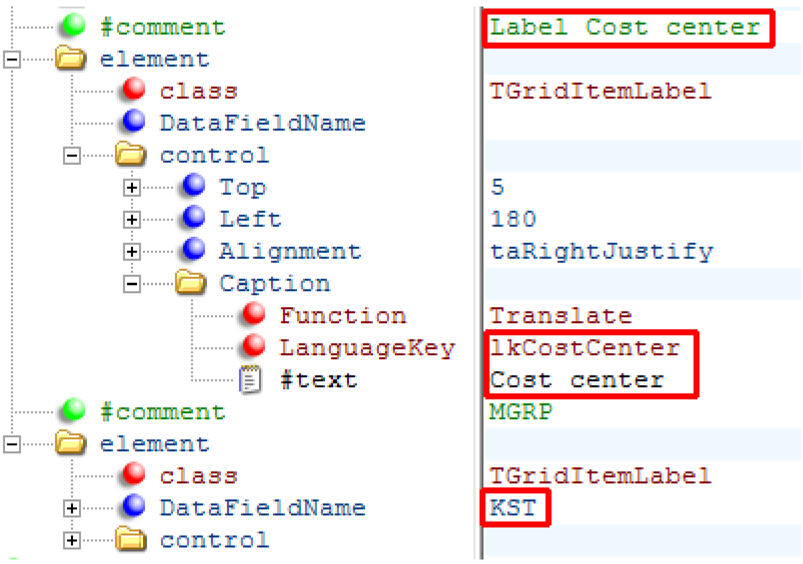
10.11.1 Change existing fields

Replace the field "group" with "cost center" in the displayed workplace data.





You need to change the entries for "label group" and "MGRP" (machine group) in the parameter **a_data_mnr.xml** as follows:

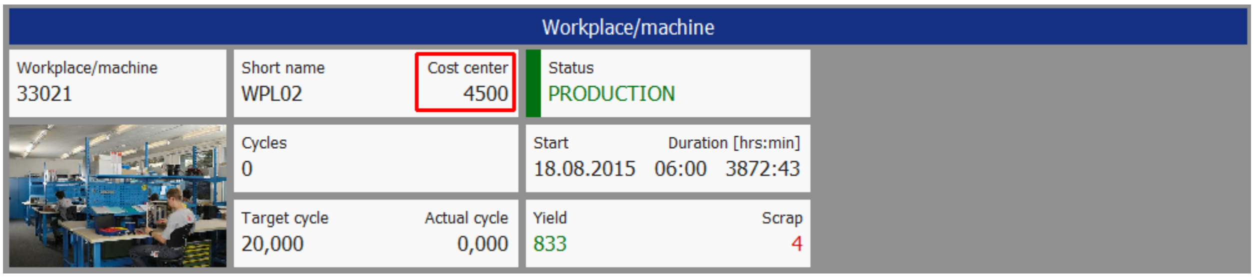


Entries with the description "#comment" are comments which do not affect processing.

The first element in the red frame shows the description above the data. The language key IkWorkplaceGroup is replaced by *IKCostCenter*. You can directly insert the text in the field "#text" if there is no language key available for the description. Both entries "Function" and "LanguageKey" must be deleted in this case.

The description is displayed towards the right hand side at position 180 ("Left": 180; "Alignment": taRightJustify).

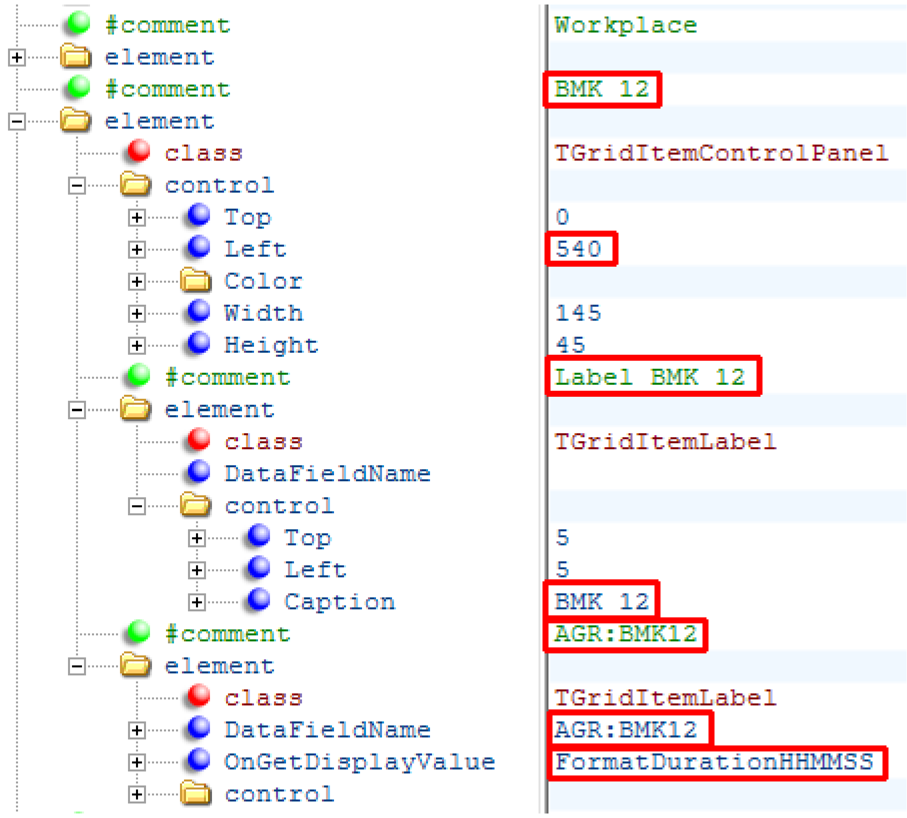
The second element is responsible for the display of the data field. Change the entry "DataFieldName" from *MGRP to KST.*



10.11.2 Add a new field

Display the duration booked on RPA 12 to the right of the status display.

This is done by copying the element with the comment *Workplace*. This element specifies the light gray space and includes 2 other elements including name and data.



The comment located above is also copied and changed on BMK12.

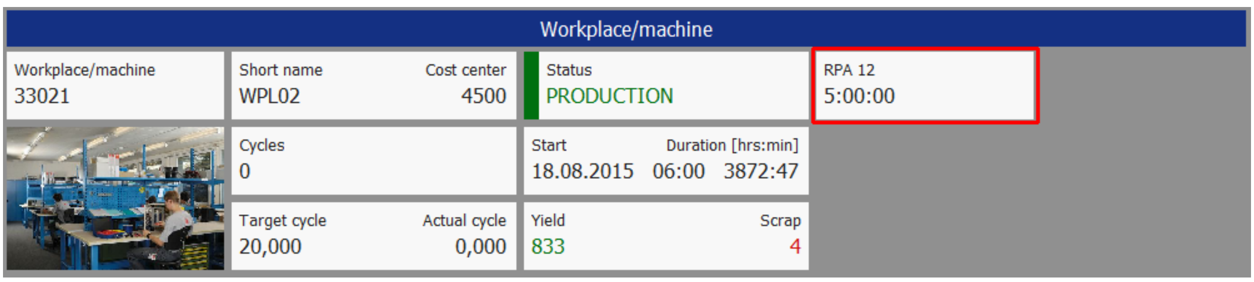
The position of the light gray area ("left") is made up of position ("left") and the width of the element *Workplace Status* plus a distance of 5 dots (345 + 190 + 5= 540). Both fields are located below the entry "Control".

The elements *Label BMK12* and *AGR:BMK12* are located on the light gray area. The position of both elements ("Top" and "Left") refer to the top left corner of the light gray space.

The entry "Caption" specifies the displayed text. Here, language keys have not been used so the text is not translated.

The entry DataFieldName below the comment *AGR:BMK12* was changed to the field name *AGR:BMK12*.

The formatting function FormatDurationHHMMSS shows the duration in hours:minutes:seconds.



10.11.3 Add user fields

Add a user field to the interface by adding a new field as described in the previous chapter. Use the the acronym of the user field (e.g. ANR_FU_65).

If you want to format user fields for dates, times, or durations, you can use the formatting functions. See section "10.9 Formatting functions".

Load user field with cataiplay.ini

Note that the AIP lists that serve as data providers do not contain user fields in the standard system. In order for the user fields to be added to the list, you must configure it in the **ctaiplay.ini** file. As long as the user fields in the **ctaiplay.ini** file are not configured correctly, they remain empty on the user interface.

In the customer-specific terminal directory (e.g. if user fields are to be added at terminal group level: \mip\<systemnr>\custom\aip2\tgrp_xxx\) a **ctaiplay.ini** is created, which contains the different section.

- Activate the additional loading of the user fields for operation- or order-related XML files in the section [Custom Userfields ANR].
- Activate the additional loading of user fields for machine-related XML files in the [Custom Userfields MNR] section.

You can find examples on how to do it further along.

The activated fields are then available in all XML files connected to the DataProvider ANR or MNR.

Available user field in machine and order lists

All identifiers of user fields and also other fields that can be reloaded are located in the headers.dat file in the "spool" directory of the terminal. It consists of four lines:

Start of the row	Content
10|…	Machine list: Fields that are always included in the list.
*10|…	Machine list: Fields that can be reloaded.
11|…	Order list: Fields that are always included in the list.
*11|…	Order list: Fields that can be reloaded.

The following user fields can be reloaded:

Machine list

FU:1 to FU:66

Machine user fields

Order list

ANR_FU_1 to ANR_FU_66

Operation user fields

AUNR_FU_1 to AUNR_FU_66

Order user fields

MNR_FU_1 up to MNR_FU_66

Machine user fields

VERARBCODE_FU_1 up to VERARBCODE_FU_66

Processing code user fields

AGR_FU_1 to AGR_FU_66

User fields of the operation status (cannot be used in the standard system, reserved for Customizing!)

Example 1: User fields in the operation list

- User field 1 of the operation should be entered in the order list with the name " Order date ".
- User field 66 of the machine with the name "My long user field" should be added to the order list.

Field definition in section [Custom user fields ANR]

```
[ Custom usernfields ANR ]
GRID_LIST_TYP=ANR
; additional fields of the order list
ANR_FU_1= ; User field 1 of operation, MyDate FU:1 [operations list]
MNR_FU_66= ; User field 66 of machine, My long user field [operations list]
```

Example 2: User field in the machine list

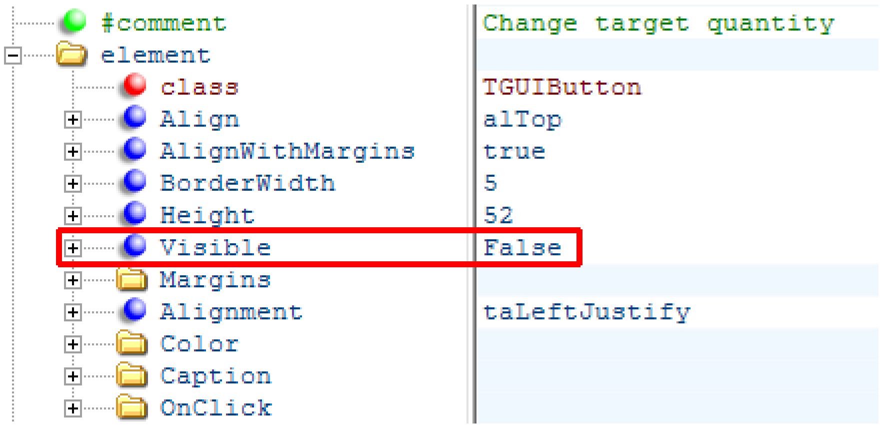
User field 66 of the machine with the name "My long user field" should be added to the machine list.

Field definition in the section [Custom user fields MNR]

```
[ Custom userbfields MNR ]
GRID_LIST_TYP=MNR
; Additional fields in the machine list
FU:66= ; User field 66 of machine, My long user field [machine list]
```

10.11.4 Remove button

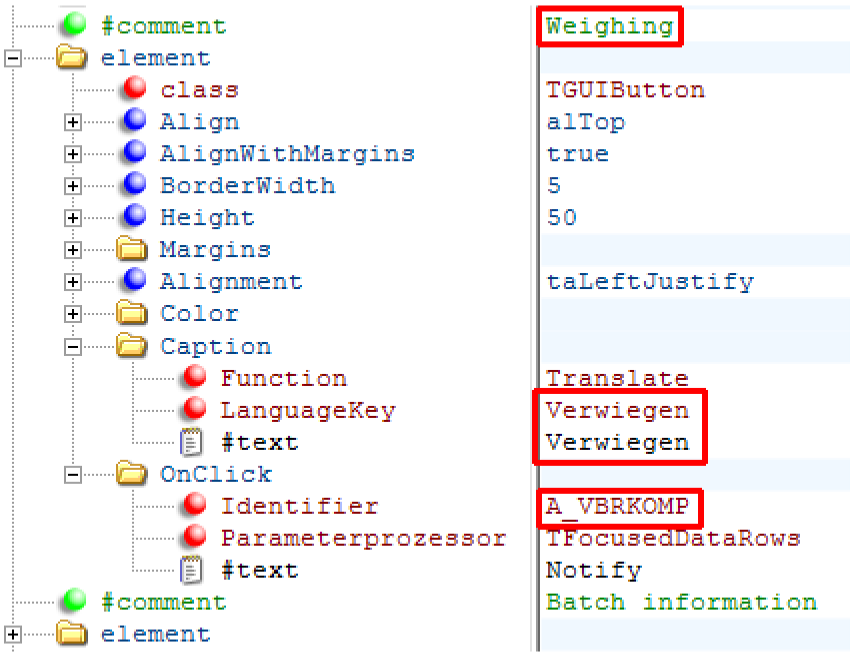
Delete the button *Change target quantity* from the operation layout **(l_anr.xml).**



The new entry Visible=False hides the button. Optionally, you can also delete the comment and the element.

10.11.5 Add button

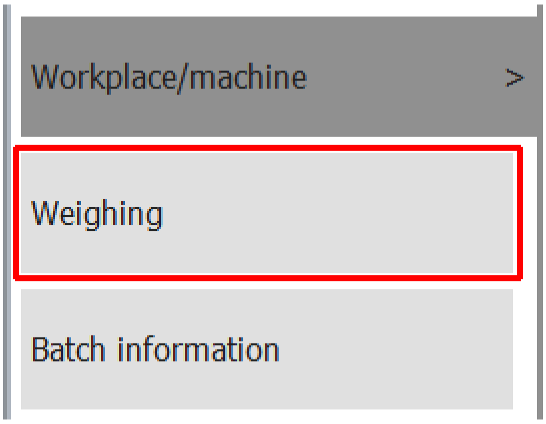
You want to add a new button "Weighing" in the layout for "input material", "output batch" and **(I_mat.xml).**



First of all, copy an existing button including the corresponding comment. In this case the button "Batch information" was copied. Change the comment in order to easily find the new button in the list of elements.

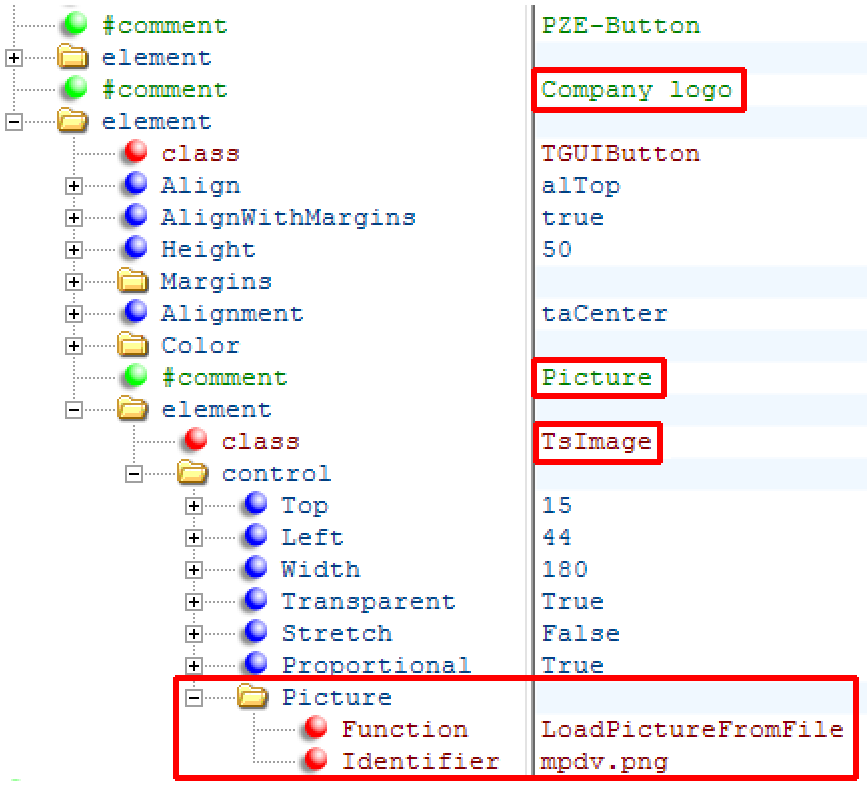
The entry "Caption" specifies the displayed text. In the example, the English text "Weigh" is used as language key.

The "Identifier" specifies which dynamic dialog is requested.



10.11.6 Integration of a picture

The task is to have a logo displayed in the main view (**l_main.xml**) below the button "PZE".



Copy the button "PZE" and the comment. Change the comment.

As the button has no labeling, delete the entry "Caption". Also delete the entries "Visible" and "OnClick".

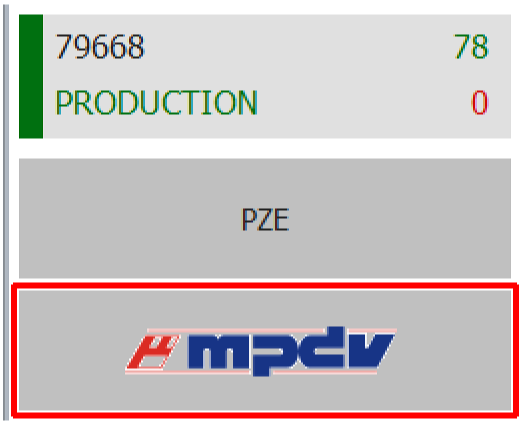
You need a new element of the class "TsImage" in order to display the new picture. This element was copied from the staff list (a_list_pnr.xml) and has the class "TGridItemImage", as it is not located on a button but in a list. Change the class to "TsImage" after copying. Delete the entry "Visible".

The file name of the picture is entered in the field "Identifier" of the entry "Picture“. There are two options to load the picture:

- **LoadPictureFromFile** reads the file from the spool directory.
- **LoadPictureFromAIP** uses pictures included in the AIP2 in the file "pict.zip" or "pict_cust.zip". This information is more efficient as these picture are stored in a buffer. This method only supports images of type PNG and BMP.

Different settings are available to display the picture.

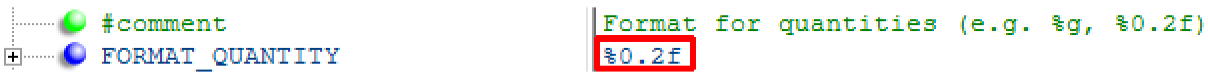
- **Transparent** – For example, PNG files support transparent areas where the background of the picture is visible. Functionality can be switched off using the value *False.*
- **Stretch** specifies if the picture is shown in its original size (value *False*) or if *Height* and *Width* are adjusted (value *True*).
- **Proportional** controls whether the ratio of the width of the image and the height of the image is maintained (value *True*) or not (value *False*) when the image size is adjusted to the specified *Height* and *Width.*



10.11.7 Change quantity format

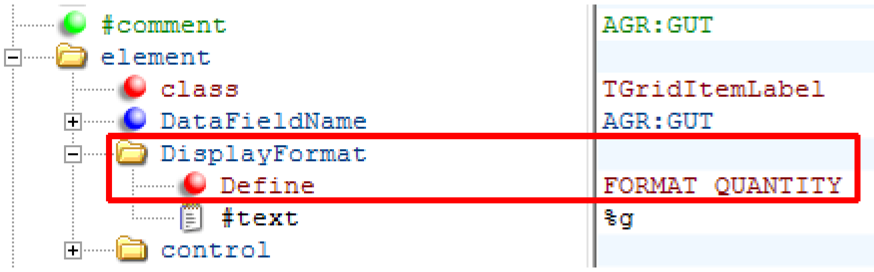
Generally show the quantity format with 2 decimal places.

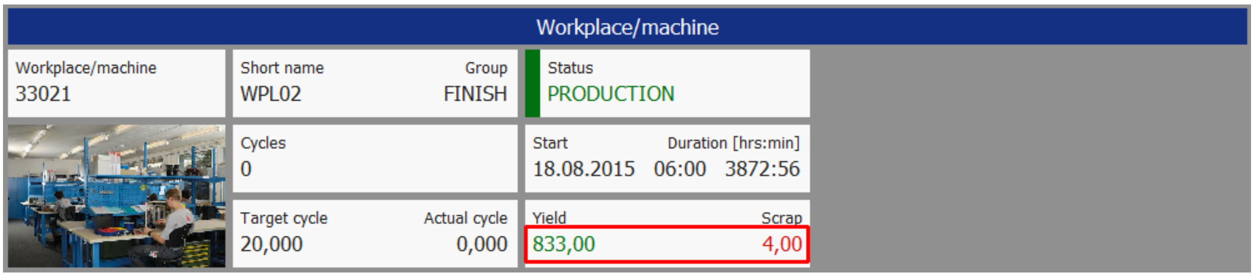
The quantity format is configured in the file **globaldefines.xml** using the constant *FORMAT_QUANTITY:*



The value "%0.2f" ensures that quantities are displayed with 2 decimal places.

You can find an example for this constant (Define) when yield is displayed for data of the workplace (**a_data_mnr.xml**):



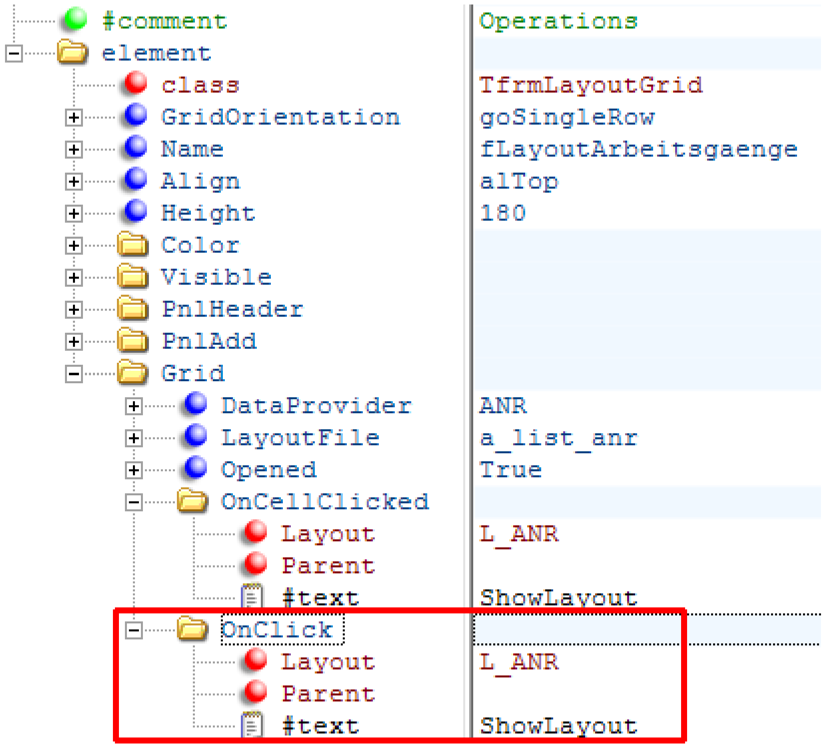


The set format only affects configured layouts and not dynamic dialogs.

10.11.8 Postings for operations not logged on

The workplace configuration in the MOC has a button called "Posting of operations not logged on". If this button is activated, you can interrupt or log off the operations not logged on (posting to the server). You have to extend the configuration if you would like to carry out these postings in the AIP2.

The operation layout only opens in the AIP by using the buttons *Interrupt* and *Logg off* if you select the logged operation. If the following extension in the file l_main.xml is carried out, this layout also opens if you click the empty space in the list of operations.



The dynamic dialogs must also be customized in order to interrupt and logg off operations. Unless so-called simple dialogs are used.

10.12 Index

11 AIP2 - Local Configuration File keyboard.ini

You configure the virtual keyboard of the AIP2 terminal in the keyboard.ini file in the directory c:\mpdv\aip2 for the specific terminal.

To activate the changes in the configuration file, you must restart the terminal software.

Logic enabling the virtual keyboard:

The AIP2 terminal shows the keyboard if an input field is focused. The keyboard is placed with reference to the field as described below:

Logic for placing the virtual keyboard:

It is tried to place the keyboard directly below the input field. If there is not enough space to the bottom of the screen, it is tried to place the keyboard directly above the input field. If the space above the control element is not sufficient for the keyboard, the keyboard is placed at the bottom of the screen.

These are the priorities for horizontal alignment:

- to the right of the control
- to the left of the control
- to the edge of the screen that is further away from the control

If the “VirtScreenSize“ option is enabled, the virtual keyboard is not aligned on the virtual screen but still on the real screen. Consequently, the keyboard may also reach beyond the terminal program.

The virtual keyboard can be configured in the local keyboard.ini file on the terminal. Example:

```
[Keyboard]
HideTime=10
ScaleMultiplier=0.9
FixNumbers=ON
Configuration=ON
;Logging=ALL
;Processes=ctaip.exe
;ClassesForLetters=TVtEdit
;ClassesForNumbers=TMPDVSimpleNumericField
```

<u>HideTime</u>

The set value specifies for how many seconds the keyboard is invisible if you click on the key showing the icon  on the left hand side. This key is not visible if the value "0" is entered.

ScaleMultiplier

The keyboard size can be reduced and increased. The value range is between 0.9 and 4.0. A dot is used as decimal separator.

The default value is 1.0.

FixNumbers

Allowed values: ON|OFF

If FixNumbers=On is set, the number keys located in the top row of the virtual keyboard remain visible even if the Shift key or CapsLock key is pressed. ON is set by default.

Configuration

Allowed values: ON|OFF

The keyboard layout, which is installed and activated in the Windows language settings, specifies the layout of the virtual keyboard. You can activate different keyboards in the operating system. For the virtual keyboard, you can then switch between the different activated keyboards.

The entry *Configuration=ON* activates the button . Use this button to open the dialog to select one of the keyboards activated in the operating system.

Default is OFF.

Logging

Allowed values: OFF|ON|ALL

Logging can be enabled using this entry. The advanced logging is configured by setting ALL.

OFF is set by default.

Processes

The entry "Processes" specifies for which additional processes the virtual keyboard will be used. The separate entries are separated by comma (e.g. processes=notepad.exe.explorer.exe). If this entry is not included, the keyboard for these processes is available in *ctaip.exe* und *iniedit.exe.*

ClassesForLetters

This entry defines for which additional classes the alpha-numeric keyboard should be displayed. The current classes for AIP2 (TMPDVSimpleField, TsEdit, TsMemo, TMPDVTypEdit, TMPDVSimpleEditField, TMPDVPictureField,TEdit, TButtonedEdit, TEditControl) are fixed in the source code. This entry can be used to extend the list.

ClassesForNumbers

This entry defines for which additional classes the numeric keyboard should be displayed. The current classes applicable for the AIP2 (TMPDVNumericField, TPagerNumField, TMPDVSimpleNumericField, TVTEdit) are fixed in the source code. This entry can be used to extend the list.

The classes that are fixed in the source code cannot be overridden using a different entry in the configuration file. If you want to display the other keyboard for a field, you can change the input type of the field in Dialog Configuration.

Dialog-specific configuration

There is the option from version 1.6.0.0 of the keyboard.exe to configure the location of the virtual keyboard per dynamic dialog. The user has to extend the configuration file keyboard.ini accordingly.

Sample configuration:

[WF_AA_QUA]	=> Name of the dynamic dialogs
X-Position=50	=> Distance in pixels from the left edge of the screen
Y-Position=50	=> Distance in pixels from the top edge of the screen

- Specifying the X- and Y-position is mandatory.
- The configuration is only available for dynamic dialogs that are configured on the MOC.

The virtual keyboard can also be switched off if the terminal is connected to a real keyboard. This can be configured in section [SYSTEM] of the local ctaip.ini file.
Example:

```
[SYSTEM]
Parameters=-t
```

Syntax:

+t/-t --> enables/disables the virtual keyboard; irrespective of the terminal type

12 Extended Application Configuration

12.1 Overview of INI configuration files

INI file	Configurations
ctaip.ini	Host name, terminal number, virtual keyboard on/off, inputs/outputs
ctaip.mld	Translation of labels
ctaipbut.ini	Configuration of buttons: order, positioning, icons, if necessary licenses
ctaiplay.ini	Configuration of grid layout; basic screen: height of tables and buttons; layout of BDE comments; OP info, machine info
dialog.ini	Configuration of font type/size in dialogs and of tab sizes in workflow dialogs
keyboard.ini	Configuration of the virtual keyboard: size and behavior

12.2 General

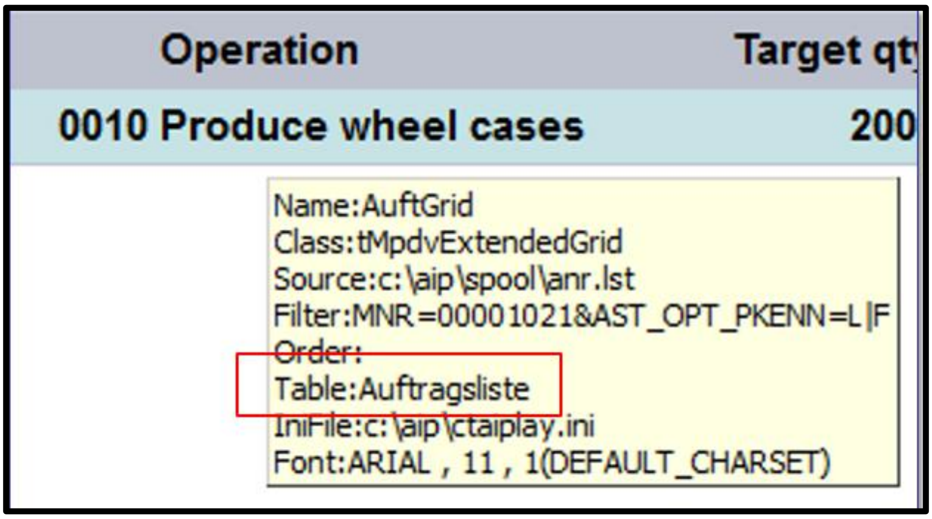
12.2.1 Identification of lists / elements in the terminal

The following shortcut activates information about available lists or elements in the terminal:

CTRL + ALT + F6
or in
AIP DEBUG menu: Further debug functions → Activate hints (scroll down)

A tooltip is shown when hovering the mouse pointer over a table or element.

The value "table" identifies the list:



12.3 Modifications to ctaipbut.ini

12.3.1 General

The complete standard INI files are located on the server directory \mip\ctnet\win\aip2

Any deviations are developed in specific, customized directories.
e.g. for customized terminal groups \mip\<SystemNo>\custom\aip2\tgrp_xxx

(xxx = number of terminal group)

An <u>empty</u> file (ctaipbut.ini) is generated in these directories. All customized sections e.g. [ANR-ALL-Page1] are copied to this file. Then the configuration is performed in this file.

After restarting the terminal, files from the main directory \mip\ctnet\win\aip2 are merged with files from customized directories (e.g. \mip\<SystemNo>\custom\aip2\tgrp_xxx). The merged file is then transferred to the local terminal directory C:\MPDV\AIP2.

The directory \mip\ctnet\win\aip2 must not be changed, otherwise AIP2 might no longer work properly. In addition, default files are stored there and any changes made will be lost after updating (e.g. service pack)!

12.3.2 Modifications to the toolbar

A ctaipbut.ini file including the modified section is stored in the customized terminal directory (e.g. if the toolbar is changed for terminal groups: \mip\<SystemNo>\custom\aip2\tgrp_xxx\).

Example:

The order of buttons should be changed in the "machines" section of the AIP2 basic screen (position button "change status" first and then the "lock production status" button).

Initial configuration

[MNR-ALL-Page1]
1=M_INFO,L,,InfoRed.png
; switch basic screen from list view to single machine view:
2=VIEW,L,,VirtualTourSmall.png
3=P_SPERRE,L,Produktionsstatus sperren,Security Risk.png
4=M_MST,L,Status ändern,Status Flag Yellow.png
5=A_AN*,R,Arbeitsgang anmelden,SyBluPly.png

New configuration

[MNR-ALL-Page1]
1=M_INFO,L,,InfoRed.png
; switch basic screen from list view to single machine view:
2=VIEW,L,,VirtualTourSmall.png
3=M_MST,L,Status ändern,Status Flag Yellow.png
4=P_SPERRE,L,Produktionsstatus sperren,Security Risk.png
5=A_AN*,R,Arbeitsgang anmelden,SyBluPly.png

12.3.3 Modifications to button labeling

A ctaipbut.ini file including the modified section is stored in the customized terminal directory (e.g. if the toolbar is changed for terminal groups: \mip\<SystemNo>\custom\aip2\tgrp_xxx\).

The required sections can just be inserted if a customized ctaipbut.ini file already exists, e.g. due to changes to the order of buttons.

Example:

Labeling of buttons in the "operation" section should be changed as follows:

- "Partial confirmation" --> "Part. conf."
- "Interrupt operation" --> "Interrupt OP"
- "Log off operation" --> "Log off OP"

Initial configuration

[ANR-ALL-Page1]
1=A_INFO,L,,InfoBlue.png
2=A_TR,R,Teilrückmeldung,SyBluAdd.png
3=A_UN*,R,Arbeitsgang unterbrechen,SyBluPau.png
4=A_AB*,R,Arbeitsgang abmelden,SyBluStp.png

New configuration

[ANR-ALL-Page1]
1=A_INFO,L,,InfoBlue.png
2=A_UN*,R,AG unterbrechen,SyBluPau.png
3=A_TR,R,Teilrück.,SyBluAdd.png
4=A_AB*,R,AG abmelden,SyBluStp.png

12.3.4 Modifications to icons

General

Generate the file pict_cust.zip in the customer specific directory \mip\<SystemNo>\custom\.

Enter customer-specific icons (e.g. custom tools.png) to the file pict_cust.zip.

Note:
The file pic.zip contains all icons used at the terminal.

The file name for the button icon can be changed in the customized section of ctaipbut.ini.

Example:

Changing the icon for the button "Log on operation" from SyBluPly.png to Custom Tools.png.

Initial configuration

[MNR-ALL-Page1]
1=M_INFO,L,,InfoRed.png
; switch basic screen from list view to single machine view:
2=VIEW,L,,VirtualTourSmall.png
3=P_SPERRE,L,Produktionsstatus sperren,Security Risk.png
4=M_MST,L,Status ändern,Status Flag Yellow.png
5=A_AN*,R,Arbeitsgang anmelden,SyBluPly.png

New configuration

[MNR-ALL-Page1]
1=M_INFO,L,,InfoRed.png
; Switch basic display between list display and single machine display:
2=VIEW,L,,VirtualTourSmall.png3=P_SPERRE,L,Lock production status,Security Risk.png
4=M_MST,L,Change status,Status Flag Yellow.png
5=A_AN*,R,Logon operation,Custom Tools.png

12.4 Modifications to ctaiplay.ini

12.4.1 General

The complete standard INI files are located on the server directory \mip\ctnet\win\aip2

Any deviations are developed in specific, customized directories.
e.g. for customized terminal groups \mip\<SystemNo>\custom\aip2\tgrp_xxx

(xxx = number of terminal group)

An empty file (ctaiplay.ini) is generated in these directories. All customized sections e.g. [ANR-ALL-Page1] are copied to this file. Then the configuration is performed in this file.

After restarting the terminal, files from the main directory \mip\ctnet\win\aip2 are merged with files from customized directories (e.g. \mip\<SystemNo>\custom\aip2\tgrp_xxx). The merged file is then transferred to the local terminal directory C:\MPDV\AIP2.

The directory \mip\ctnet\win\aip2 must not be changed, otherwise AIP2 might no longer work properly. In addition, default files are stored there and any changes made will be lost after updating (e.g. service pack)!

12.4.2 Enter user fields in a table

Overview

A ctaiplay.ini file including the modified section is stored in the customized terminal directory (e.g. if the toolbar is changed for terminal groups: \mip\<SystemNo>\custom\aip2\tgrp_xxx\).

ctaiplay.ini is configured in two steps:

- Activate the additional loading of the user fields for operation- or order-related XML files in the section [Custom Userfields ANR]. Activate the user fields for machines in the [Custom Userfields MNR] section.
- Configure the field to be displayed in the grid in the section (e.g. to display the additional field in the order list), e.g. [Order list].

Below both steps are explained in detail.

If the user field is only to be displayed in the GUI with XML configuration, the configuration for display in the grid in the [Order List] or [Machine List] can be omitted.

Available user field in machine and order lists

All identifiers of user fields and also other fields that can be reloaded are located in the headers.dat file in the "spool" directory of the terminal. It consists of four lines:

Start of the row	Content
10|…	Machine list: Fields that are always included in the list.
*10|…	Machine list: Fields that can be reloaded.
11|…	Order list: Fields that are always included.
*11|…	Order list: Fields that can be reloaded.

The following user fields can be reloaded:

- Machine list:
  - FU:1 to FU:66:
    Machine user field
- Order list:
  - ANR_FU_1 to ANR_FU_66:
    Operation user fields
  - AUNR_FU_1 to AUNR_FU_66:
    Order user fields
  - MNR_FU_1 to MNR_FU_66:
    Machine user fields
  - VERARBCODE_FU_1 to VERARBCODE_FU_66:
    Processing code user fields
  - AGR_FU_1 to AGR_FU_66:
    Operation status user fields (reserved for customizations)

Example 1: User fields in the operation list

User field 1 of the operation should be entered in the order list with the name " Order date ".

User field 66 of the machine with the name "My long user field" should be added to the order list.

Step 1 ➔ Field definition of the section [Custom Userfields ANR]

```
[ Custom Userfields ANR ]
GRID_LIST_TYP=ANR
; Additional fields in list of operations
ANR_FU_1= ; User field 1 of operation, MyDate FU:1 [operations list]
MNR_FU_66= ; User field 66 of machine, My long user field [operations list]
```

Step 2 ➔ add the field to the grid [order list]

If the user field is only to be displayed in the GUI with XML configuration, the configuration in the [Order List] can be omitted.

```
[Order list]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite
GRID_LIST_TYP=ANR
EXAMINE_BITMAP1=B1,OPT_INFOAN,T=Attach Notes.png
EXAMINE_BITMAP2=B2,OPT_INFOAI,T=Text Document.png
ATK=C25,100,L,Article
ANR_FU_1=dd.mm.yyyy,90,L,MyDate FU:1
MNR_FU_66=C40,150,L,My long user field of machine
```

Example 2: User field in the machine list

User field 66 of the machine with the name "My long user field" should be added to the machine list.

Step 1 ➔ Field definition of the section [Custom Userfields MNR]

```
[ Custom Userfields MNR ]
GRID_LIST_TYP=MNR
; Additional fields in list of machines
ANR_FU_66= ; User field 66 of machine, My long user field
```

Step 2 ➔ Enter the field for the grid [Machine list]

If the user field is only to be displayed in the GUI with XML configuration, the configuration in the [Machine List] can be omitted.

```
[Machine list]
GRID_FONT=Arial
GRID_FONTSIZE=9
GRID_COLOR=clBlack
GRID_BACKGROUND=clWhite
GRID_LIST_TYP=ANR
EXAMINE_BITMAP1=B1,OPT_INFOAN,T=Attach Notes.png
EXAMINE_BITMAP2=B2,OPT_INFOAI,T=Text Document.png
ATK=C25,100,L,Article
FU:66=C40,150,L,My long user field
```

Explaination of the syntax using the order list

ANR_FU_1 = user field of of the operation

dd.mm.yyyy = formatted as date

90 = number of pixel for the column

L = aligned to the left

MyDate FU:1 = column header

MNR_FU_66 = user field 66 of the machine

C40 = alphanumeric field with 40 digits

150 = number of pixel for the column

L = left aligned

My long user field = column header

12.4.3 Change order of columns in AIP2

A ctaiplay.ini file including the modified section is stored in the customized terminal directory (e.g. if the toolbar is changed for terminal groups: \mip\<SystemNo>\custom\aip2\tgrp_xxx\).

The required sections can just be inserted if a customized ctaiplay.ini file already exists, e.g. due to changes to the order of buttons.

Example:

The "order" column should be displayed in the first place and then the "article" column.

Initial configuration

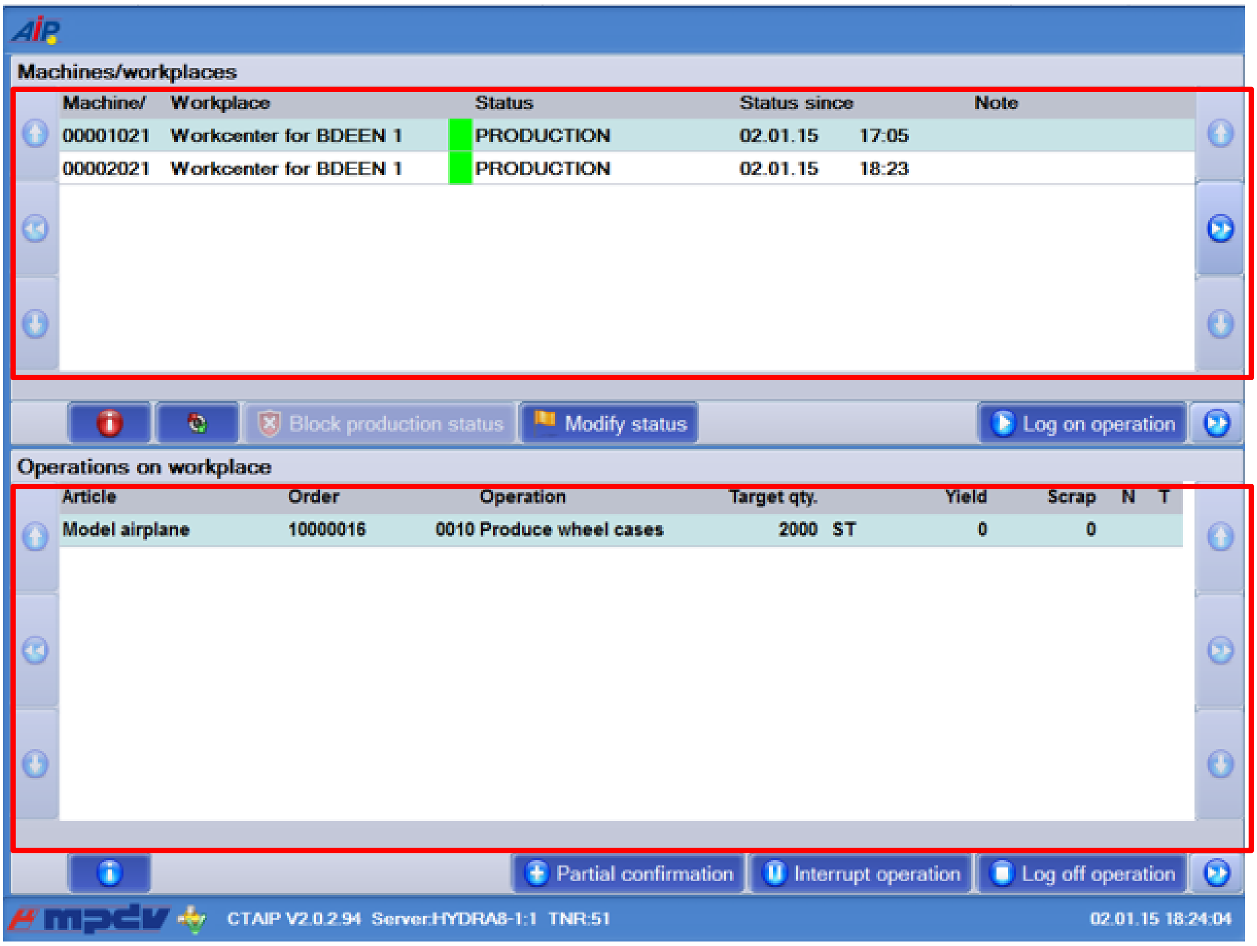
```
[Order list]
…
ATK=C25,100,L,Artikel
AUNR=C10,85,L,Auftrag
ANR_FU_65=C30,150,L,Artikelbezeichnung 2
AGNR=C4,39,R," "
```

New configuration

[Order list]
…
AUNR=C10,85,L,Auftrag
ATK=C25,100,L,Artikel
ANR_FU_65=C30,150,L,Artikelbezeichnung2
AGNR=C4,39,R," „

12.4.4 Changing the height of AIP2 lists

Changing the height of lists (operation and machine list) in the basic screen [MainView1] of AIP2.



The configured heights are scaled to the current height. Consequently, the total sum of entered heights is irrelevant.

The height of lists and elements of the basic screen (machines, order grid, 3rd list, toolbar) can be configured in section [MainView1] of the customized ctaiplay.ini file.

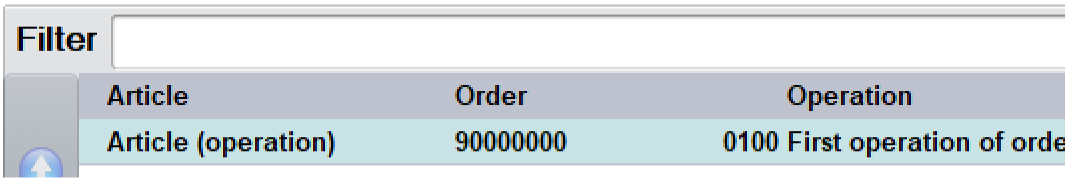
```
[MainView1]
; Values are scaled to match the current resolution
; in order to use the full screen. Percentage values can also be entered
OrderGridHeight=440
MachineGridHeight=380
;List3GridHeight=200
ButtonBarHeight=50
```

Explanation: SYNTAX of order list

OrderGridHeight = height of the order list (indicated in pixels)
MachineGridHeight = height of the machine list (indicated in pixels)
List3GridHeight= height of the third list (tools, staff, batches,…) (indicated in pixels)
ButtonBarHeight= height of the toolbar (indicated in pixels)

12.4.5 Changing the filter function in tables

The filter function is activated for many tables. The filter function You can configure on which column the filtering is to take effect.



Requirements:

- The field for filtering is activated for the table in the dialog configuration (field attribute "AUTOFILTERFIELD").
- There is an entry "AUTOFILTERCOL" in the configuration file ctaiplay.ini which specifies the column to be filtered.

Proceed as follows:

1. Use the shortcut "Ctrl+Alt+F6" in the AIP2 to activate the tooltip.
2. Display the required table on the AIP that already has a filter field.
3. Use the tooltip to identify the affected section in the ctaiplay.ini file by moving the mouse pointer over the desired table. E.g.„… Cfg: **WF@AGNR** …“.
4. Find the section in the standard configuration file <server>\mip\ctnet\win\aip2\ctaiplay.ini and copy the section to your customized global file <server>\mip\<SystemNo>\custom\aip2\ctaiplay.ini or to a terminal group specific file <server>\mip\<SystemNo>\custom\aip2\tgrp_<TerminalGrpup>\ctaiplay.ini

5. Change the value of the existing AUTOFILTERCOL property in the copied customer-specific section. You can find information on availale columns in other attributes of the section or in the file headers.dat. If the property AUTOFILTERCOL is not available, the filtering cannot be changed by configuration. If the filtering is not activated, you first need to check further requirements and activate the filtering via a customizing.

The setting AUTOFILTERCOL=<ALL> ensures that the filter value applies to all columns of the table. Following, a row is displayed if the column contains filter text.

Example<server>\mip\<SystemNo>\custom\aip2\ctaiplay.ini:

```
;******************************************************************************************
;
; ctaiplay.ini (customer's)
;
; ----------------------------------------------------------------------------------------

[WF@ANR]
CMD=DLG=LIST;11|MOD=V|MNR=<MNR>|
MODE=CMD:MODE=#LOCKED#|DATALOCKUNTILSHOW=TRUE|ADDCALCULATEDFIELDS=A|FOCUSLISTITEM=ANR=<ANR>|
FILTER=
SECTION=Sequencing List (Auto)
DATAFIELDS=ANR & AGNR & ATK & ATKBEZ & AGBEZ & *CHPFL=CHPFL & ANR.ATK=ATK & ANR.ATKBEZ=ATKBEZ & ANR.AGBEZ=AGBEZ &
ANR.SGR:GUTP=SGR:GUTP & ANR.EGR:GUTP=EGR:GUTP & ANR.EGR:AUSP=EGR:AUSP & ANR.AUNR=AUNR & RMNR=ANR_RMNR & ANR.FERTIG=FERTIG
FILE=vlist.<MNR>.lst
AUTOFILTERCOL=AGNR

;*******************************************************************************************
```

12.4.6 Cyclic reload of the sequencing list

The sequencing list is cyclically updated on the AIP. The setting of the cycle takes place in ctaipnet.ini

mdereloadvorgabeliste=600

If the sequencing list is to be reloaded each time the operation logon dialog is called, the parameter CMD:MODE=#LOCKED#| must be removed from the section [WF@ANR] in the MODE entry of the ctaiplay.ini. If you want to make a custom implementation, you should copy this section into a new (empty, if not already existing) file ctaiplay.ini and delete the above entry.

12.5 Changes to ctaip.ini

12.5.1 General

The file ctaip.ini is not merged with a file stored in the server. The file must be edited locally in the terminal.

12.5.2 Start Third-Party Application from AIP

Starting a third-party application is configured as follows:

- Configure a button in the ctaipbut.ini file

- Configuration of the function

AIP allows for the integration of buttons starting third-party applications in all toolbars. These buttons

- start third-party applications provided they are not running
- bring third-party applications to the front when they are running

Configuration of buttons

The first button starting a third-party software is configured as "USER1" in ctaipbut.ini.

Further buttons starting third-party software can be configured as "USER2" to "USER9" in the ctaipbut.ini file.

Example:

Configuration of a new button. The button is to be displayed for a specific terminal group in the "machines" section of the AIP2 basic screen. It is configured in the ctaipbut.ini file specific to terminal groups in the server.

\mip\<SystemNo>\custom\aip2\tgrp_xxx\ctaipbut.ini)

```
[MNR-ALL-Page1]
1=M_INFO,L,,InfoRed.png
; Switching the basic screen between list view and presentation of individual machines:
2=VIEW,L,,VirtualTourSmall.png
3=P_SPERRE,L,Produktionsstatus sperren,Security Risk.png
4=M_MST,L,Status ändern,Status Flag Yellow.png
5=A_AN*,R,Arbeitsgang anmelden,SyBluPly.png
6=USER1,R,Notepad
```

Configuration of the function

The function is configured in section [ext. software] of the local terminal configuration file "ctaip.ini":

Example:

```
[ext. software]
Button=Notepad
WindowName=Notepad
ProgFileName=C:\Program Files (x86)\Notepad++\notepad++.exe
SearchParts=On
```

Please note:
"**SearchParts**=“ ➔ If this entry is set, it is sufficient to enter the program name only partly in WindowName.

12.5.3 Remember staff badge number

The person memorized by the terminal only changes if the person logs on with the order. (A_P_AN instead of A_AN)

The memorized person is removed from the memory when the person explicitly logs off from the machine (the same applies for "log off all").

This pre-assignment can be suppressed. Configure "default=0" and set the field attribute SETVALUE in the dialog configuration.

The number is pre-assigned in all dialogs for order postings, status changes and when batches are posted C_UMB, C_GEN, CA_WL.

The memorized persons are deleted in the memory for all machines when shifts change or at the beginning of a new shift (**N**ote: If the shift changes at one machine of the terminal, the persons at the other machines of the terminal are also deleted)!

This can be configured via the entry "HoldPersonInfo=on" in section [SYSTEM] of the ctaip.ini file.

Example:

```
[System]
…..
HoldPersonInfo=on
```

12.6 Hide virtual keyboard

The virtual keyboard can also be switched off if the terminal is connected to a real keyboard. This can be configured in section [SYSTEM] of the <u>local</u> ctaip.ini file. Example:

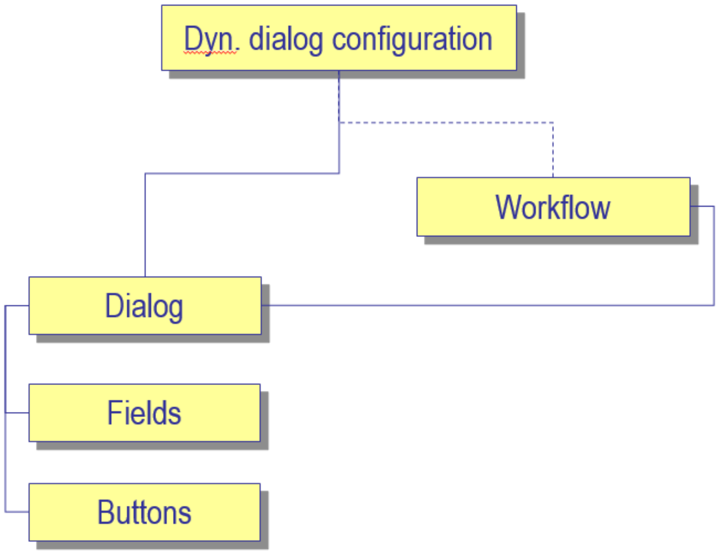
```
[SYSTEM]
Parameters=-t
```

Syntax:

+t/-t --> enables/disables the virtual keyboard; irrespective of the terminal type

12.7 Dynamic dialogs

12.7.1 Overview



12.7.2 AIP2 dialog types

AIP2 provides the following dialog types:

- AIPDEF – Default dialogs (customization)
- AIPTGRP – Dialogs for specific terminal groups (configuration)
- AIPTNR – Dialogs for specific terminals (customization)

You can only create/change dialogs for specific terminal groups.You can change existing dialogs for a specific terminal.
But default dialogs **cannot** be changed.

The terminal has to be rebooted after dialogs were changed.

12.7.3 Dialogs for specific terminal groups

You can make configurations for specified terminal groups.

Before starting the configuration, make a backup copy of the concerned dialogs in a backup group (e.g. AIPTGRP 999). (In case old backups exist, delete them beforehand).

Activate the new dialogs and reboot the terminal.

Example:

Special posting dialogs should be used for terminal group xxx. The workflows and dynamic dialogs are assigned to this terminal group.

How to proceed

1. Assign terminal to terminal group
2. Copy workflows to terminal group xxx
3. Copy dynamic dialogs to terminal group xxx
4. Activate dialogs

Assign terminal (MOC)

Menu: System administration --> Terminals --> Terminal groups

Copy workflows (MOC)

Menu: System administration --> Terminals --> Workflow

Copy complete workflow configuration from AIPDEF 0 to AIPTGRP xxx.

Copy dynamic dialogs (MOC)

Menu: System administration --> Terminals --> Dynamic dialogs

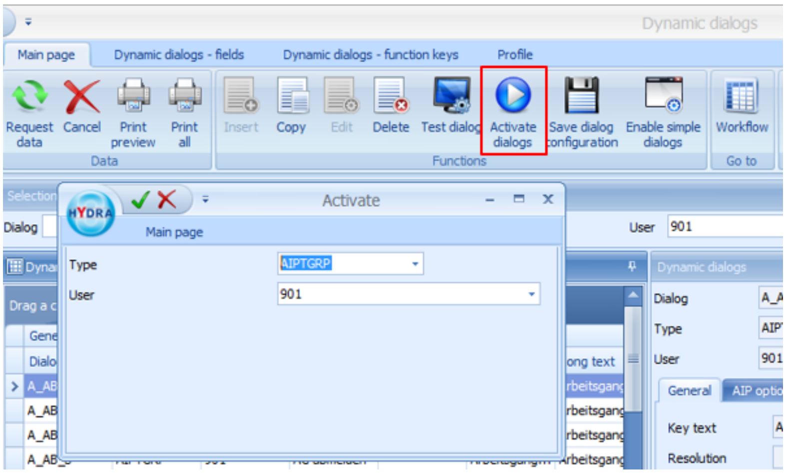
Copy the complete dialog configuration from AIPDEF 0 to AIPTGRP xxx

Activate dialogs (MOC)

Menu: System administration --> Terminals --> Dynamic dialogs

Button "Activate dialogs"

Dialog input: Type =AIPGRP ; User=xxx



Note:

You can delete all dialogs of a terminal group if you select all rows of the terminal group (AIPTGRP).

12.7.4 Hide fields (for specific terminal groups)

Identify the dialogs used on the AIP2 terminal.

Using the shortcut Ctrl + ALT + F6, a tooltip indicating the dialog name is shown.

General procedure:

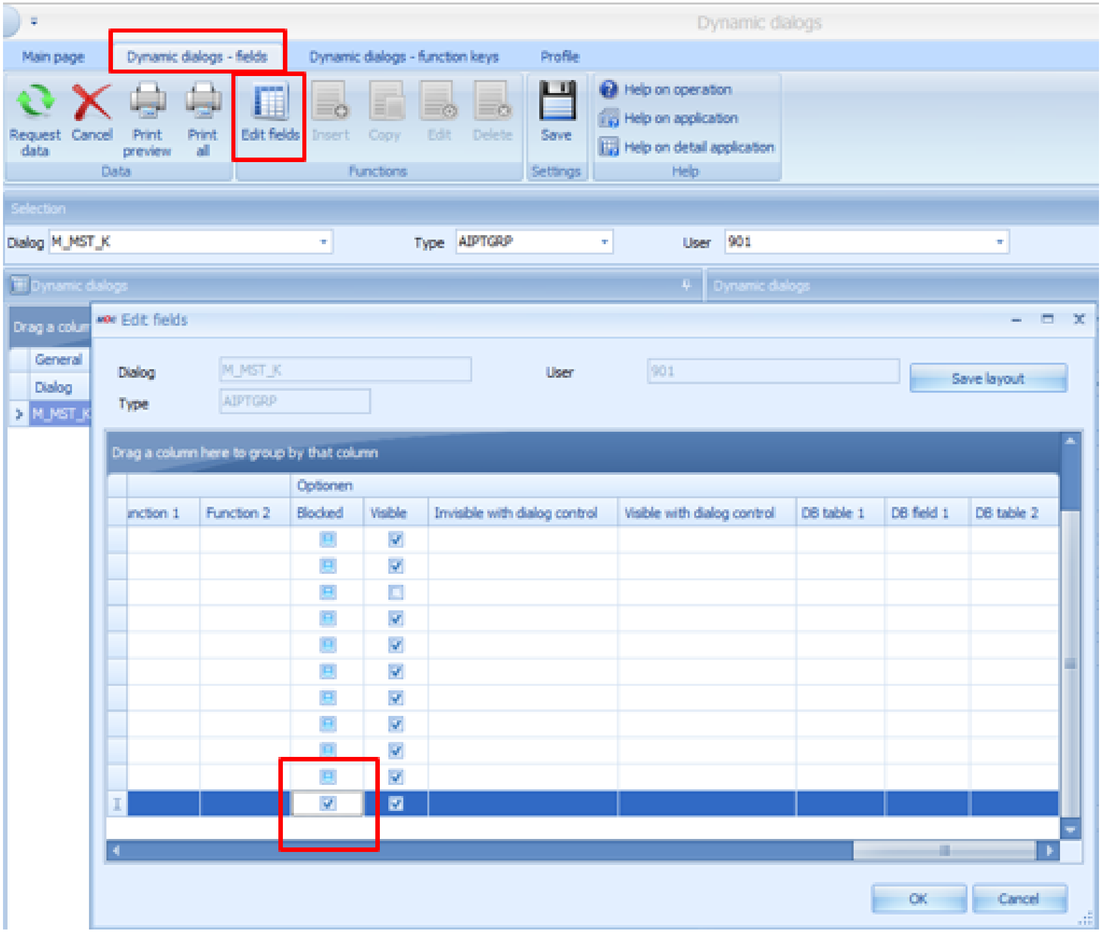
- Identify the dialog where a field should be hidden
- Change and activate dynamic dialogs of terminal group xxx.

Edit dynamic dialogs (MOC)

Menu: System administration --> Terminals --> Dynamic dialogs

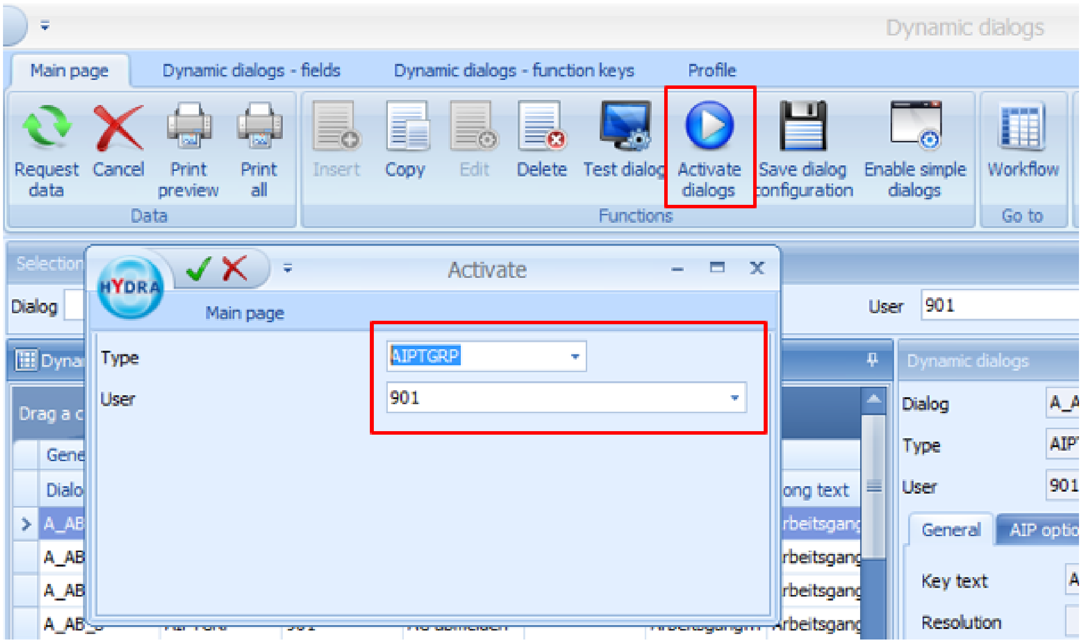
Select the dialog for a specific terminal group and start the edit mode via the menu tab "dynamic dialogs - fields" and the button "edit fields"

Choose the required field and check the option "blocked"



Activate dialog (MOC)

Activate dialogs for specific terminal groups



12.7.5 Default assignment in dialog fields (for specific terminal groups)

General procedure:

- Identify the dialog where a field should be completed with default values
- Change and activate dynamic dialogs of terminal group xxx.

Edit dynamic dialogs (MOC)

Menu: System administration --> Terminals --> Dynamic dialogs

Select the dialog for a specific terminal group and start the edit mode via the menu tab "dynamic dialogs - fields" and the button "edit fields"

Set field "field attribute 2" to "SETVALUE"

Add field "default"

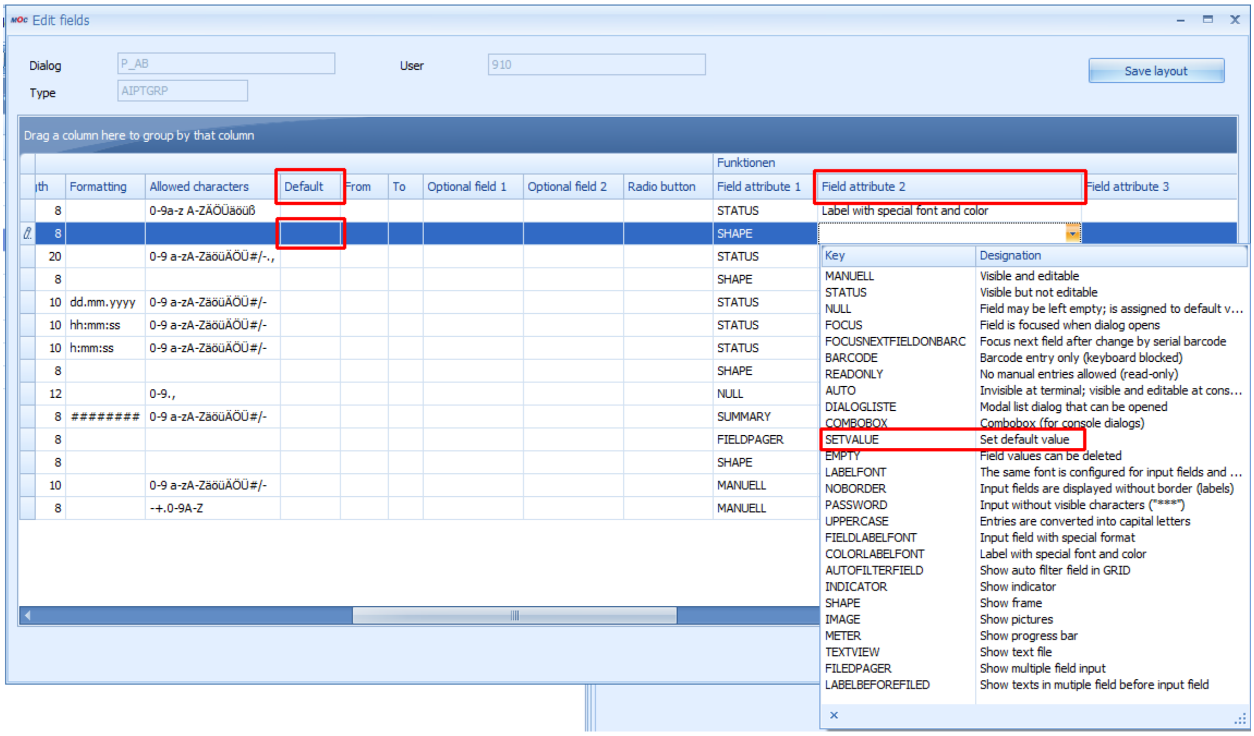
Allowed characters for dynamic dialog fields

The minus character "-" must always be placed at the end to prevent it from being mistaken for the character used for the definition of "from" - "to" ranges.

Example:

a-z A-Z/-,. is interpreted as range from a to z and A to Z but in this case also as range from "/" to ","

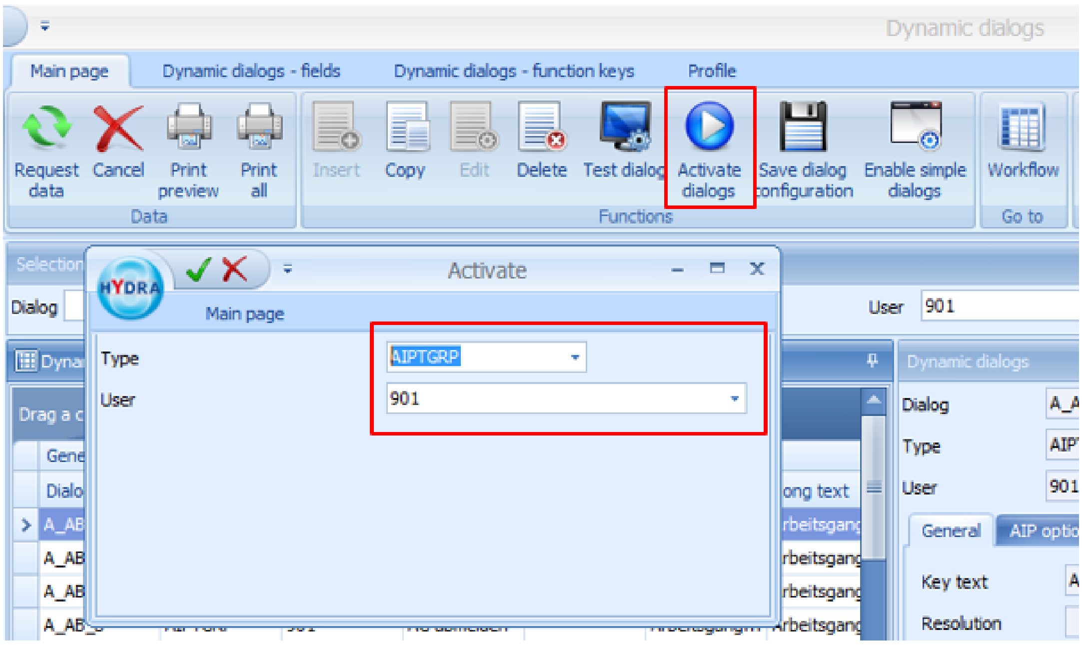
a-z A-Z/,.- is interpreted as range from a to z and A to Z and as the allowed characters / , . and -



Activate dialog (MOC)

Activate dialogs for specific terminal groups

Change field name (for a specific terminal group)



General procedure:

- Identify the dialog where a field name should be changed

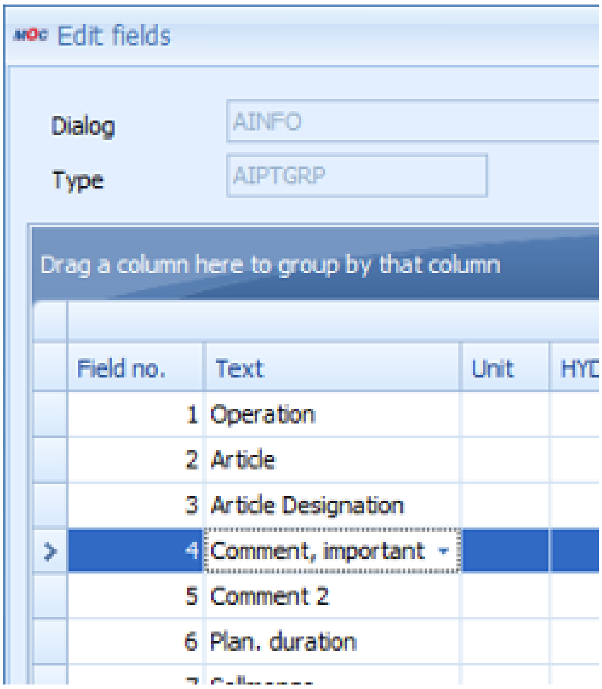
- Change and activate dynamic dialogs of terminal group xxx.

Edit dynamic dialogs (MOC)

Menu: System administration --> Terminals --> Dynamic dialogs

Select the dialog for a specific terminal group and start the edit mode via the menu tab "dynamic dialogs - fields" and the button "edit fields"

Change field contents of the column "text".



Activate dialog (MOC)

Activate dialogs for specific terminal groups

12.7.6 Activate simplified dialogs

There are simplified dialogs for: logging on operations, reporting partial quantities for operations, interrupting operations, logging off operations.

Use the button Enable simple dialogs to store the simplified dialogs for the standard AIPDEF 0 in the workflow.

How to proceed:

- Menu: System administration --> Terminals --> Dynamic dialogs --> Button "Enable simple dialogs"
- Enable dialogs for AIPDEF 0

Only one dialog is entered in the workflow if simple dialogs are in use.

Once simplified dialogs have been activated, it **cannot** be undone by way of configuration. This can only be changed by customizing the system, which has to be ordered from MPDV.

Activation via the standard dialogs AIPDEF 0.

12.8 Customizing files

12.8.1 Terminal script files

File names and directories of default/customized terminal scripts must be named as follows.

	AIP 2	Description
MPDV	**.\ctnet\win\aip2\etc\aip_mpdv.zip**	MPDV standard (not used)
MPDV	**.\ctnet\win\aip2\etc\mpdv-aip.zip**	MPDV standard
CUST	**.\custom\userexit\aip2_<customer number>.zip**	Customization with customer number
CUST	**.\custom\userexit\aip2_<project>.zip**	Customization with project abbreviation

“aip_” is added as prefix to terminal script files for AIP2.

	PRIO	AIP 2	Description
MPDV	**1**	**.\aip_system_mpdv.scr** **.\aip_<dialog>_mpdv.scr**	MPDV standard (not used)
MPDV	**2**	**.\aip_mpdv-system.scr** **.\aip_mpdv-<dialog>.scr**	MPDV standard
CUST	**1**	**.\aip_system_<customer no.>.scr** **.\aip_<dialog>_<customer no.>.scr**	Customization with customer number
CUST	**2**	**.\aip_system_<project>.scr** **.\aip_<dialog>_<project>.scr**	Customization with project abbreviation

ZIP files are only unpacked in live operation, once they have been successfully DOWNLOADED from the server. In DEMO mode there is no unpacking of terminal script ZIP files.

13 Rework and Open Quantity on the AIP2

13.1 Purpose

Use the following instruction to collect not only yield and scrap, but also rework quantities and/or open quantities on the AIP2.

This instruction describes how to collect additional quantities with the following order-related dialogs:

- Upload/post partial quantity for operation
- Interrupt operation
- Log off operation

13.2 Requirements

For the present instruction, the requirements listed below were respected. But also with different technical conditions, you can still collect rework and open quantities (e.g. with AIP 8.1 and MW 3.1). In some of the workflow steps, you might have to make changes, e.g. change the place of fields.

- The screen of the AIP2 has a width of 1280 pixels. If the screen is smaller, parts of the input fields can be cut off.
- MW 4.0pe with service pack 15. Only from service pack 15 onwards, you can create new fields in dynamic dialogs without development license MDS-AIS.
- Unchanged dialog configuration as in the new installation of MW 4.0pe at the beginning of 2020.

The instruction is for persons with technical know-how. They must know how to configure dynamic dialogs and how to activate and test these dialogs on the AIP (basic knowledge).

For detailed information on the configuration of dynamic dialogs and the GUI configuration of the AIP2, refer to the document EAT-AIP_82.pdf.

The instruction describes the collection of a rework quantity. Use the same configuration steps for open quantities. To configure open quantities
- use the identifier "PRB" instead of "NCH"
- define reasons for open quantities instead of reasons for rework and
- change the label texts accordingly.

13.3 How to proceed

13.3.1 Defining reasons for rework

Use the application *Reasons* on the MOC to define one or severel reasons for rework (type=rework).

Tip: As a test, only create two or three reasons to not overload the display in the dialog.

13.3.2 Dynamic dialogs: Copy

Copy the dynamic dialogs from AIPTNR with user 0 to a specific terminal or terminal group. In the present instruction, the dialogs were copied to terminal 110.

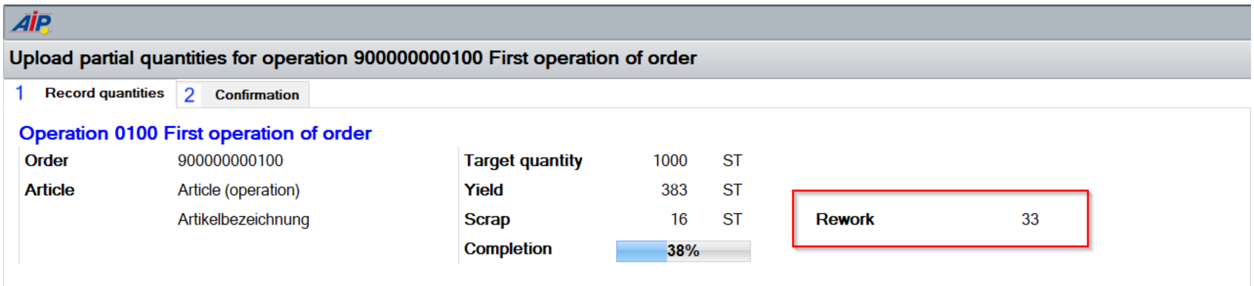
Perform the steps in the following in the new copy of the dialogs for the terminal or terminal group selected. This way, the currently productive dialogs are not changed and you can return to the standard version, if required.

When the tests are completed, you can copy the changes to the productive dialog configurations.

13.3.3 Dynamic dialogs: WF_AA_QUA

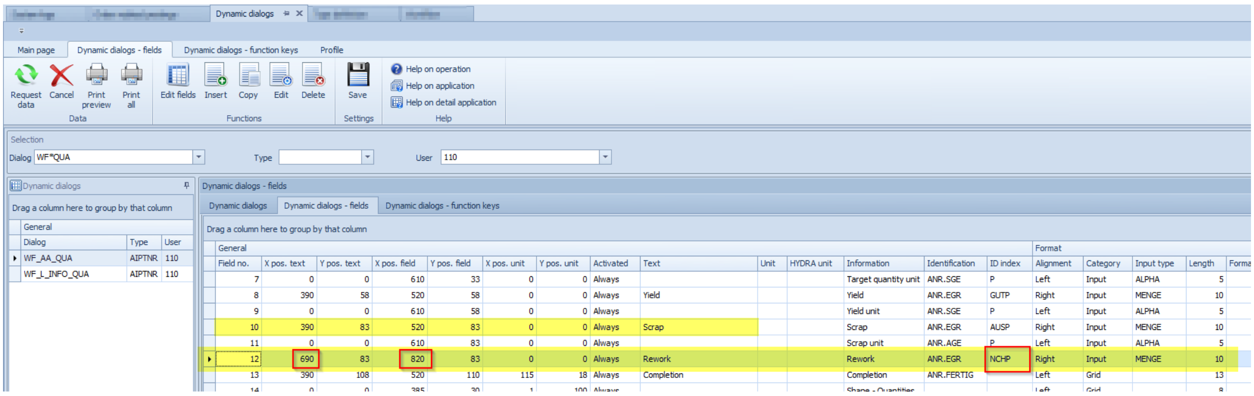
The dialog WF_AA_QUA is used in all order-related workflows to collect quantities. If you integrate new quantity fields in the dialog WF_AA_QUA, the new quantity fields are then automatically available in the workflows A_TR, A_UN and A_AB.

13.3.3.1 Adding status field in the header of the input dialog



- Copy the field for scrap (field number + 2, the numbers of the following fields are automatically incremented).
- Edit the copied field.
    - Change the text to "Rework".
    - Change the field *ID index* to NCHP.

- Increment the X-positions for text and field by 300. This way, the *Rework* field is placed to the right of the *Scrap* field.



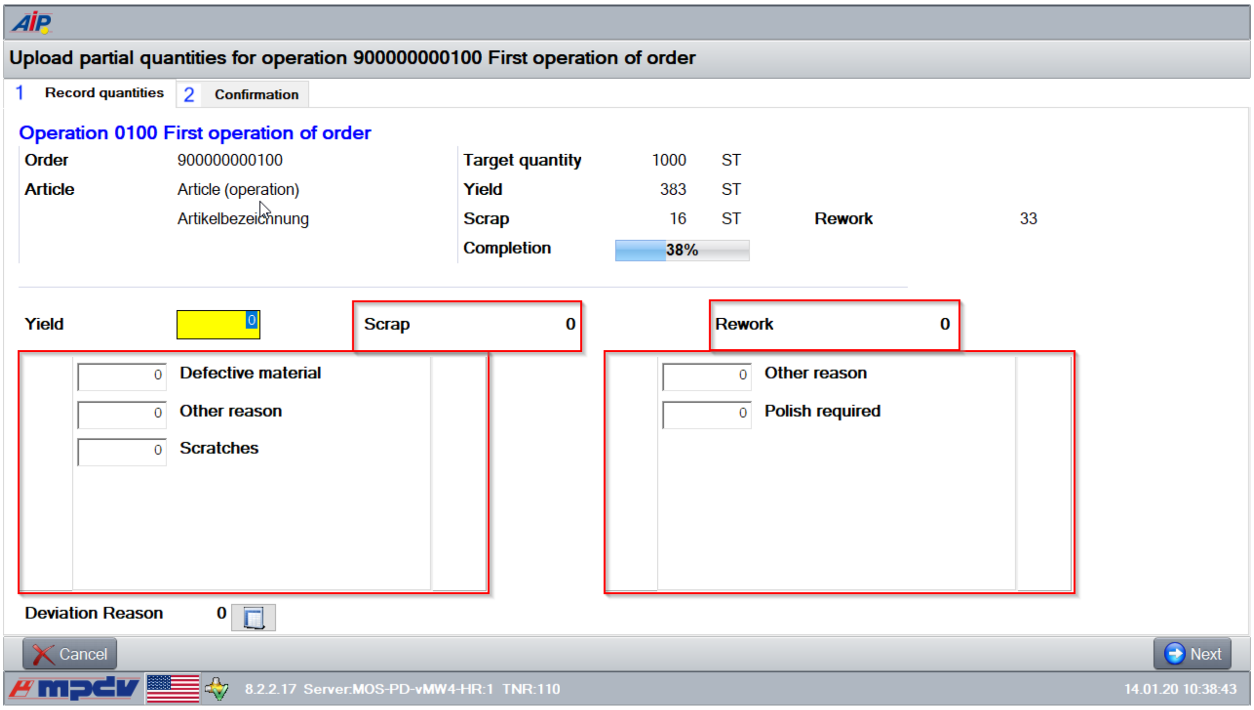
13.3.3.2 Adding input fields

You can optionally collect quantities for several scrap reasons on the AIP. The system then automatically creates a posting record as posting of a part quantity for each scrap reason collected.

You require a so-called FIELDPAGER to display several reasons on the AIP. The fieldpager is configured like a single input field in the configuration of the dynamic dialogs. But here, the dialog layout on the AIP is different. A larger area is reserved and not only a single field. In this area, a separate input field is provided for each scrap reason that is valid at the machine.

In the example below, we also use a FIELDPAGER for the rework quantities.

For details on the configuration of the FIELDPAGER, refer to the document MOC_DialogField.



- You must ensure that there are not too many valid scrap reasons for the terminal used, because the input area for scrap is otherwise overloaded.
- First, decrease the width of the FIELDPAGER for scrap to create space for the new FIELDPAGER for rework quantity. In the example, we have decreased the width of the area using the field for the X-position of the unit (field *X pos. unit*). The width was decreased from 770 to 400.

- Copy the totals field of the scrap to the field number +2.
- Change the copied totals field for scrap.
  - Change the texts from scrap to rework.
  - Change the identification (replace AUS with NCH).
  - Increment the X-positions of the new field by 300. This way, the field is placed to the right of the totals field for scrap.

- Copy the FIELDPAGER for scrap to the field number +2.
- Change the copied FIELDPAGER for scrap:
  - Change the texts from scrap to rework.
  - Change the identification (replace AUS with NCH).
  - On the "Functions" tab, change the "Dialog list function" to "PAGER-AGRD-NCH.LST".
  - Increment the X-positions of the field by 500. Do not change the X-position of the unit because the X-position of the unit specifies the width of the area in case of a FIELDPAGER. The FIELDPAGER for the rework quantity is now displayed to the right of the FIELDPAGER for scrap.

General											
Field no.	X pos. text	Y pos. text	X pos. field	Y pos. field	X pos. unit	Y pos. unit	Activated	Text	Information	Identification	ID index
14	0	0	385	30	1	100	Always		Shape - Quantities		
15	0	0	10	150	760	1	Always		Shape - Horizontal 1		
16	15	173	145	170	0	0	Always	Yield	Yield	EGR	GUT
17	305	173	425	173	0	0	Always	Total quantity	Total quantity	A	
18	305	173	425	173	0	0	Always	Scrap	Scrap sum	$CT.AUS:SUM	
19	0	0	10	210	400	200	Always		Scrap quantities	$CT.AUS:	
20	605	173	745	173	0	0	Always	Rework	Sum Rework	$CT.NCH:SUM	
21	0	0	510	210	400	200	Always		Rework quantities	$CT.NCH:	
22	15	420	145	420	0	0	Always	Deviation Reason	Deviation Reason	EGG	GUT
23	10	15	160	12	220	15	Always	Workplace	Workplace	MNR	

	Format						
Field no.	Alignment	Category	Input type	Length	Formatting	Field attribute 1	Allowed
14	Left	Grid		8		SHAPE	
15	Left	Grid		8		SHAPE	
16	Right	Input	MENGE	12		NULL	0-9.
17	Left	Input	MENGE	12		STATUS	0-9.
18	Left	Input	FLIESS	8	########	SUMMARY	0-9
19	Left	Grid		8		FIELDPAGER	
20	Left	Input	FLIESS	8	########	SUMMARY	0-9
21	Left	Grid		8		FIELDPAGER	
22	Right	Input	NUMERISCH	4	########	MANUELL	0-9
23	Left	Input	ALPHA	8		MANUELL	-+.

Export the dialogs for the selected terminal number.

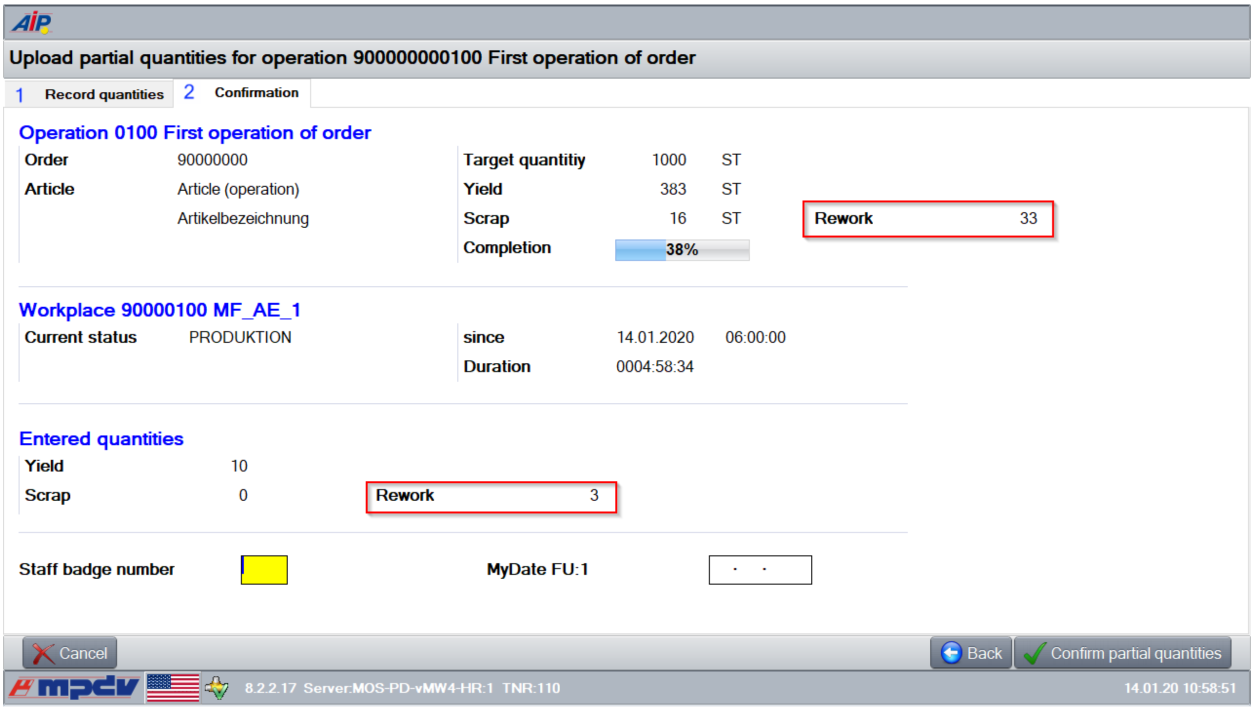
Test the changes on the AIP. To this end, restart the AIP or reload the dialogs.

13.3.4 Dynamic dialogs: WF_*_CHK

In the workflows A_TR, A_UN and A_AB, different dialogs are used in the standard for the final check by the user. In these dialogs, the collected rework quantities must be entered:

- **WF_ATR_CHK**: Workflow step "confirmation" with upload of a part quantity
- **WF_AUN_CHK**: Workflow step "confirmation" with interrupt OP
- **WF_AUN_CHK**: Workflow step "confirmation" with OP logoff

Find below a description of how to proceed with the dialog WF_ATR_CHK as an example. Proceed similarly with the other dialogs.



13.3.4.1 Adding status field in the header of the dialog

- Copy the status field for the former scrap (approx. field number 10 to field number +2)
- Edit the new field.
  - Change the texts to "Rework".
  - Change the field *ID index* (replace AUS with NCH).
  - Increment the X-positions for text and field by 300.

13.3.4.2 Adding an input field

- Copy the field for the collected scrap (approx. field number 27 to field number + 1)
- Edit the new field.
  - Change the texts to "Rework".

- Change the field *Identification* (replace AUS with NCH).
- Increment the X-positions for text and field by 300.

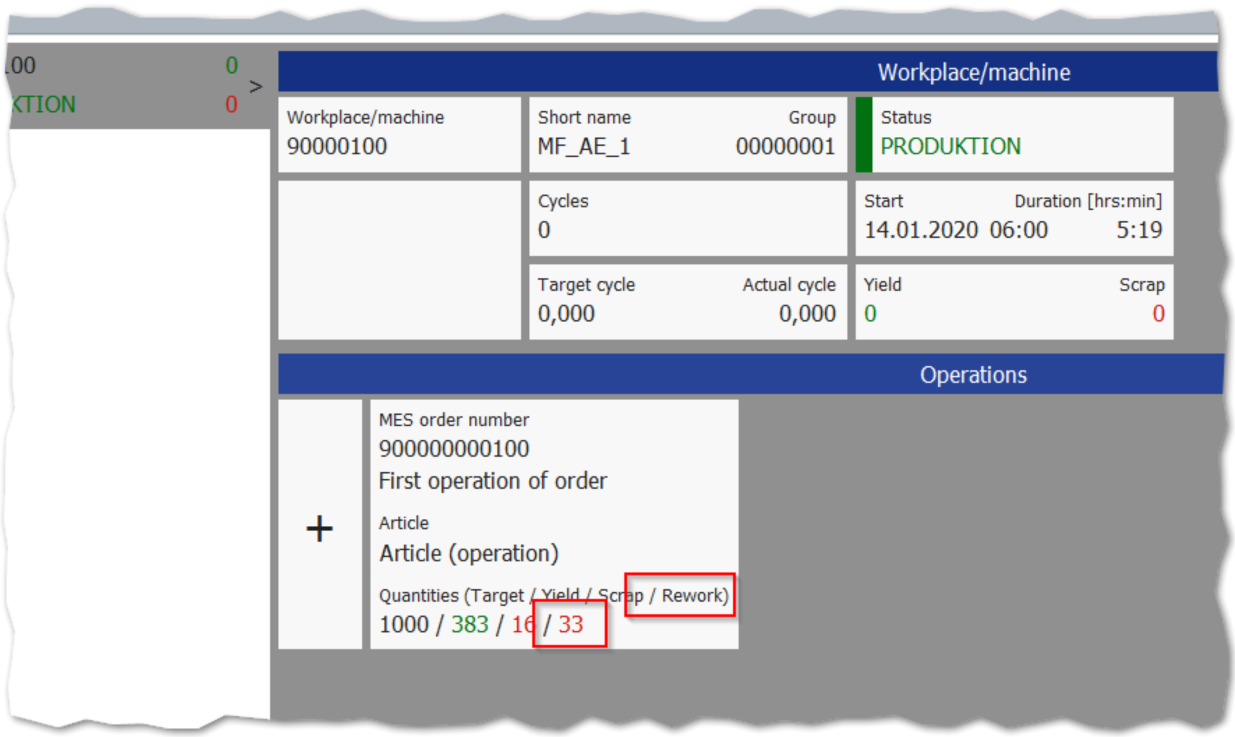
Export the dialogs for the selected terminal number.

Test the changes on the AIP. To this end, restart the AIP or reload the dialogs.

Make similar changes for all three dialogs WF_ATR_CHK, WF_AUN_CHK and WF_AUN_CHK.

13.3.5 AIP layout: Rework quantity in main view

You want to additionally show the collected rework quantity in the main view of the AIP.



Make the configuration in the XML file for the GUI of the AIP2.

- Create a copy of the file a_list_anr.xml in the local scope. Copy the following file on the server

  $INSTALLDIR$\ctnet\win\aip2\gui\a_list_anr.xml

  to

  $INSTALLDIR$\1\custom\aip2\gui\a_list_anr@local.xml
- Change the caption for the quantities in the file in the local scope (a_list_anr@local.xml). In the example, the translation of the caption is deliberately neglected to keep the example simple.

```
...
</element>
<!--Label Quantities-->
```

```
<element class="TGridItemLabel">
<DataFieldName></DataFieldName>
<control>
  <Top>110</Top>
  <Left>5</Left>
  <!-- Caption Function="Translate" LanguageKey="lkQuantitiesTargetYieldScrap">Mengen (Soll / Gut / Ausschuss)</Caption -->
  <Caption>Quantities (Target / Yield / Scrap / Rework)</Caption>
</control>
</element>
...
```

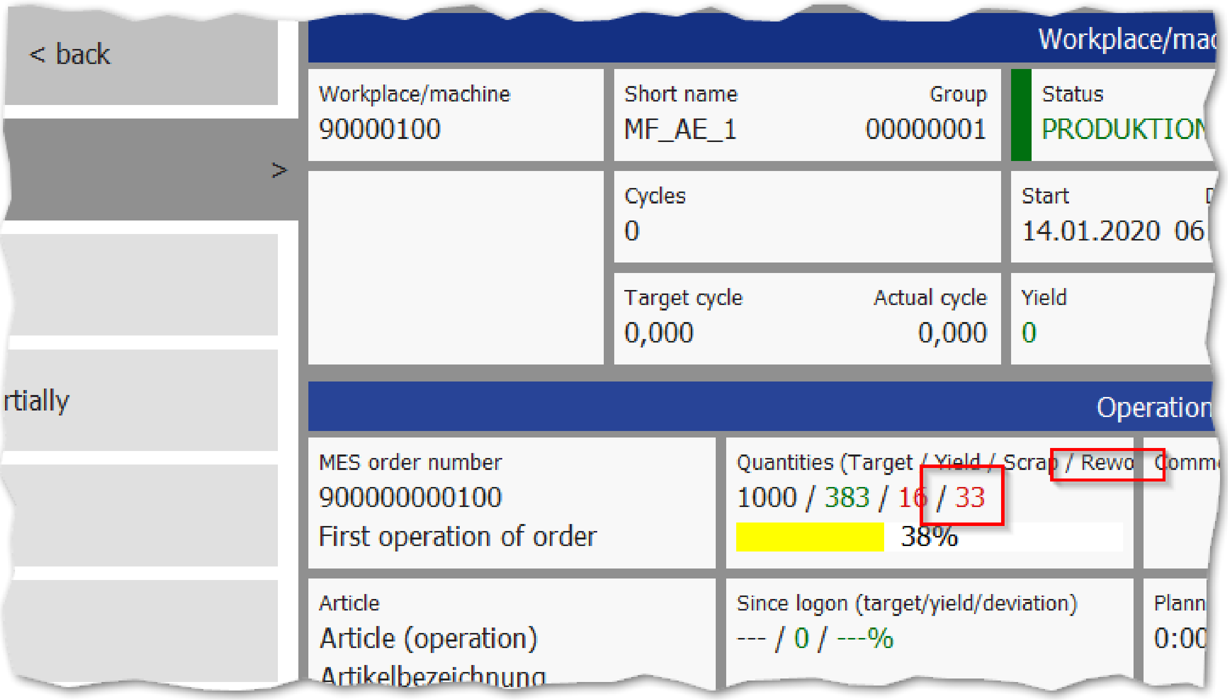
- Copy the sections for the scrap quantity and the slant line in front and replace AUS with NCH in the new sections.

```
...
<!--EGR:AUSP-->
<element class="TGridItemLabel">
  <DataFieldName>EGR:AUSP</DataFieldName>
  <DisplayFormat Define="FORMAT_QUANTITY">%g</DisplayFormat>
  <control>
  <Align>alLeft</Align>
  <AlignWithMargins>False</AlignWithMargins>
  <Font>
    <Color Define="COLOR_SCRAP">$0000DB</Color>
  </Font>
  </control>
</element>
<!--/-->
<element class="TGridItemLabel">
  <control>
  <Align>alLeft</Align>
  <AlignWithMargins>False</AlignWithMargins>
  <Caption> / </Caption>
  </control>
</element>
<!--EGR:NCHP-->
<element class="TGridItemLabel">
  <DataFieldName>EGR:NCHP</DataFieldName>
  <DisplayFormat Define="FORMAT_QUANTITY">%g</DisplayFormat>
  <control>
  <Align>alLeft</Align>
  <AlignWithMargins>False</AlignWithMargins>
  <Font>
    <Color Define="COLOR_SCRAP">$0000DB</Color>
  </Font>
  </control>
</element>
</element>    <!--Panel for icons longtext and notes-->
...
```

Test the changes on the AIP. To this end, restart the AIP so that the AIP loads the new file from the local scope.

13.3.6 AIP layout: Rework quantity in *Operation* layout

You want to additionally show the collected rework quantity in the *Operation* layout of the AIP.



Make the configuration in the XML file for the GUI of the AIP2.

- Create a copy of the file a_list_anr.xml in the local scope. Copy the following file on the server

  $INSTALLDIR$\ctnet\win\aip2\gui\a_data_anr.xml

  to

  $INSTALLDIR$\1\custom\aip2\gui\a_data_anr@local.xml

Perform the same steps in the new XML file that you performed for the main view.

Test the changes on the AIP. To this end, restart the AIP so that the AIP loads the new file from the local scope.

13.4 Result

When all configurations are made, the following functions are available:

- In the main view and in the *Operation* layout, the AIP displays the rework quantities uploaded so far for the operation.
- If you post a part quantity, interrupt or log off an operation, the AIP displays the quantities uploaded so far.
- The users can collect rework quantities with different reasons.
- The users can check the collected rework quantities on the AIP before the final confirmation of a dialog.
- The system books the collected rework quantities for the operation.

- The system automatically creates several order-related postings as postings of part quantities for the rework quantities entered with different rework reasons.
- The system automatically books the total of the rework quantity in the order-related postings for interrupt or logoff.

14 Collecting Order-Related User Fields on the AIP2

14.1 Purpose

Use the following instruction if you want to collect user fields for order-related postings and optionally for operations on the AIP2.

The instruction is helpful for the following dialogs:

- Upload/post partial quantity for operation
- Interrupt operation
- Log operation off

The processing has not been tested explicitly for other dialogs such as *Log staff off*, but the collection of user fields probably works in other dialogs as well.

The following order-related user fields are available in the system. The data type and the possible field length is different for each user field.

Identification	Data type	Description
FU:1 to FU:6	Date	User fields for a date
FU:7 to FU:22	N	Integer with value range from -2147483647 to 2147483647
FU:23 to FU:28	DECIMAL(18,6)	Decimal number with a value range of 12 digits before decimal separator and a precision of 6 digits after decimal separator
FU:29 to FU:44	C1	User field for 1 character
FU:45 to FU:50	C10	User field for a text with a maximum of 10 characters
FU:51 to FU:64	C20	User field for a text with a maximum of 20 characters
FU:65 to FU:66	C40	User field for a text with a maximum of 40 characters

The instruction is made for the user fields in the *Operation* and *Order-related postings*. It also applies for user fields of batches. You must then use the field IDs CNR.FU:1 to CNR.FU:66.

14.2 Requirements

- The screen of the AIP2 must have a width of 1280 pixels. If the screen is smaller, parts of the input fields can be cut off.
- MW 4.0 pe with service pack 15. Only from service pack 15 onwards, you can create new fields in dynamic dialogs without development license MDS-AIS.

- The dialog configuration is unchanged as in a new installation of MW 4.0pe at the beginning of 2020.

The instruction is for persons with technical know-how. They must know how to configure dynamic dialogs and how to activate and test these dialogs on the AIP (basic knowledge).

The screenshots are only available in English.

For detailed information on the configuration of dynamic dialogs and the GUI configuration of the AIP2, refer to the document EAT-AIP_82.pdf.

14.3 How to proceed

14.3.1 Overview

The instruction below describes how you can collect a user field with a date when you post a part quantity for an operation, for example.

It is also described how the user field must be configured on the MOC so that the collected data is also displayed on the MOC.

14.3.2 Dynamic dialogs: Copy

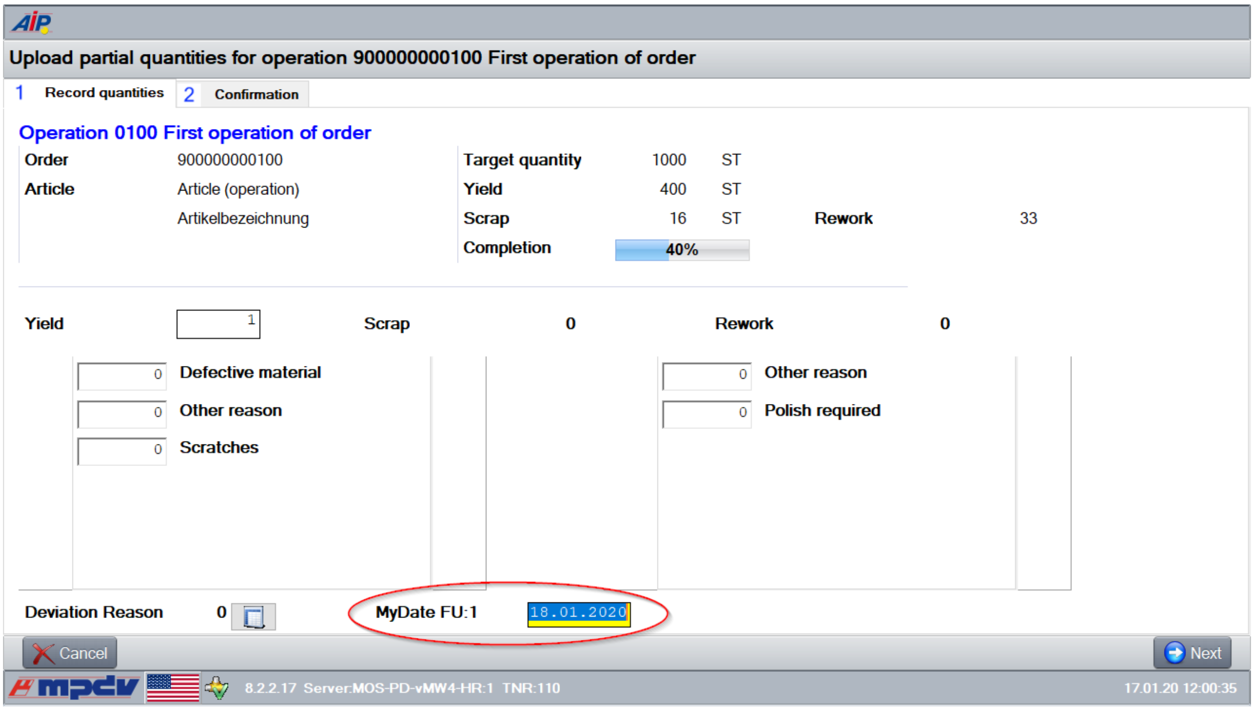
Copy the dynamic dialogs from AIPTNR with user 0 to a specific terminal or terminal group. In the present instruction, the dialogs were copied to terminal 110.

Perform the steps in the following in the new copy of the dialogs for the terminal or terminal group selected. This way, the currently productive dialogs are not changed and you can return to the standard version, if required.

When the tests are completed, you can copy the changes to the productive dialog configurations.

14.3.3 Dynamic dialogs: WF_AA_QUA

The dialog WF_AA_QUA is used in the order-related workflows A_TR, A_UN and A_AB to collect quantities. We now want to collect an additional user field of type date.



- Note down the field values below from field *Deviation reason* with the ID EGG:GUT:

Field	For your notes (enter value)
Field no.	
Y pos. text	
Y pos. Field	

- Create a new field with the following data (do not change the default value for settings that are not listed in the table):

Tab	Field	Value (comment)
General	Field no.	➔ Value of EGG:GUT incremented by 1
	Text	MyDate FU:1
	Unit	
	Information	MyDate FU:1
	Identification	FU
	ID index	1
Position	X pos. text	315
	Y pos. text	➔ Value of EGG:GUT

	X pos. field	445
	Y pos. field	➔ Value of EGG:GUT
Format	Alignment	Left
	Category	Input
	Input type	DATUM ("date")
	Length	10
	Allowed characters	0-9
	Default	+1 (this results in the pre-allocation with the date of tomorrow. You can use positive and negative offsets. Note: To pre-allocate the field with the default value, you must additionally set the field attribute SETVALUE; see below. If you do not specify a default value, the field remains empty.)
Functions	Field attribute 1	MANUELL
	Field attribute 2	NULL (field may remain empty. If you do not configure a field attribute with NULL, the user field is a mandatory field.)
	Field attribute 3	SETVALUE (use SETVALUE to take over the default value to the field. If you do not configure a field attribute with SETVALUE, the user field remains empty.)
Options	Visible	Activated

The following input types are available for the different user fields in the field configuration:

Identification	**Data type**	**Input type**
FU:1 to FU:6	Date	DATUM ("date")
FU:7 to FU:22	N	NUMERISCH ("numeric") Using the numeric user fields, you can also collect times. Use the input type ZEIT ("time") or DAUER ("duration") then.
FU:23 to FU:28	DECIMAL(18,6)	FLIESS ("flow")
FU:29 to FU:66	C1, C10, C20, C40	ALPHA

For details on the field configuration and the formatting of input and output, refer to the document MOC_DialogField.

Export the dialogs for the selected terminal number.

Test the changes on the AIP. To this end, restart the AIP to reload the dialogs.

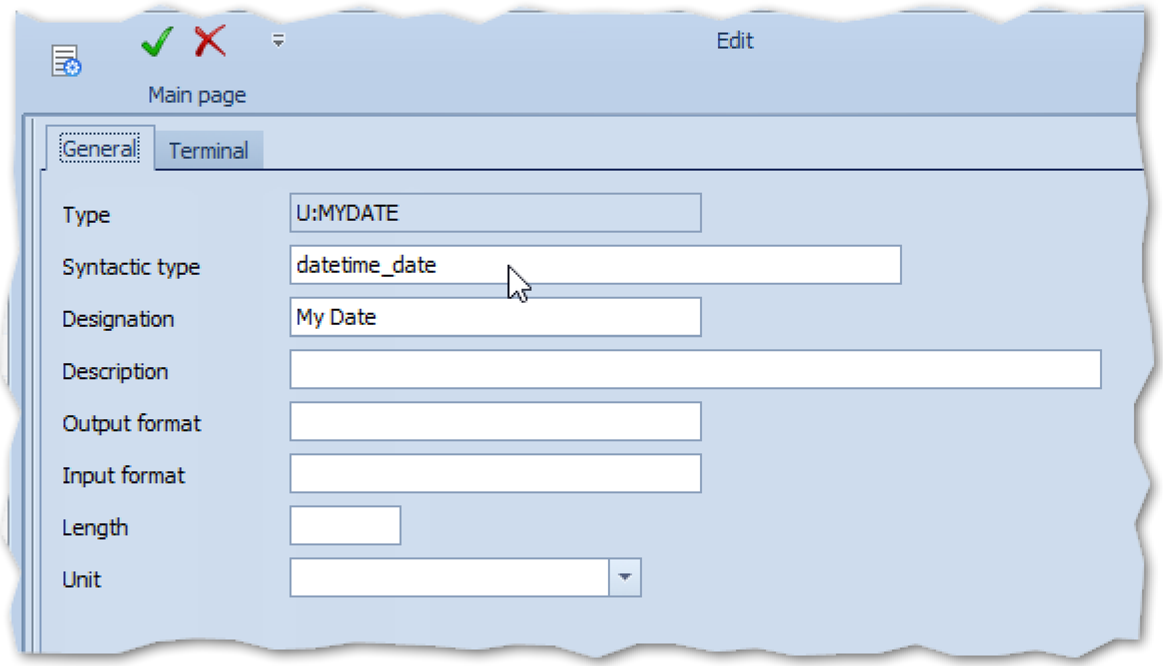
14.3.4 User field configuration for the MOC

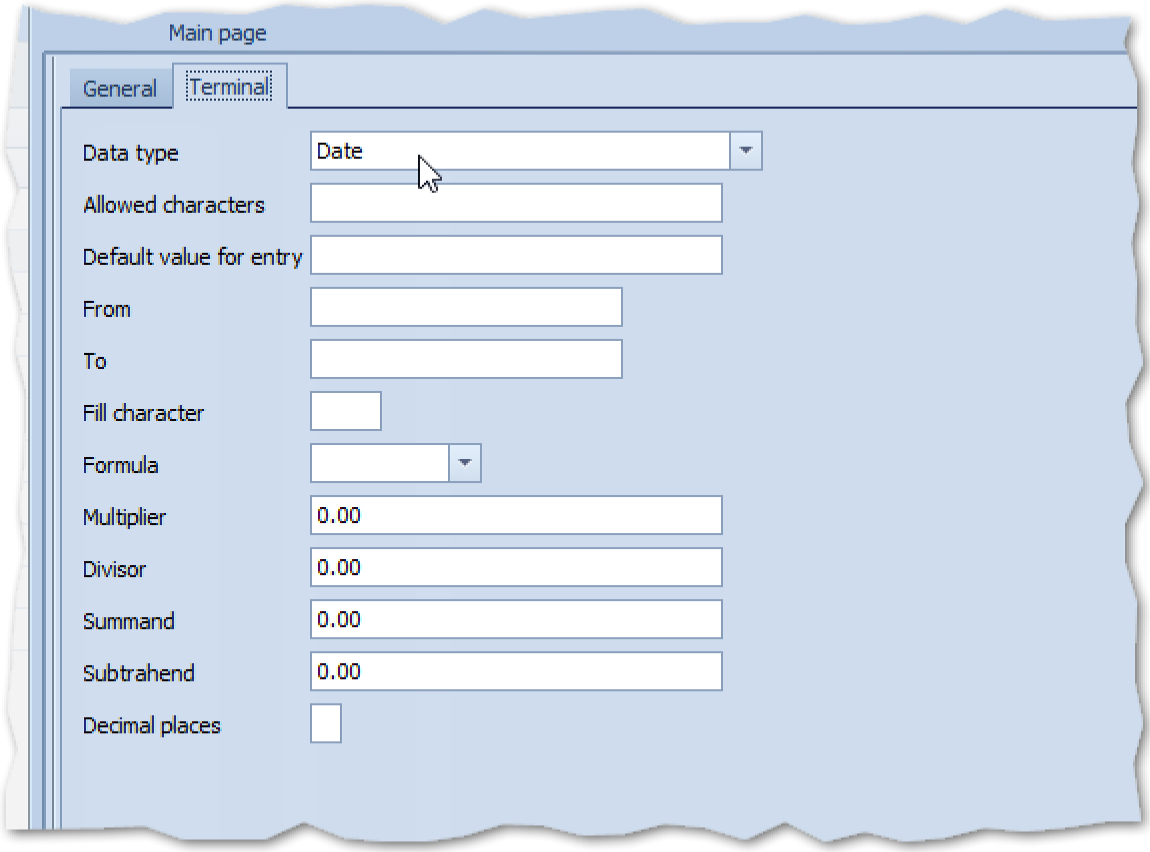
Create a user field configuration on the MOC. For further information on the configuration of user field keys and user fields, refer to the documentation Configuration_Userfields.

14.3.4.1 Type definition

Check if there is a suitable type definition for your user field.

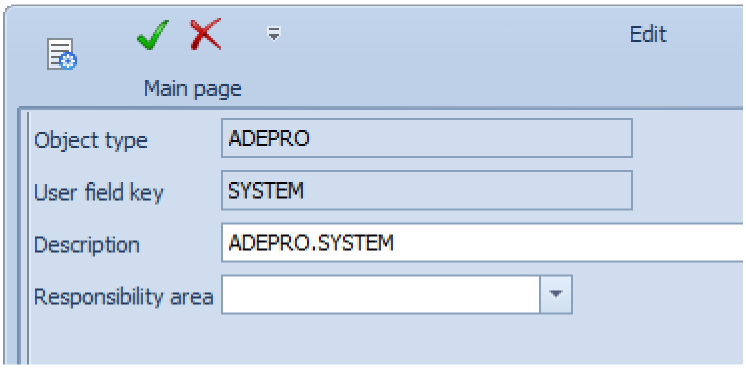
In the example, we create an own type definition "U:MYDATE". For further information, refer to the online help in the application *Type definition* or to the document MOC_UserFieldDefinition, in particular the section "Sample definition for MOC user fields (input and output)".





14.3.4.2 User field key

Check if the two user field keys ADEPRO/SYSTEM and AGNR/SYSTEM are already available. If not, create the user field keys. Enter any description.

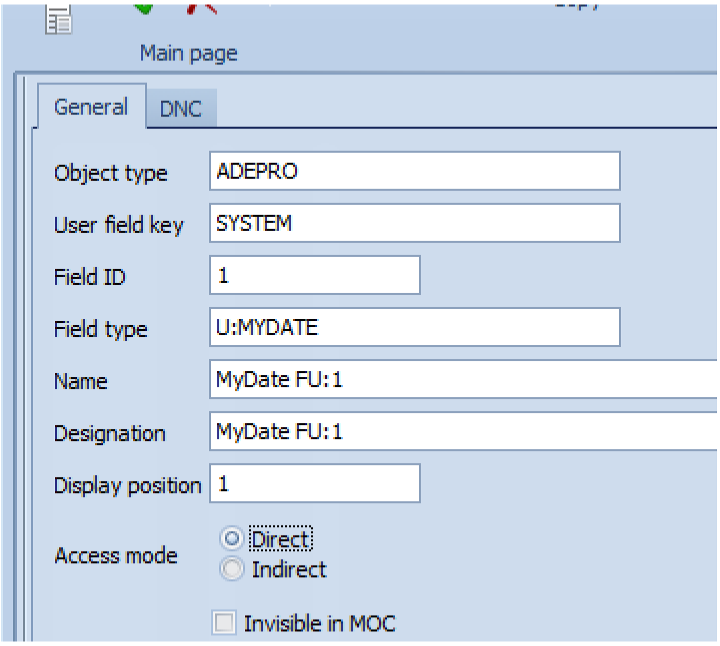


14.3.4.3 User field for *Order-related postings* and *Operation*

Create a user field for *Order-related postings* with the following data:

Field	Value
Object type	ADEPRO

User field key	SYSTEM
Field ID	1 (number of collected user field)
Field type	U:MYDATE
Name	Any
Designation	Any
Display position	1 (if you have several user fields per object type and user field key, the display position specifies the order of display on the MOC)



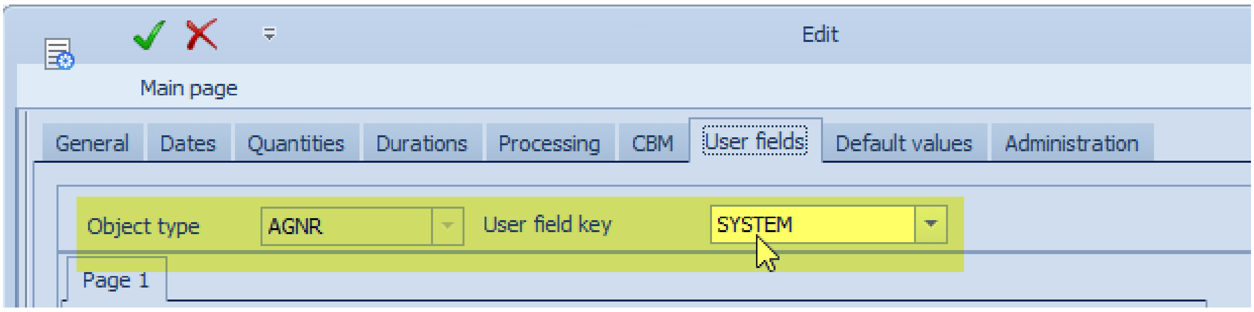
Create another user field for *Operations* with the following data:

Field	Value
Object type	AGNR
User field key	SYSTEM
Field ID	1 (number of collected user field)
Field type	U:MYDATE
Name	Any
Designation	Any
Display position	1 (if you have several user fields per object type and user field key, the display position specifies the order of display on the MOC)

14.3.4.4 Assigning user field key in *Operation*

Restart MOC to load the new user field configuration.

Assign the user field key AGNR/SYSTEM (edop) to the operation used for test purposes.

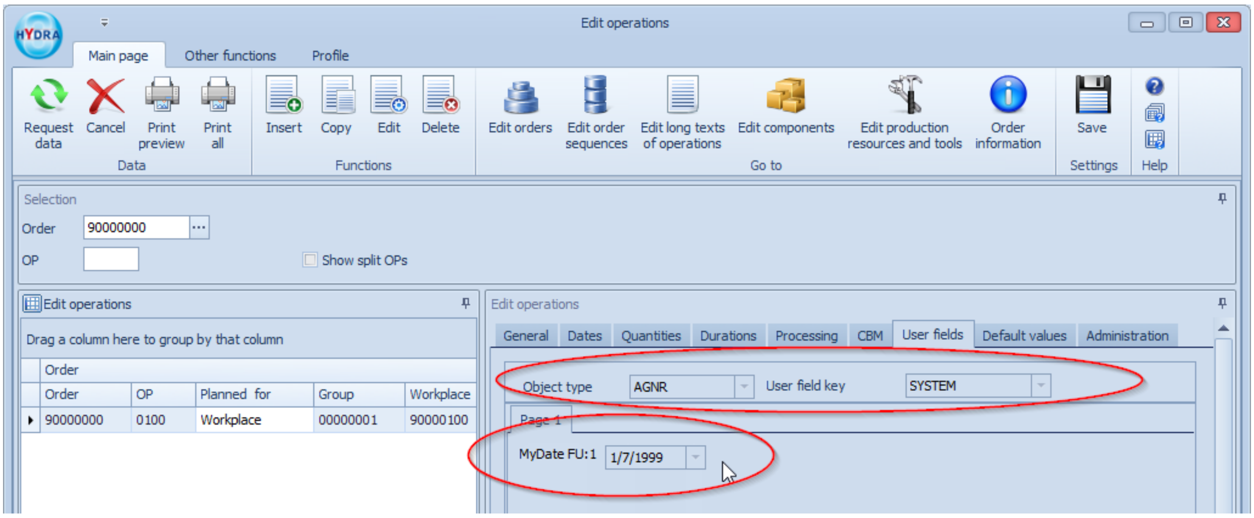


14.3.4.5 Checking the configuration

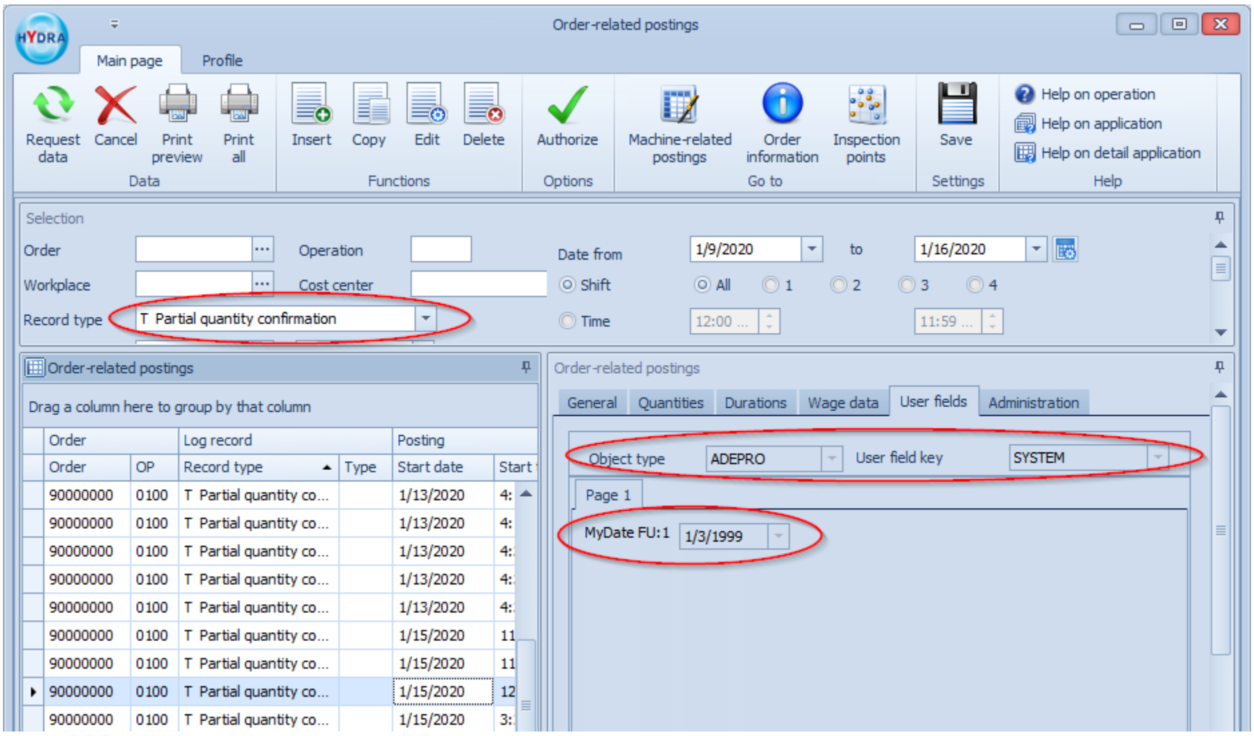
When the user field configurations are changed, you must restart the MOC to enable the change.

Check the display of the collected user field using the data previously noted down.

Edit operations (edop)



Order-related postings (oboo)



14.3.5 User field is not stored for OP

Sometimes it is unwanted that the collected user fields overwrite the user fields of the operation. If the AIP transmits the field identification ANR:SETUSRFLD with the value "N", the system does not use the collected user fields in the *Operation*, but only in the *Order-related postings*.

To this end, create a hidden field in the field configuration with the identification ANR:SETUSRFLD and the default value N. Result: The field is not visible on the AIP when collected, but is sent to the server and avoids that the user field in the operation is overwritten.

The collected user field is then only stored in the order-related posting.

Create a new field with the following data (do not change the default value for settings that are not listed in the table):

Tab	Field	Value (comment)
General	Field no.	Highest field number up to now in dialog + 1
	Text	Disable FU update of operation
	Unit	
	Information	Disable FU update of operation
	Identification	ANR:SETUSRFLD

	ID index	
Position	X pos. text	10
	Y pos. text	100
	X pos. field	400
	Y pos. field	100
Format	Alignment	Left
	Category	Input
	Input type	ALPHA
	Length	1
	Allowed characters	JN
	Default	N
Functions	Field attribute 1	MANUELL
	Field attribute 2	SETVALUE (use SETVALUE to take over the default value to the field. If you do not configure a field attribute with SETVALUE, the user field remains empty.)
Options	Visible	Deactivated (the field is not visible on the AIP, but is transmitted to the server)

14.4 Result

When all configurations are made, the following functions are available:

- When you post part quantities, you can enter a date in a user field in the workflow step *Confirmation*.
- The user field is saved in the *Order-related postings*.
- You can control if the collected user field is also written to the operation.
- The collected user field can be displayed on the MOC.